



Federal Committee on Statistical Methodology

2026 Annual Conference

March 11-12, 2026
Suitland Federal Center
Suitland, MD

Program

Hosted by:



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2026 FCSM Annual Conference

The Federal Committee on Statistical Methodology (FCSM) was founded in 1975 by the Statistical and Science Policy Branch (formerly, the Office of Statistical Policy) in the Office of Information and Regulatory Affairs in the Office of Management and Budget (OMB) to assist in carrying out SSP/OMB's role in setting and coordinating statistical policy.

The FCSM serves as a resource for OMB and the federal statistical system to inform decision making on matters of statistical policy and to provide technical assistance and guidance on statistical and methodological issues. The FCSM sponsors regular conferences (in partnership with the Council on Professional Associations on Federal Statistics), hosts seminars and workshops (often in collaboration with partner organizations such as the Washington Statistical Society), writes reports aimed at the federal statistical system and the community, and creates subcommittees and interest groups focused on topics relevant to the federal statistical community. Recent news and resources from FCSM are available at www.fcsm.gov.

The FCSM Annual Conference helps the committee achieve their major goals, which are to:

- Communicate and disseminate information on statistical practice among all federal statistical agencies;
- Recommend the introduction of new methodologies in federal statistical programs to improve data quality; and
- Provide a mechanism for statisticians in different federal agencies to meet and exchange ideas.

The conference provides a forum for experts and practitioners to discuss and exchange current methodological knowledge and policy insights about topics of current and critical importance to federal agencies as well as the Federal Statistical System as a whole.

Each day of the conference will offer papers on a wide range of topics relevant to the production, quality and use of federal statistics. Attendees from a range of backgrounds will find sessions of interest, including statistical methods, administrative data, questionnaire design, artificial intelligence (AI), and more. All sessions will include an open discussion and some sessions will include a formal discussion. Presentations will be made available on the conference website following the conference.

2026 FCSM Annual Conference

Program Committee

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Federal Committee on Statistical Methodology

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Council of Professional Associations on Federal Statistics

The Council of Professional Associations on Federal Statistics (COPAFS) is devoted to educational activities and preserving the public good represented by federal statistical collections. Since 1981, COPAFS has provided an open dialog between those who use federal statistics in professional contexts and the Federal statistical agencies that produce those statistics for the public good. Supporting organizations include professional associations, businesses, research institutes, and others that help to produce and/or use federal statistics. Our Goal: Advancing Excellence in Federal Statistics.

COPAFS' objectives are to:

- Increase the level and scope of knowledge about developments affecting Federal statistics;
- Encourage discussion within and among professional organizations to respond to important issues in Federal statistics and bring the views of professional associations to bear on decisions affecting Federal statistical programs.

In support of these objectives, COPAFS:

- Obtains information on developments in statistics through discussions with officials, attendance at congressional hearings and meetings of statistical advisory committees, engaging with the broader statistical community, and reviewing recent reports or directives affecting the Federal statistical system;
- Disseminates information and encourages discussion and action on developments in federal statistics through correspondence and presentations at COPAFS and professional association meetings, direct calls for action via email, and announcements on social media; and
- Plans and presents educational programs on uses of statistics in policy formulation, public and private decision- making, research, the distribution of products, and the allocation of resources.

COPAFS helps:

- Professional associations and other organizations obtain and share information about developments affecting federal statistical programs;
- Federal agencies to disseminate information on developments of interest to the professional community and to obtain advice about professional societies' concerns and priorities;
- Congressional offices to clarify issues and questions about the federal statistical system, to plan hearings related to federal statistical programs, and to identify experts to testify; and
- The public to learn more about the federal statistical agencies, to communicate views of data users concerning Federal statistical activities, and to obtain a better understanding of how policy and budget are likely to affect the availability of federal statistics.

2026 FCSM Schedule Overview

Wednesday, March 11

8:30 am – 10:00 am **CONCURRENT SESSIONS A**

[A-1. Improving Access to Data Through Strategic Collaborations and Innovation](#)

[A-2. Using AI to Improve Operations and Products](#)

[A-3. Innovations in Linking Health Records Across Federal Agencies](#)

[A-4. Testing New Survey Instruments, Modes, and Methods for Sensitive Topics in BJS Surveys](#)

[A-5. Building Better Questions: Insights from Cognitive Interviews, Paradata, and Experiments](#)

[A-6. Innovations in Regional Economic Measurement: Capital, Consumption, and Productivity](#)

[A-7. What is the Value? Coping with Outliers and Missingness](#)

[A-8. From Costs to Care: Economic and Population Perspectives on Health](#)

10:00 am - 10:15 am **Break**

10:15 am - 11:45 am **CONCURRENT SESSIONS B**

[B-1. Statistical Communication: Improving Value, Engagement, and Public Trust](#)

[B-2. From Freight to Tax Filings: AI Classification and Applications for Federal Data Collections](#)

[B-3. Data Linkage: Increasing Quality and Information](#)

[B-4. Mode Matters: How Design Choices Shape Survey Results](#)

[B-6. Show Me the Monetary: New Metrics for Measuring Value](#)

[B-7. SAE at Census: Bridging Data Gaps with Flagship Models and New Innovations](#)

[B-8. Methodological Advancements in Health Measurement](#)

[B-X. Signal & Noise: AI in Federal Statistics](#)

11:45 am – 1:15 pm **Lunch on Your Own**

1:15 pm – 2:45 pm **CONCURRENT SESSIONS C**

[C-1. NSDS Use Cases: Advancing Federal Mission-Critical Priorities Through Shared Services](#)

[C-2. Making Federal Data AI-Ready](#)

[C-3. Next Steps in Record Linkage: Adding Value to Existing Data and Hard-To-Measure Concepts](#)

[C-4. Modernizing the Survey of Income and Program Participation](#)

[C-5. Evaluating the Unseen: Tackling Bias and Error in Federal Statistics](#)

[C-6. State-Level Statistics at BEA: New Approaches to Trade, Government, and Gross Output Meas.](#)

[C-7. Innovative Solutions for Small Area Problems: From Machine Learning to Novel Modeling](#)

[C-8. Connecting the Dots: Linking Data for Smarter Social and Economic Measures](#)

[C-X. Meet a Member of the Interagency Council on Statistical Policy \(ICSP\)](#)

2:45 pm - 3:00 pm **Break**

3:00 pm – 4:30 pm **CONCURRENT SESSIONS D**

[D-1. Open for Collaboration and Innovation: Partnerships Across the Federal Statistical System](#)

[D-2. Keeping Secrets while Telling the Truth: Synthetic Data, Methods, and Applications](#)

[D-3. Assembling the Mosaic: Linking Data for Coverage, Poverty, and Consent](#)

[D-4. Building the Future of Juvenile Facility Data: Modernizing the Census Program](#)

[D-5. Defining, Measuring, and Monitoring Data Quality](#)

[D-6. Where in the World? Defining and Applying Measures of Geography](#)

[D-7. Precision at Scale: Modern Approaches to Small Area Estimation](#)

2026 FCSM Schedule Overview

Thursday, March 12

8:30 am – 10:00 am **CONCURRENT SESSIONS E**

[E-1. Updates on the 2030 Census Disclosure Avoidance Research Program](#)

[E-2. Estimating in an Imperfect World: Addressing Error in Federal Statistical Estimates](#)

[E-3. The 2030 Administrative Data Frame and Use Case](#)

[E-4. Modernizing a Longstanding Federal Survey: Overview of Updates to the CPS](#)

[E-6. Recent Research on Nonresponse and Stratification for the CES and CPI](#)

[E-7. Using Advanced Computational Methods for Complex Statistical Problems](#)

[E-8. Engaging Every Voice: Strategies for Understanding and Reaching All Respondents](#)

10:00 am - 10:15 am **Break**

10:15 am - 11:45 am **CONCURRENT SESSIONS F**

[F-1. How Safe Is Safe Enough? Disclosure Avoidance and Differential Privacy](#)

[F-2. Reading Between the Lines: Using Text Analysis in Bureau of Labor Statistics Surveys](#)

[F-3. Addressing Addresses: Solutions to Enterprise-Wide Address Management and Matching](#)

[F-4. From Interviewer to Interface: Experimental and Qual Research to Inform Web Surveys](#)

[F-5. Navigating Nonresponse: Modeling and Mitigation](#)

[F-6. Economic Measurement in Practice: From Consumer Finance to Land Use](#)

[F-7. Strengthening Federal Statistics Through Weighting Strategies and Adjustment Techniques](#)

[F-8. Improving Efficiency and Tackling Rising Survey Costs](#)

[F-X. Welcome to the Federal Statistical Community: Mentorship & Networking for New Federal Staff](#)

11:45 am – 1:15 pm **Lunch on Your Own**

1:15 pm – 2:45 pm **CONCURRENT SESSIONS G**

[G-1. Practical Approaches to Data Linkage and Disclosure Avoidance](#)

[G-2. Taming the Wild Text: NLP Models for Text Classification](#)

[G-3. Using Proprietary Data to Address Information Gaps in Food Statistics](#)

[G-4. Pilot Studies for Modernization: Methods, Findings, and Lessons Learned](#)

[G-6. Indigenous Data and Tribal Sovereignty: Best Practices from Across the Federal Government](#)

[G-7. From Frames to Estimates: Establishment Survey Sampling and Weighting](#)

[G-8. Optimizing Contact Strategies to Increase Response Rates](#)

[G-X. FCSM Geospatial Interest Group \(GIG\) Un-Session](#)

2:45 pm - 3:00 pm **Break**

3:00 pm – 4:30 pm **CONCURRENT SESSIONS H**

[H-1. Synthetic Data: Applications Across the Federal Statistical System](#)

[H-2. Statisticians Don't Code, They Prompt: Modernizing Methods with AI](#)

[H-4. Modernizing Survey Instruments: Evaluations of Visual Design Decisions](#)

[H-6. Innovation Before Inference – New Steps Along the Way to Estimation](#)

[H-7. Creating Accuracy: Advancing Sample Coverage, Sampling Frames, and Sample Outcomes](#)

[H-8. Engaging Respondents: Tools to Improve Participation and Data Quality](#)

Abstract Listings for Wednesday, March 11

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|-------------------------|---------------------|
| • Concurrent Sessions A | 8:30 am – 10:00 am |
| • Concurrent Sessions B | 10:15 am – 11:45 am |
| • Concurrent Sessions C | 1:15 pm – 2:45 pm |
| • Concurrent Sessions D | 3:00 pm – 4:30 pm |

Session A-1: Improving Access to Data Through Strategic Collaborations and Innovation

Organizer: Melissa Braybrooks, *Department of the Interior*

Chair: Pedro Almoguera, *Government Accountability Office*

Census Auditorium 1

Strategic Partnerships to Improve Household Income and Expenditure Surveys

Anya Stockburger, *Bureau of Labor Statistics*

Melissa Braybrooks, *Department of the Interior*

Rebecca Morrison, *Bureau of Labor Statistics*

The US Bureau of Labor Statistics measures labor market activity, working conditions, price changes, and productivity in the US economy. The US territories share the critical need for these data. The BLS produces limited statistics in partnership with some territories. The territories produce some statistics independently, such as the Household Income and Expenditure Surveys (HIES). These data on household income and spending are critical for many government and private activities such as calculating consumer inflation and conducting market research. The BLS meets this need by conducting the Consumer Expenditure (CE) Surveys on an on-going basis, resulting in annual estimates to provide a current snapshot of income and spending. By contrast, the territories conduct the HIES roughly every 10 years. This presentation begins with an overview of the BLS-produced territorial statistics and a further exploration of the labor market statistical landscape for Guam. Finally, the presentation takes a deeper dive on the HIES by comparing it to the CE. Along the way, the presentation identifies potential areas for improvement in the future.

Filling Gaps in Government Finance Data through the Survey of Native Nations

Vanessa Palmer, *Federal Reserve Bank of Minneapolis*

The Survey of Native Nations, conducted by the Center for Indian Country Development (CICD) at the Federal Reserve Bank of Minneapolis, aims to address data gaps in Indian Country to help advance the economic self-determination and prosperity of Native nations through actionable data and research. In 2022, CICD initiated a survey offering comparable information to what state and local governments receive through the U.S. Census Bureau's Census of Governments. The goal of the survey is to provide tribal governments with revenue and expenditure data that can be comparable to other tribal and local/state geographies, and summarized to inform discussions with local governments, lawmakers, and citizens. Since the launch CICD has refined and expanded the scope of the survey. The Federal Reserve Bank of Minneapolis's CICD will present information on the approach taken, observed benefits, lessons learned, and potential opportunities for extrapolating the experience to help further data gaps.

Evaluating Territorial GDP: A Comparison of the Expenditure and Production Approaches Based on an Inventory of Available Source Data

Melissa Braybrooks, *Department of the Interior*

Gross Domestic Product is a key indicator of the health of National and local U.S. economies, including the U.S. territories, whose economic sustainability is critical to achieving territorial goals and supporting U.S. National Security priorities. However, without high-quality source data, this GDP estimates may be challenged to fully serve their intended purpose. The research catalogs and evaluates available territorial source data – specifically accuracy and relevance – to identify the most suitable methodologies for producing GDP in the U.S. territories of American Samoa, Guam, the Commonwealth of the Northern Mariana Islands, and the U.S. Virgin Islands. The evaluation of territorial GDP is structured around two comparative approaches: the current U.S. expenditure method and the System of National Account's value-added (production) method used throughout several Pacific Island nations. The objective is to explore medium- and long-term opportunities for developing more robust territorial GDP estimates to serve as a high-quality economic indicator. Finally, the study provides

targeted recommendations for a coordinated federal interagency strategy to address territorial data needs and enhance the quality of GDP estimates.

Bridging Data Availability and Data Use for Native Communities

Emily Silverman, *Department of the Interior*

Ann Miller, *Department of the Interior*

The Census Bureau provides a variety of data tables from the American Community Survey (ACS) summarized to American Indian and Alaska Native geographies. These data can support analyses that (1) describe the overall economic conditions at the tribal level in relation to adjacent economies; (2) provide an understanding of the economic interconnections among tribal, regional, and national economies; and (3) assess the impacts of federal government investment on tribal economies. But due to imperfect alignment of geographies and the communities of interest, as well as problems of data reliability for small geographies, caution is needed in the use and interpretation of the data. The Department of the Interior is developing guidance, reference tables, and documentation to support and simplify appropriate use of ACS and other data sources for describing Native populations. This effort will empower users to make informed decisions and conduct accurate analyses. Bringing awareness and transparency to how federal data sets represent, or may be limited in representing, specific communities is essential for helping decision makers draw appropriate conclusions from federal data sets.

Opportunities to Improve Statistics through Governance Procedures for Alternative Source Data

Melissa Braybrooks, *Department of the Interior*

Ann Miller, *Department of the Interior*

Emily Silverman, *Department of the Interior*

An objective of producing federal statistics is to provide reliable, relevant information that informs efficient and effective decision-making. To achieve this, federal agencies must carefully balance key data quality criteria - accuracy, timeliness, consistency, and validity - when selecting source data and designing methodologies. For example, alternative administrative data sources may provide timeliness, lower costs, or address an existing data gap, but often lack the methodological rigor typically required for the incorporation into official statistics. This discussion emphasizes the importance of establishing clear criteria for putting forward, reviewing, and incorporating alternative source data, which often involve trade-offs in data quality. The aim is to help navigate the 'grey area' of balancing statistical standards and the opportunity to incorporate alternative source data, to ultimately meet data users' needs for high-quality information.

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Wednesday, March 11, 2026

8:30 AM

Session A-2: Using AI to Improve Operations and Products

Organizers: Ellen Galantucci, *Veterans Administration* & Brandon Kopp, *Bureau of Labor Statistics*

Chair: Mark Xu, *Internal Revenue Service*

Census Auditorium 2

StateChat – How the State Department Released the First Enterprise AI Chatbot for Sensitive Documents

Matthew L Key, *U.S. State Department*

Ellis S. Obrien, *General Services Administration*

The U.S. Department of State released StateChat, the first enterprise GenAI chatbot approved to handle sensitive documents, including Sensitive but Unclassified (SBU) information. StateChat leverages a cutting-edge software as a service (SaaS) technology to enhance operational efficiency while maintaining robust security and governance standards.

The SaaS platform provides a secure foundation for StateChat through inherited FedRAMP security controls, granular access permissions and other features. These ensure comprehensive security and monitoring, enabling safe use of AI in sensitive contexts.

StateChat is designed to provide contextual responses based on authoritative and proprietary State Department resources, such as the Foreign Affairs Manual, State Department cables, Active Directory, and official news briefings. Users can also upload their own documents for tailored interactions, further expanding the tool's utility while focusing LLM context windows on curated information to reduce hallucination.

To ensure responsible AI deployment, StateChat undergoes rigorous testing aligned with the National Institute of Standards and Technology's (NIST) Responsible AI Framework. This includes assessments for model confabulation and bias captured in a comprehensive risk register. These efforts reflect the Department's commitment to trustworthy and transparent AI practices.

The development of StateChat represents a significant milestone in the Department's broader AI strategy, and propels innovation and efficiency as a Department and Administration priority in line with Executive Order 14179 and America's AI Action Plan. By leveraging FedRAMP-approved SaaS products, the Department is able to approve, build, and release AI applications quickly. The Department is developing crisis management applications, classified AI chatbots, and other advanced applications to further meet the needs of its workforce. This presentation will explore the technical infrastructure, security features, and testing protocols behind StateChat and the SaaS platform, as well as the Department's vision for expanding AI capabilities to promote, sustain, and enhance American diplomacy and national security.

Implementing the Generic Statistical Information Model at the Census Bureau in Support of Automated Data Ingestion

Chris Carrino and Benjamin Schultz, *U.S. Census Bureau*

Taylor Wilson, *Reveal Global Consulting*

Amaia Mayes, *U.S. Census Bureau*

Michael Padgett, *U.S. Census Bureau*

Anthony Rosati, *U.S. Census Bureau*

Traditionally, the Census Bureau has stored and governed metadata on a program-by-program basis, implementing individual solutions for individual programs. These programs rely on internal and external data providers who use disparate methods, terminologies, standards and levels of effort to define their metadata. Over time, this has produced siloed systems that cause redundancies and work as a roadblock to organizational standardization. The Generic Statistical Information Model (GSIM) provides a conceptual framework to manage data and metadata used in the production of official statistics. We have begun implementing GSIM across the collection and processing stages of the data life cycle for both Demographic and Economic statistics. This presentation will discuss how we have implemented and extended GSIM to provide further quality to our data products, to add efficiency through the reuse of metadata, and to enable the reproducibility of our results.

Machine Learning using Tax Information: Improving Entity Classification in IRS Form 1041

Sudeshna Roy, *Internal Revenue Service*

The IRS' Statistics of Income (SOI) division publishes data from Form 1041, U.S. Income Tax Return for Estates and Trusts, which provides statistical information on the income, deductions, and tax liability of various types of trusts and estates (collectively known as entities) that file this form. This data is widely used by SOI stakeholders, researchers and policymakers to study wealth and income concentration and understand effects of the tax code on the filing population. Since entity type dictates tax treatment, reporting requirements and tax burden allocation to beneficiaries, there has been an increase in requests for entity information to be included in SOI samples and tables.

The current method of identifying the filing entity type relies on a code-lookup system that suffers from issues such as code overlaps, errors and unclear assignment instructions, resulting in 22-25% of returns lacking entity

information. Furthermore, complex structures of entities make it difficult to distinguish between them using rule-based methods. To address these limitations, this research explores the use of supervised machine learning models that leverage information from taxpayer-entered line entries on the return to predict and classify entity types. Preliminary results show that the models can capture complex relationships between form fields to distinguish between entities with high accuracy and identify important variables that further our understanding of entity structures.

Importantly, this work demonstrates our ability to use machine learning to harness information from non-text entries in tax returns to characterize the filing population, which will have widespread research benefits. In this talk, we will describe the methodology of the classification, discuss results and insights gained, and explore how this methodology has the potential to enhance and expand future SOI products and offerings.

AI is My Copilot: The Bureau of Economic Analysis (BEA) Experience With an AI Coding Assistant

Brian Quistorff, *Bureau of Economic Analysis*

Amanda Lyndaker, *Bureau of Economic Analysis*

Artificial Intelligence (AI) holds much promise for fundamentally shifting the “business” of government. The Bureau of Economic Analysis (BEA) is exploring AI’s potential to increase efficiency and improve workloads while upholding our fundamental responsibilities as a federal statistical agency. This presentation describes our experiences with GitHub Copilot, an AI tool that suggests code and functions in real time, and how this work sets the stage for implementing AI more widely.

RATIONALE. BEA is undertaking a multi-year initiative to adopt Python for all desktop production of official statistics. This requires 300 economists, statisticians, accountants, and data scientists to re-engineer hundreds of applications. While BEA has vast experience in FAME, SAS, STATA, and other similar technologies, most staffers are new to Python. Using AI tools to close this knowledge gap could supercharge code conversion and free up valuable time for analysis and innovation that improves our statistics.

RESEARCH. Recently, BEA launched a groundbreaking pilot to answer the question of whether GitHub Copilot could enhance analysts’ ability to reconfigure legacy code. This effort also explored operational implications like licensing, security, and the broader AI governance framework.

RESULTS. Pilot participants reported significant time savings (25-50 percent) and better quality of work life (e.g. reducing frustrations with error handling). Nearly all (19 of 20) users agreed that BEA should make GitHub Copilot more widely available to assist with Python conversion. Based on the success of the pilot, we have now deployed GitHub Copilot to nearly 150 staff.

RELEVANCE. The lessons learned from the pilot and ongoing implementation continue to inform BEA’s evolving AI practices and policies and better position us to embrace new technologies. This work also provides critical insights for other agencies looking to implement AI successfully while maintaining strict data integrity and security standards.

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Wednesday, March 11, 2026

8:30 AM

Session A-3: Innovations in Linking Health Records Across Federal Agencies

Organizer: Victoria Udalova, *U.S. Census Bureau*

Chair: Victoria Udalova, *U.S. Census Bureau*

Census Auditorium 3

Results and National Estimates from Linkage of the National Ambulatory Medical Care Survey Health Center Component to the Transformed Medicaid Statistical Information System

Lello Guluma, *National Center for Health Statistics*

Cordell Golden, *National Center for Health Statistics*

Brian W. Ward, *National Center for Health Statistics*
Sonja N. Williams, *National Center for Health Statistics*
Cindy Zhang, *National Center for Health Statistics*

The National Ambulatory Medical Care Survey Health Center (NAMCS HC) Component is a nationally representative establishment survey that collects data on ambulatory medical care visits to health centers throughout the 50 U.S. states and District of Columbia. In 2021, the NAMCS HC Component modernized its data collection and began collecting clinical visit data from health centers using electronic health record (EHR) submission. With this shift from manual, in-person, data abstraction to data collection via EHR submission, for the first time in its survey history the NAMCS HC Component is now collecting personally identifiable information (PII). The collection of these PII data has allowed for the National Center for Health Statistics to link NAMCS HC Component data with external data sources at the patient level. Data from the survey have previously been linked to the National Death Index and U.S. Housing and Urban Development administrative data. However, this past year a linkage to the Transformed Medicaid Statistical Information System (T-MSIS), a resource for comprehensive and quality Medicaid data, was performed. In this presentation the process and methodology behind the matching techniques used to link the NAMCS HC Component and T-MSIS are discussed. An overview of the linkage methodology and results of the linkage process will also be presented. Finally, using the linked data, nationally representative estimates of health center visits, by patients with and without Medicaid coverage, are presented.

How Representative Are Electronic Health Records? A Record Linkage Study Using Individual-Level Census Data

Victoria Udalova, *U.S. Census Bureau*
Aubrey Limburg, *U.S. Census Bureau*
Tim Carey, *University of North Carolina at Chapel Hill*
Anisha Ganguly, *University of North Carolina at Chapel Hill*
John P. Powers, *University of North Carolina at Chapel Hill*
Emily Pfaff, *University of North Carolina at Chapel Hill*
Barbara Entwisle, *University of North Carolina at Chapel Hill*

Electronic health records (EHRs) may be a promising alternative to traditional health surveys for population health surveillance due to their detailed health and patient information, low cost, and minimal respondent burden. However, concerns about the population representativeness of EHRs raise questions about their validity for public health monitoring and tracking social determinants. This study addresses these concerns by evaluating the representativeness of EHRs from UNC Health, a large integrated health delivery system in North Carolina, by linking individual-level EHRs (2018-2022; $n = 2.12$ million unique patients) with individual-level microdata from the nationally representative American Community Survey (2018-2022). Specifically, we evaluate how demographic factors (age, sex, race/ethnicity), socioeconomic factors (education, employment, poverty, food stamps, public assistance), and health insurance impact the likelihood that a NC ACS respondent will appear in the UNC Health EHRs. Results indicate that although UNC Health patients are not fully representative of the state population, selection biases are generally modest and align with known patterns of healthcare utilization (e.g., overrepresentation among females, older, adults, and individuals with health insurance). Moderate selection is observed by race/ethnicity and socioeconomic status, with overrepresentation at both the high and low ends of the socioeconomic spectrum. Since key demographic and insurance variables are available in the EHRs, weighting adjustments can and should be implemented to mitigate potential bias. These findings provide cautious reassurance for the use of EHR data in population health monitoring while demonstrating the value of evaluating and improving the utility of EHRs in public health research through linkages with individual-level nationally representative data.

The opportunities and challenges tied to the NIA LINKAGE Program with CMS

Theresa Kim, *National Institutes of Health*
Irim Azam, *National Institutes of Health*

Since 2021, the Centers for Medicare & Medicaid Services (CMS) and the National Institute on Aging (NIA) have shared a data use agreement (DUA) called LINKAGE where NIA awardees may access datasets from CMS

through a secure data enclave. One of the greatest opportunities with the LINKAGE program is to provide a real-world data (RWD) option for more comprehensive information about patients in clinical trials, NIA-supported survey data, and electronic health records. The talk will discuss how LINKAGE allows for comparisons across trials where longer-term follow-up is necessary for measuring patient-centered outcomes, the opportunities for comparisons between intervention and treatment groups in aging clinical trials, and how we may understand any reasons for attrition in aging research clinical trials yet affirm that such RWD are enhancements but not a full substitute for clinical trial data. Examples will come from NIA-supported research.

Using Augmented Twin Neural Networks to Match Occupational Injury and Illness Data

Elan Segarra, *Bureau of Labor Statistics*

When record linkage efforts involve complex characteristics there is potential for machine learning (ML) techniques to succeed where traditional probabilistic linkage methods (e.g. Fellegi-Sunter) might fall short. However, there can still be pre-processing (e.g. geocoding) and hand-picked metrics (e.g. edit distances) that can further improve linkage outcomes beyond ML models' abilities. We present a fusion of these sides we are calling an Augmented Twin Neural Network. This approach leverages the nonlinear flexibility of Twin Neural Networks while adding additional layers to allow for hand curated comparators that may be difficult for ML optimizers to implicitly identify without sufficiently large, labeled data sets. The framework is used to match businesses from the BLS Survey of Occupational Injuries and Illnesses to businesses in the OSHA Injury Tracking Application data. Difficulties in matching company names and addresses and the existence of multi-establishment firms make this a prime application for testing. Linkage outcome metrics of this augmented method are compared with the results from both probabilistic and standard ML methods to illustrate the added benefits.

Frameworks for Streamlining Decisions about Data Linkage

Rebecca F. Rosen, *National Institutes of Health*

Valerie A. Cotton, *National Institutes of Health*

Elizabeth M. Clerkin, *National Institutes of Health*

The Eunice Kennedy Shriver National Institute for Child Health and Human Development (NICHD) Office of Data Science and Sharing (ODSS) is leading an effort to develop frameworks and tools to support responsible individual-level record linkage for research with NICHD populations. The first framework is a checklist of governance and technical decisions that must be made prior to designing and implementing a strategy for linking data from multiple sources and sharing and using the linked datasets for research. The checklist was developed from a qualitative assessment of 13 extant record linkage implementations and has been incorporated into several NIH funding announcements. The second framework is a metadata schema that enabled the development of tools to collect, exchange, and visualize dataset-level governance information in a standard way to inform whether certain datasets can be linked, and if so, what rules and controls apply to the resulting linked dataset. The data governance metadata schema is based on the Open Digital Rights Language (ORDL) information model and provides a new data governance profile based on an analysis of data governance information from federally funded clinical and administrative datasets relevant to biomedical research. Data governance profile terms were mapped to existing data standards where possible. The ultimate goal of this work is to protect participant privacy, manage risks, address ethical considerations, and respect participant trust while maximizing the value of shared data.

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Wednesday, March 11, 2026

8:30 AM

Session A-4: Testing new survey instruments, administration modes, and methods for measuring sensitive topics in criminal justice data collections: Cognitive interviews for Bureau of Justice Statistics' surveys

Organizer: Alexandra Thompson, *Bureau of Justice Statistics*

Chair: Alexandra Thompson, *Bureau of Justice Statistics*

Census Conference Rooms 1-2

Access to Justice Design and Testing Program: Identifying the nexus of civil and criminal legal needs

Emilie J. Coen, *Bureau of Justice Statistics*

Many individuals who interact with the criminal justice system face related civil legal issues both because of the interaction or due to connected circumstances. The Access to Justice Design and Testing Program (AJDTP) was launched to support efforts to identify and fill gaps in its statistical coverage of access to justice for civil legal needs and ultimately inform an understanding of the intersection between the criminal justice and civil justice systems. Part of AJDTP's objective is to design and pilot a household based Civil Legal Needs Survey (CLNS). Upon developing the CLNS instrument, cognitive and usability testing were conducted to assess respondent understanding of questions and how the survey could be improved to maximize validity and minimize measurement error across respondents. The presentation will discuss the process of drafting the instrument and cognitive testing protocol, respondent recruitment, testing procedures, and preparation for the development of a sampling strategy for a nationwide pilot test.

Campus Climate Victimization Survey: creating a survey to measure the victimization experiences and attitudes of all postsecondary students

Alexandra Thompson, *Bureau of Justice Statistics*

The 2022 Violence Against Women Act Reauthorization Act directed that the "Secretary of Education, in consultation with the Attorney General ... shall develop, design, and make available [an] online survey tool regarding postsecondary student experiences with domestic violence, dating violence, sexual assault, sexual harassment, and stalking." In response, the Bureau of Justice Statistics (BJS) is developing a survey instrument that measures the prevalence of these crimes at postsecondary institutions, in addition to the characteristics of these incidents and students' attitudes on these subjects. After drafting an instrument, cognitive testing was completed to ensure the questions were being understood as intended. This presentation will cover the drafting of the survey instrument, development of cognitive testing probes, strategies for recruitment, lessons learned, and next steps.

Tribal Crime and Justice: Improving Law Enforcement, Court, and Jails data in Indian Country

Steven Perry, *Bureau of Justice Statistics*

Carol Ann Sharo, *Bureau of Justice Statistics*

There is a paucity of reliable and timely data on tribal crime and justice in Indian country. The Tribal Law and Order Act of 2010 requires the Bureau of Justice Statistics (BJS) to establish and implement a tribal crime data collection system. BJS cognitively tested the next Census of Tribal Law Enforcement Agencies (CTLEA) and Census of Tribal Courts Systems (CTCS), the first web-based survey collection for these topics, among a sample of tribal nations. BJS prioritized improving information on criminal case counts and court filings, juvenile justice and delinquency, and staffing and personnel retention in tribal justice agencies. This presentation will discuss consultation strategies and data needs for the CTLEA and CTCS.

Testing the Item Count Technique for Estimating Perceived Innocence: Cognitive Interviews with Incarcerated Participants

Emily D. Buehler, *Bureau of Justice Statistics*

Melissa A. Cidade, *Bureau of Justice Statistics*

In recent testing to support updating the Survey of Prison Inmates (SPI), a cross-sectional survey of the state and sentenced federal prison populations, the Bureau of Justice Statistics (BJS) embedded an experiment to understand the challenges and opportunities in using Item Count Technique (ICT) Methodology to estimate self-reported innocence or guilt of incarcerated participants. This presentation will provide an overview of ICT and the special considerations for using this method with a population of incarcerated adults. We will describe the experimental conditions used to test two sensitive items – one that articulated guilt as having committed one or more crimes for which they are currently serving a sentence and the other a statement of innocence as having not committed any of the crimes for which they are currently serving a sentence. We will describe the cognitive interview protocol used to measure respondents' comprehension and comfort with the ICT format generally and the sensitive items of interest. Finally, we will discuss implications for future use of ICT in surveys of incarcerated populations.

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Wednesday, March 11, 2026

8:30 AM

Session A-5: Building better questions: Insights from Cognitive Interviews, Paradata, and Experiments

Organizer: Aleia Clark Fobia, *National Oceanic and Atmospheric Administration*

Chair: Aleia Clark Fobia, *National Oceanic and Atmospheric Administration*

Census Conference Rooms 3-4

An Examination of Construct Validity When Survey Items Omit Information Essential to the Question-Response Process

Stephanie Willson, *National Center for Health Statistics*

Through the use of cognitive interviewing, staff at the Collaborating Center for Question Design and Evaluation Research recently evaluated questions for inclusion on the Rapid Surveys System at the National Center for Health Statistics. One series of questions focused on signs and symptoms of stroke. The main purpose of this series was to assess the action respondents would take if they believed themselves to be having a stroke. This was operationalized as a battery of physical symptoms, many of which – but not all – are associated with stroke. However, survey items did not specifically link any of the symptoms to a possible cause, stroke or otherwise. Respondents were simply asked what they would do if they experienced the symptom. Response options ranged from calling 911 immediately to waiting a day before deciding what to do. Cognitive interviews revealed that the questions lacked essential information, thereby requiring speculation on the part of respondents. This presentation will describe the question-response process undertaken by respondents when answering these questions. It will discuss the meaning behind the survey answers and demonstrate how the underlying construct relates less to symptoms and more to symptom causes. The presentation will conclude by arguing that, to the extent that respondents arrived at different conclusions about possible causes associated with the symptoms in question, there was systematic interpretive variation of the construct.

Using Paradata to Examine Response Changes to the Hispanic Origin Question

Ana I. Sánchez-Rivera, *U.S. Census Bureau*

Rae Ellis, *U.S. Census Bureau*

Internet paradata are data collected to support instrument functioning during the response process that record

respondent actions (e.g., clicks, keystrokes, logins) as individuals complete an online survey. While survey research typically emphasizes participants' final responses, paradata provide a window into the processes by which those responses are formed. These data offer a unique perspective on how respondents interact with survey instruments and navigate demographic questions.

Internal Census Bureau usability tests and internet self-response paradata research has shown that the Hispanic origin question was among the most difficult items for respondents in the 2020 Census. Challenges included longer median response times, frequent use of help text and higher rates of edit warnings across at least seven languages, including Spanish. These are also supported by cognitive testing interviews with Hispanic origin respondents in both English and Spanish. Building on these findings, the present study uses demographic survey paradata to analyze patterns of changes in responses to the Hispanic origin question.

Analyses focus on identifying when and how respondents revise their answers and instances of uncertainty indicated by selecting and reselecting response categories, extended response times, returning to the question or use of help text. By highlighting patterns of response change, this research demonstrates the potential of paradata to inform the study of demographic data collection and to enhance our understanding of how respondents engage with complex survey items.

Reference Period and Employment Reporting: Evidence from the 2022 American Community Survey Content Test Linked to Administrative Records

Robert Munk, *U.S. Census Bureau*

Matthew Pesner, *U.S. Census Bureau*

Clayton Gumber, *U.S. Census Bureau*

The American Community Survey (ACS) provides national, state, and local labor-market statistics used for policymaking and resource allocation. Yet declining response rates are straining the survey and increasing reliance on imputation. Administrative records can strengthen ACS's imputations, but a key obstacle is reference-period misalignment: the ACS uses the prior 12 months in its annual employment measure, while administrative sources use the calendar year, hindering integration. In this project, we evaluate whether aligning the ACS employment reference period to the calendar year—matching most administrative records—improves administrative alignment at the possible cost of reduced reported data quality.

We use the 2022 ACS Content Test, which randomized the annual employment reference period to the prior calendar year (treatment) or the prior 12 months (control). This change poses a salience-recall tradeoff with an uncertain net effect: calendar-year wording may better match how people conceptualize employment (higher salience), but it lengthens recall and may reduce accuracy. We link ACS-CT respondents to Longitudinal Employer-Household Dynamics (LEHD) data which is derived from Unemployment Insurance records. We assess agreement across treatment groups at the extensive margin (employment reported in both sources or in neither) and the intensive margin (ACS-reported weeks versus LEHD employment quarters). We examine heterogeneity by demographic and economic characteristics and work intensity, and we account for the two reference periods that span different phases of the COVID-19 recovery.

This study provides the first nationally representative evidence on the quality of reported ACS employment data using linked administrative data and the first experimental test of an ACS employment reference-period change. For the ACS, we deliver estimates of how calendar-year wording would affect reported data accuracy. More broadly, our results inform survey design when integrating administrative records, balancing respondent recall burden against administrative alignment.

The Iterative Survey Design Process for Data Collection of Emerging Economic Topics and Trends

Katie Beardall, *U.S. Census Bureau*

Temika Holland, *U.S. Census Bureau*

The U.S. Census Bureau's Business Trends and Outlook Survey (BTOS) collects survey responses every two weeks from a representative sample of all single location, non-farm businesses in the United States, Puerto Rico, and Washington D.C. This high frequency survey allows for more timely, granular data products that provide responsive measures of business conditions and outlooks as social and economic conditions change, creating new challenges and opportunities for businesses. For example, Work from Home (WFH) increased during and following the Covid-19 pandemic, however, more comprehensive and complete data of its

prevalence/use and the policies surrounding its implementation remain unclear. As such, there was the need to plan, test, and field a supplement of questions about WFH, and using the BTOS is an effective way to gather novel data. Moderated and unmoderated cognitive testing techniques were used to evaluate and revise proposed WFH content. Following the deployment and data collection of the WFH supplement, debriefing interviews were conducted with survey respondents to assess its performance in the field.

This presentation will discuss the results of the iterative evaluations used to inform the design of the final Work from Home supplement. Additionally, we will share results from the survey instrument's deployment as well as feedback from respondents to outline the overall performance of the supplement in the field. This presentation intends to highlight and share lessons learned for using a multi-method, iterative design process for the development of effective, relevant questions about emerging economic topics in business surveys.

Meet Touchpoints: The New Customer Satisfaction Measurement Platform on the Bureau of Labor Statistics Website

William Wiatrowski, *Bureau of Labor Statistics*

The Bureau of Labor Statistics (BLS) has obtained customer feedback on its website for two decades. In 2024, BLS transitioned to Touchpoints after the previous vendor discontinued service. Touchpoints is a shared service provided by GSA, which provides a streamlined and highly flexible system to collect satisfaction measures from website visitors. Either through a pop-up survey from a sample of visitors or through user-initiated entries, data users answer up to 11 questions, which is a large decrease from the previous service with over 30 questions. The new questions focused largely on satisfaction and usability. Respondents are asked to categorize what they are looking for, such as latest numbers or career information, and to classify themselves, such as businessperson or student. The last question asks: Based on your visit to BLS.gov today, how much would you agree or disagree that the Bureau of Labor Statistics is a trusted source of information, followed by an open-ended text box to suggest website improvements.

The presentation will touch on the planning that went into transitioning to Touchpoints, decisions on which questions to ask, and how the data is useful to BLS. Aided by a nifty dashboard created by a BLS data scientist, the presentation will include a variety of cross-tabulations and focused results. We show how the system can provide data for specific web pages or groups of pages, such as the Consumer Price Index pages, (overcoming challenges when information and data span multiple webpages) or specific dates such as Employment Situation release day. We'll also look at some of the more "interesting" open-ended comments – providing additional context and challenges to our quantitative measures.

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Wednesday, March 11, 2026

8:30 AM

Session A-6: Innovations in Regional Economic Measurement: Capital, Consumption, and Productivity

Organizer: Kyle Hood, *Bureau of Economic Analysis*

Chair: Kyle Hood, *Bureau of Economic Analysis*

BLS Auditorium A

The Role of R&D in the Economy: Insights from BEA's Experimental R&D Satellite Account

Ledia Guci, *Bureau of Economic Analysis*

Christian Awuku-Budu, *Bureau of Economic Analysis*

Gabriel Medeiros, *Bureau of Economic Analysis*

Dirk van Duym, *Bureau of Economic Analysis*

With the 2013 comprehensive revision of the National Income and Product Accounts (NIPAs), the Bureau of

Economic Analysis (BEA) reclassified research and development (R&D) from an intermediate service input to a capital asset, a change that increased both investment and GDP. This change recognized the long-lasting contribution of R&D to the economy, enabling economists to measure its impact on productivity and economic growth within a framework consistent with other forms of capital.

More recently, BEA, in collaboration with the National Science Foundation (NSF), has developed a new R&D satellite account that focuses on R&D production. Experimental statistics from this account, available for 2012–2023, include measures of value added, compensation, and employment associated with the production of R&D output.

This presentation will discuss several methodological contributions such as integrating disparate R&D data sources into a cohesive national and geographic framework, improving the establishment allocation for R&D, and building up estimates of R&D value added based on its components consistent with NIPA concepts. It will also showcase results for the 2012–2023, highlighting R&D trends at the national, state, and sector level. These new data offer valuable insights for economic development, local government planning, business investment and other decision making.

Towards County Personal Consumption Expenditures

Mahsa Gholizadeh, *Bureau of Economic Analysis*

Kyle Hood, *Bureau of Economic Analysis*

This project introduces a new framework for producing county-level Personal Consumption Expenditures (PCE) statistics. The paper outlines the data sources and methodological innovations used to construct these statistics, emphasizing the integration of diverse county-level datasets. Key inputs include administrative data on receipts and wages, county-level expenditures on medical care, housing cost data from the American Community Survey, and other relevant local economic indicators.

A central methodological contribution is the incorporation of a residency adjustment based on inter-county spending flows. This adjustment leverages transaction-level data—primarily from card purchases—where both the purchaser's and merchant's locations are known, allowing for a more accurate attribution of consumption to the county of residence rather than the point of sale.

In addition to describing the construction of the statistics, we present a set of exploratory "research" estimates to illustrate the utility and potential insights of the new series. These preliminary results highlight geographic variation in consumption patterns and demonstrate the feasibility of producing consistent, granular PCE estimates at the county level.

Evaluating Disaggregated Capital Stock Statistics for U.S. States

Gonca Senel, *Bureau of Economic Analysis*

Christian Awuku-Budu, *Bureau of Economic Analysis*

Kyle Hood, *Bureau of Economic Analysis*

Daniel Kwon, *Bureau of Economic Analysis*

Daniel Thang, *Bureau of Economic Analysis*

The Bureau of Economic Analysis (BEA) publishes national statistics on fixed capital stocks, yet no official measures exist at the state level, limiting regional economic research. This paper explores alternative methods to construct state-level capital stock series consistent with national totals and evaluates their performance. We develop three measures that vary in their allocation strategy: (1) allocating national industry capital stocks to states using value added shares, (2) allocating investment using value added shares and applying the perpetual inventory method (PIM), and (3) allocating investment using state-level capital spending from the Annual Survey of Manufactures and constructing stocks via the PIM. We assess these measures against theoretical and empirical benchmarks, focusing on (i) the joint dynamics of capital, employment, and compensation, and (ii) the predictive relationship between Tobin's Q and subsequent labor market outcomes. The results provide mixed evidence: while no method is uniformly superior, the third approach more closely aligns with theory and demonstrates stronger predictive validity. Notably, high Tobin's Q is associated with rising compensation but a weaker and less persistent employment response, highlighting an unresolved puzzle in the interaction of capital, labor, and productivity at the regional level. These findings underscore both the feasibility and the challenges of constructing state-level capital stock measures, and they point to the importance of using data tied directly to capital purchases in order to capture underlying economic dynamics.

Revisions to States Gross Domestic Product and Personal Income

Eva de Francisco, *Bureau of Economic Analysis*

Mauricio Ortiz, *Bureau of Economic Analysis*

During all periods of the business cycle, including crisis such as the COVID-19 pandemic, the private sector and policy makers rely on the Regional Economic Accounts that the Bureau of Economic Analysis (BEA) produces to provide a timely, comprehensive, and accurate picture of the economic conditions within the different regions of the U.S. and inform their economic decisions.

The two main measures the public and policy makers rely on to grasp the pulse of the economy at the regional level are gross domestic product (GDP) and personal income (PI). GDP by state measures the value of goods and services produced in a state by industry, while PI by state measures the income received by all persons in a state from all sources – from participation as laborers in production, from owning a business or property, from ownership of financial assets, and from government and business in the form of transfers. Since both the industry composition of GDP and the different sources of personal income varies by state, both measures complement each other. Moreover, the timing of when BEA receives source data for the different components of each estimate also varies. For example, for wages and salaries, preliminary estimates use monthly data from the Current Employment Statistics (CES) program of the Bureau of Labor Statistics (BLS), while later vintages use the more comprehensive Quarterly Census of Employment and Wages (QCEW) also from BLS. Meanwhile, to measure the value of production in manufacturing or the personal income from financial assets, the BEA has to wait for more than a year to incorporate data from the Census Bureau and the Internal Revenue Service (IRS).

This study focuses on the reliability of the different vintages of both measures through the fourth quarter of 2023. Throughout this study, we will systematically analyze the quarterly growth rates of states GDP and PI data to assess the reliability of preliminary estimates in relation to later vintages.

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Wednesday, March 11, 2026

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Session A-7: What is the Value? Coping with Outliers and Missingness

Organizer: Brad Rhein, *Bureau of Labor Statistics*

Chair: Julie Gershunskaya, *Bureau of Labor Statistics*

BLS CTC 2A

Determining Edit Limits for an Agricultural Survey Using Outlier Detection Methods

Darcy Miller, *National Agricultural Statistics Service*

Yumiko Siegfried, *National Agricultural Statistics Service*

Megan Lipke, *National Agricultural Statistics Service*

Kay Turner, *National Agricultural Statistics Service*

The United States Department of Agriculture's (USDA) National Agricultural Statistics Service (NASS) conducts over 300 surveys each year. Many of these surveys rely on pre-assigned upper and lower limits to identify reported values that are too large or too small and must be edited. These hard limits are currently assigned manually by subject matter experts, and when values fall outside of the limit range, they are flagged to be reviewed and manually changed by statisticians. NASS seeks to minimize this manual editing by automating most edit and imputation actions. Currently, research is being conducted to evaluate several outlier detection methods to create hard edit limits using data-driven methods. This paper evaluates the application of four outlier detection methods – Cook's Distance, local outlier probability (LoOP), isolation forest (IF), and fuzzy HRT in determining edit limits using a case study from the September 2024 Crops APS (Agricultural Production Survey). We summarize the characteristics of each method; review the current edit limits; present the application results; and discuss next steps.

Hierarchical Bayesian spatio-temporal models for mass imputation of the CPI in small areas from sparse survey data

Vladislav Beresovsky, *Bureau of Labor Statistics*

Terrance D. Savitsky, *Bureau of Labor Statistics*

Jeffrey M. Gonzalez, *Bureau of Labor Statistics*

There is an interest in estimating Consumer Price Indexes (CPI) for small Core-Based Statistical Areas (CBSA) and states. Currently, consumer prices are sampled in a geographically sparse set of CBSA with the goal of providing reliable index estimates for nine Census Divisions and 23 large metropolitan areas. We use spatial-temporal hierarchical Bayesian models with the idea that accounting for local spatial-temporal correlations will compensate for the geographical sparseness of the collected data and will allow for reliable estimation in small areas. Our research presented at the 2024 JSM demonstrated the utility of accounting for spatial correlations. In this paper we investigate if learning temporal trends from time series will further improve robustness of nowcasting in CBSA and states. We compare model-based estimates of fuel prices from the CPI survey with estimates from the large administrative dataset.

Who Does What Now? Payroll-Based Occupation Imputation to Track AI-Driven Transitions

Michael Dalton, *Bureau of Labor Statistics*

We build a multi-state worker-firm quarterly panel from Unemployment Insurance (UI) wage records to measure how exposure to generative AI assigned at the occupation level relates to earnings, mobility, and employment dynamics. A central contribution is a machine-learning framework to recover longitudinal worker occupations in administrative wage records that typically lack reliable occupation codes. The approach is semi-supervised: (i) we construct worker and job-spell features from wage trajectories, job histories, and employer context; (ii) we cluster or model workers' latent "types" using these features; and (iii) we probabilistically map latent types to SOC occupations using external anchors: OEWS occupation wage profiles and employer-specific occupation priors implied by OEWS employer-occupation distributions and QCEW-linked employer characteristics. This yields time-consistent worker-level occupation labels (with uncertainty) that support occupation-to-occupation transition measurement within and across firms. We then merge publicly available occupation-level AI-exposure indices and estimate event-study/difference-in-differences designs, leveraging within-employer comparisons to absorb firm shocks. Outcomes include worker earnings, separations, job-to-job moves, and heterogeneity by tenure and wage rank, alongside firm headcount, wage bills, entry wages, and wage dispersion. The inferred occupation histories allow tests of mechanisms such as within-firm task/role reshuffling versus net employment contraction, with particular attention to early-career roles.

Nonresponse Imputations and Related Measurement Issues in the CPI for Shelter

Hugh Montag, *Bureau of Labor Statistics*

Lara Loewenstein, *Cleveland Federal Reserve*

Randal Verbrugge, *Cleveland Federal Reserve*

Rent underpins 32% of the Consumer Price Index. However, in recent years around 35% of rent changes in the housing survey are imputed, either due to non-response or vacancy. Using the housing survey microdata, we show that observations are not missing at random and observed rent changes are correlated with characteristics not used in the imputation. Performing non-response imputations based on renter tenure length, rather than the current approach with the rent level of the unit, yields markedly lower year-on-year changes in the rent index. We find that the current rent level approach generally overestimates rent changes in units, while the tenure length approach is more accurate.

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Session A-8: From Costs to Care: Economic and Population Perspectives on Health

Organizer: Brian Ward, *National Center for Health Statistics*

Chair: Brian Ward, *National Center for Health Statistics*

BLS Auditorium C

Regional Price Level Estimates for Medical Services in the United States

Calvin Ackley, *Bureau of Economic Analysis*

This paper estimates regional price levels for medical services in the United States using two of the largest available sources of commercial health claims. I provide estimates for inpatient, outpatient, and professional services, as well as aggregate expenditure-weighted estimates. By using two independent data sources, this paper addresses representativeness concerns and sheds light on the interchangeability of two widely used commercial claims databases. I utilize the estimated price indices to examine the relationship between medical prices and total health care spending per beneficiary, and also provide a novel state-level comparison of medical and non-medical price levels. I find that, Alaska, Wisconsin, Wyoming, Oregon, and California tend to have the highest health care prices, while Alabama, Arkansas, Kentucky, Michigan, and Louisiana have the lowest, although there is considerable heterogeneity across service categories. Medical prices are significantly more disperse than non-medical prices, and the correlation between the two is weak across states. I find that variation in the medical price level explains about one-half of the variation in health care spending per person.

Integrating Multimodal Data to Advance Childhood Cancer Research

Johanna Goderre, *National Cancer Institute*

Subhashini Jagu, *National Cancer Institute*

The Childhood Cancer Data Initiative (CCDI) has pioneered innovative informatics solutions for collection and dissemination of clinical, research, and population health data to advance investigations into the causes, effective treatments, and long-term effects of cancer in children, adolescents, and young adults. The CCDI Data Ecosystem is a connected network of tools and resources that enable the collection, sharing, and analysis of data. CCDI Hub Explore allows researchers to build large datasets across multimodal data from multiple research studies and public health. We unite federated resources with a Participant Index that enables researchers to map multimodal data for the same participant across different datasets. The CCDI Molecular Characterization Initiative (MCI) offers a model for integrating comprehensive genomic and transcriptomic characterization into clinical care and research. The CCDI's methodologies and insights are relevant to building capacity for comprehensive, high-quality research in pediatric cancer, rare diseases, adult cancers, and other health conditions. We will describe how this program contributes to federal statistical systems in data collection (genomics, electronic health record and laboratory systems, claims, etc.), semantic harmonization, interoperability, and release mechanisms of real-world data; active and proposed data analyses; and methods to protect the privacy of patients represented in the data.

The Mortality Gap: Unveiling Mortality Patterns and Differences of the HUD-assisted Population

David Pritchard, *U.S. Census Bureau*

Rachel M. Shattuck, *U.S. Census Bureau*

Rachel M. Shattuck, *U.S. Census Bureau*

Thomas B. Foster, *U.S. Census Bureau*

Sonya R. Porter, *U.S. Census Bureau*

Kyle Raze, *U.S. Census Bureau*

Denise Flanagan-Doyle, *U.S. Census Bureau*

Veronica Garrison, *Housing and Urban Development*

Jacqueline Bachand, *Housing and Urban Development*

This paper provides a population-based, longitudinal analysis of all-cause mortality among individuals receiving federal housing assistance from the U.S. Department of Housing and Urban Development (HUD). We link administrative records from HUD to individual-level mortality data from the Census Numident to estimate standardized mortality rates for the HUD-assisted population. Then, we compare the mortality of HUD recipients to two benchmark groups using the American Community Survey: 1) the full U.S. population and 2) the U.S. population of low-income renters not receiving HUD assistance. We find that HUD-assisted individuals face mortality rates much higher than the general U.S. population, a gap that has widened each year over the last decade. We show this widening is attributable to improvements in mortality among the general population not experienced by HUD recipients. In contrast, the mortality of HUD recipients and low-income, non-HUD renters are very similar and have remained stable over time, suggesting that broader mortality declines are largely bypassing low-income and disadvantaged populations. Disaggregated analyses show substantial variation by age, sex, race/ethnicity, disability, income, program enrollment, and geography, with non-Hispanic White HUD recipients having the highest mortality rate and largest mortality gap. Furthermore, we find HUD-assisted individuals with disabilities have noticeably lower mortality than the non-HUD, disabled population. Finally, we track mortality 15 years after individuals exit HUD. We find mortality peaks in the first year after exit that is several times higher the rate in the U.S. population, then declines steadily over time. Additionally, individuals who exit HUD for “positive” reasons have better health outcomes than low-income, non-HUD renters. Our results together highlight the large health vulnerabilities faced by HUD participants and suggest improved outcomes may emerge among the most vulnerable members of HUD.

National Spending on Mental Health and Substance Use Disorder Treatment, 2000 – 2021

Lasanthi Fernando, *Bureau of Economic Analysis*

Abe Dunn, *Bureau of Economic Analysis*

Tami Mark, *Research Triangle Institute*

The U.S. Substance Abuse and Mental Health Administration’s (SAMHSA) discontinued report on Mental Health and Substance Use Disorder (MH/SUD) is the most prolifically cited source for expenditure and prevalence estimates for MH/SUD. Prior to discontinuation, the last year for MH/SUD estimates was for 2014, over a decade ago. We use the Health Care Satellite Account (HCSA) from the U.S. Bureau of Economic Analysis (BEA) to report timelier and more recent expenditures on MH/SUD. From 2000-2021, mental health and substance use disorder (MH/SUD) nominal spending grew from \$40.8B to \$139.6B, representing growth from 4.5% of total disease spending in 2000 to 5.5% in 2021. Real per capita MH/SUD spending grew at a real average annual rate of 3.27%, one percent faster than the overall medical services (2.21%). MH/SUD spending increases were driven primarily by increases in the number of people receiving treatment (87.3% of growth) and to a much lesser extent in the cost per case (12.7% of growth). This contrasts with spending increases on all diseases where 66.3% total spending increases was due to increases in cost per case while 33.7% of the increase was from more people receiving treatment. This comparison highlights that data from the HCSA provide not only timelier MH/SUD estimates, but also side-by-side comparison to other diseases. The large growth in the number of treated cases is arguably related to expanded access to mental health services, but it also raises questions about increased prevalence of MH/SUD conditions in the population, as well as the effectiveness of MH/SUD treatment.

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Session B-1: Statistical Communication: Improving Value, Engagement, and Public Trust

Organizer: Jennifer Childs, Michael Hawes & Carla Medalia, *U.S. Census Bureau*

Chair: Vipin Arora, *Bureau of Economic Analysis*

Census Auditorium 1

Open Census: A Federal Statistical Agency's Approach to Open Science

Emily Molfino, *U.S. Census Bureau*

We are in a time of increased action to promote open science. Open scholarship, code, and data policies promote core tenets of open science such as transparency, reproducibility, and accessibility. Open Census is a groundbreaking initiative that promotes open science practices within the Census Bureau. By providing the necessary infrastructure, guidance, and standardized metadata, Open Census empowers researchers to share their work effectively, including research articles, code, and data. This presentation will offer an in-depth look into Open. We will explore how the initiative enhances research accessibility and transparency and provide practical insights for how other agencies and external stakeholders can leverage the guidance and infrastructure provided. The presentation will address the unique considerations for implementing open science within a federal statistical agency, highlighting key efforts such as:

- Developing citation guidance for both public and restricted data.
- Providing data via commercial cloud repositories.
- Expanding the Census Bureau's GitHub.com site to share research and statistical code.

The presentation will also address the unique considerations for implementing open science within a federal statistical agency and outline the next steps for the continued evolution of Open Census.

Some Practical Considerations on Disclosure Avoidance for Operational Statistics

Joanna Fane Lineback, *U.S. Census Bureau*

Eric B. Jensen, *U.S. Census Bureau*

Noah McCord, *U.S. Census Bureau*

Michael B. Hawes, *U.S. Census Bureau*

In addition to data on demographic, social, and economic trends, federal survey programs often publish statistics about the survey operation itself. For example, the decennial census operation releases return rates and their inputs, response rates, and type of enumeration area statistics, such as self-response and update-leave counts, which can be found on the Census Bureau's Planning Database along with other select American Community Survey and census statistics. Census operational statistics provide invaluable insights to other statistical agencies and organizations that use them to inform their own survey and data collection operations. To protect respondent confidentiality, Census Bureau policy requires that any publicly released substate statistics from the 2020 Census, including operational statistics, need to be protected using differentially private noise infusion. An evaluation of the potential impact that disclosure avoidance-induced error would have on the usability of the 2020 Census operational statistics, however, showed that the required noise would overly degrade the utility of the data for their intended purposes. There were many instances of implausible values, and there was evidence that noise would impact certain geographies more than others. Of particular concern was the impact noise had on certain subgroups – evidence suggested that if noisy data were used for census or survey planning, outreach or follow-up efforts might be incorrectly targeted. In light of this assessment, the team took a closer look at the disclosure risks associated with each of these operational statistics - for example, census tract level update leave area housing unit counts, internet response rates, and paper return rates - and recommended a set of disclosure avoidance rules customized to address those specific risks. Additionally, the recommendations establish a reasonable disclosure avoidance precedent for future censuses and surveys that may want to publish similar metrics.

Availability of Federal Data via Data.gov

Meghan Stuessy, *Congressional Research Service*

Policymakers, members of the public, and businesses regularly use federal data and statistical information to make public policy, programmatic, and funding decisions. Although non-federal entities may produce data, federal data is often preferred by users for its credibility, accuracy, and objectivity. In 2013, the Office of Management and Budget described openly available federal data and statistical information as “a valuable national resource and strategic asset” that, when made accessible, discoverable, and usable by the public “can help fuel entrepreneurship, innovation, and scientific discovery.”

Efforts to make federal data more readily available have evolved over time, where such data may have been stored and filed in hard and paper copies, later in software and electronic formats, to today, where certain data may be retrieved through agency websites or on data.gov. Changing public perspectives on data availability may focus on making the data available for predictable periods of time, in a variety of software-compatible formats, and with appropriate descriptive metadata for findability and usability of the information.

As part of the implementation of the Foundations for Evidence-Based Policymaking Act of 2018, on January 15, 2025, OMB issued guidance to agencies on how to comply with the law’s data availability, data management, and dissemination requirements. In late January 2025, several news outlets reported the apparent removal of data sets from data.gov. These events may raise new questions regarding whether and how current guidance sufficiently supports efforts to make agency data open and available to multiple audiences while balancing interests in prioritizing access to the most up-to-date and accurate information.

This presentation follows the historical development and policy debates that influenced the creation of data.gov. The presentation will also provide information on statutory requirements and ambiguities regarding the dissemination of quality data on data.gov.

Highlights of a GAO Expert Forum: Challenges and Opportunities Affecting the Federal Statistical System

Jared Smith, *Government Accountability Office*

Jeff Tessin, *Government Accountability Office*

Erin Villareal, *Government Accountability Office*

Sirin Yaemsiri, *Government Accountability Office*

Jason Rodriguez-Masi, *Government Accountability Office*

Scott Hiromoto, *Government Accountability Office*

Frances Tirado, *Government Accountability Office*

The federal statistical system produces data that are vital for decisions that directly impact the public, including the allocation of federal funding to states and localities and the production of key national statistics on health, demographics, and the economy. However, the system faces a critical juncture as it works to modernize and adapt to a rapidly changing data landscape, driven by increasing demand for timely, detailed, and relevant information amid declining survey response rates and rising data collection costs. On August 21-22, 2024, GAO held a forum with 29 participants to discuss the challenges and opportunities facing the federal statistical system. The participants were from the private sector; federal, state, local, and international government agencies; nonprofit organizations; academic institutions; and international organizations. This presentation will highlight key topics participants discussed: factors affecting the system’s ability to (1) build and maintain public trust, (2) meet the needs of its users, (3) use alternative data sources, and (4) engage in effective interagency coordination. In addition, we will discuss prior GAO work and current and future work areas related to the federal statistical system.

Gold Standard Science, Communication, and Public Trust

Michael Hawes, *U.S. Census Bureau*

In May 2025, Executive Order 14303 on “Restoring Gold Standard Science” affirmed the federal government’s commitment “to restoring a gold standard for science to ensure that federally funded research is transparent, rigorous, and impactful, and that Federal decisions are informed by the most credible, reliable, and impartial scientific evidence available.” The seven tenets of Gold Standard Science identified by the executive order build upon and enhance a multifaceted framework of information quality standards, scientific rigor, and professional

codes of practice that have long guided the activities of the federal statistical system. This presentation will discuss these tenets and their relevance for statistical agencies with a particular focus on the critical role of communication and stakeholder engagement in advancing and achieving these tenets and in earning and sustaining the public's trust in the value and objectivity of federal statistics.

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Session B-2: From Freight to Tax Filings: AI Classification and Applications for Federal Data Collections

Organizer: Jordan Riddle, *Bureau of Transportation Statistics*

Chair: Jordan Riddle, *Bureau of Transportation Statistics*

Census Auditorium 2

Evaluation of an AI Model for the Automation of In-Field Reinterview Clerical Resolution

Kexin Zhang, *U.S. Census Bureau*

Mike Shaw, *U.S. Census Bureau*

Kristine Roinestad, *U.S. Census Bureau*

Robert Long, *U.S. Census Bureau*

For the 2026 Census Test In-Field Enumeration Reinterview program, production interviews will be selected for reinterview. Some of these interview pairs will undergo a clerical matching of their household information. This process is critical for determining if the interviews were conducted at the correct household while following proper procedures. Traditionally, this process relies on human expertise and manual review, which is time-consuming, subjective, and prone to human error. This paper presents a methodology for evaluating the performance of a pre-trained Large Language Model (LLM), prompted to automate this clerical matching process. The evaluation framework focuses on a comparative analysis of the AI model's classification outcome against that assigned by experienced subject matter experts.

The evaluation process starts with a manual matching task performed by a panel of subject matter experts to determine a reliable ground truth classification. The consistency of their collective decisions is quantified using Fleiss' Kappa. A high Fleiss' Kappa value validates the quality of this human-generated gold standard, ensuring a dependable benchmark for model evaluation. The AI model's performance is rigorously assessed by comparing its predictions to this gold standard using inter-rater reliability, confusion matrix and other key metrics. The AI model demonstrated strong agreement with the clerks' decisions, measured by the value of Cohen's Kappa, with precision and recall exceeding [x] percent in a small test using household cases from the 2020 Census. Human-in-the-loop feedback can help AI model identify data patterns that mislead it. This feedback will improve the error analysis and guide adjustments to the model's features, prompts, or rules.

These evaluation results are pivotal for implementing AI models in future Census quality assurance programs, such as in the 2030 Census. This study provides a transparent framework for benchmarking AI model against established clerical standards, offering a clear understanding of model reliability and identifying targeted areas for refinement.

2022 CFS Commodity Codes & ML

Trent Langley, *U.S. Census Bureau*

Liz Jeninga, *U.S. Census Bureau*

The Commodity Flow Survey (CFS) is the primary source of information on the intra- and interstate movement of goods in the United States. The CFS is a shipper-based survey and collects shipment data from a variety of sampled companies across the country and is conducted every five years. The 2022 CFS collected data on

almost 100 million shipments – more data than ever before. For the first time, machine learning assisted in the classification of reported commodities. The technology was utilized in two ways – first, in real time while respondents completed the survey, and second, in the editing and imputation phase of data processing. The FCSM presentation by colleagues in October 2023 focused on the plan to use machine learning in the 2022 CFS. This presentation will describe the strategies we employed to bolster the quality and confidence of machine learning commodity code assignments for the 2022 CFS. The goal is to identify potential opportunities for improvements to incorporating machine learning into data collection and processing and may have implications for future CFS cycles and other surveys.

A Deep Dive into an LLM-Based Industry & Occupation Autocoder

Julia Beckhusen, *U.S. Census Bureau*

Jiahui Xu, *Reveal Global Consulting, LLC*

Yezzi Angi Lee, *Reveal Global Consulting, LLC*

Jackson Chen, *Reveal Global Consulting, LLC*

Lynda Laughlin, *U.S. Census Bureau*

Ana J Montalvo, *U.S. Census Bureau*

Administered by the U.S. Census Bureau, the American Community Survey (ACS) collects comprehensive demographic, social, economic, and housing data from a national representative sample of U.S. households. Within the ACS, the industry and occupation open-ended free-text responses are converted to census codes through a combination of autocoding and clerical coding. We leveraged Llama and 2024 January ACS Industry & Occupation (I&O) write-ins to develop a Large Language Model-based autocoder that jointly codes industry and occupation. We used the transfer learning technique and redesigned the prediction head with a two-head, multi-task architecture to perform the task without hallucination. For each write-in, the model predicts the industry and occupation codes and the associated probabilities for each.

In this research, beyond overall accuracy, we focus on hard cases that drive clerical workload and error, including rare or emergent titles, multi-job strings, vague descriptors, cross-domain ambiguities (e.g., “driver” across industries), mixed language entries, and inconsistent industry-occupation pairs. This analysis informs when autocoded results can be reliably accepted and identifies areas where the model requires targeted improvement. Specifically, we reviewed and identified inconsistencies between the existing autocoder and the new autocoder to further refine the new model. To inform decisions, we calibrated probabilities derived from logits and evaluated multiple strategies: (1) per-class cutoffs tuned on precision-recall to control false positives under fixed clerical capacity; (2) conformal prediction sets for selective classification with coverage guarantees; (3) hierarchical and cross-walk consistency checks to enforce industry-occupation compatibility; and (4) cost-sensitive thresholds that reflect downstream rework costs. Together, the deep dive leads to better maintenance and reliable performance in both the short-term and long-term run.

Early Response Prediction of COVID-19 Hospitalizations and Deaths Using CDC Weekly Data with Meta Prophet and Wave Dynamics Index

Daping (David) Zhang, *Centers for Disease Control and Prevention*

Michael Zhang, *Civil Air Patrol; Univer LLC*

Rajeev Srivastava, *Centers for Disease Control and Prevention*

Accurate forecasting of COVID-19 hospitalizations and deaths is vital for federal public health planning. Traditional models, like the COVID-19 Forecast Hub, face n week delays and complex ensemble methods, hindering statistical inference. This study proposes a framework using CDC published data, employing Meta Prophet—a black-box AI for time series forecasting—and a Wave Dynamics Index (WDI) for n-weeks earlier warnings, validated statistically.

The methodology leverages a CDC Data Modernization Initiative (DMI) pipeline, integrating weekly data (2020–2025) on hospitalizations, deaths, NAAT tests, ED visits, and test positivity from the COVID-19 Data Tracker. Preprocessing addresses missing values and 10-week updates for robust statistical modeling. Meta Prophet captures non-linear trends, seasonality, and holiday effects, refined by health expertise. The WDI, $i = cH/gT^2$ (H: epidemiological impact, T: period, g: gravity, c: cohesion), quantifies wave value changes (e.g., hospitalization surges), enhancing interpretability. As a static index, WDI informs AI forecasts ($i > 0.3$: red alert, $0.2-0.3$: orange, <0.2 : safe); high, persistent peaks (n weeks) may trigger near-future spikes, flagged for

response. This layer boosts transparency, validated by cross-checks. Charts show predicted and actual peak alignment (mean error <5%) with visualizations.

Statistical analysis confirms the December 2023 surge prediction (error <5%), with weekly peaks of 150,000–175,000 hospitalizations and 15,000–20,000 deaths (e.g., Jan 2021, Dec 2023), and a projected August 2024 peak of 50,000 hospitalizations, 5,000 deaths. Monthly peaks occur in January (500,000–750,000 hospitalizations, 50,000–75,000 deaths) and September, with December 2024 at 200,000 and 20,000. Quarterly peaks exceed 1,000,000 hospitalizations and 100,000 deaths (e.g., Q1 2021), projecting 600,000 and 60,000 in Q4 2024. WDI improves accuracy by 15–20% by explaining wave variability, enabling 1–2 week AI lead times.

This scalable tool enhances CDC/FED forecasting, integrating Meta Prophet and WDI for reliable predictions, supporting timely interventions on time series fed data analysis.

Keywords: AI Forecasting, Seasonal Trends, Early Warning, Statistical Analysis

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Session B-3: Data linkage: Increasing Quality and Information

Organizer: Julie Parker, *Economic Research Service*

Chair: Ronda Britt, *National Center for Science and Engineering Statistics*

Census Auditorium 3

Integrating Federal Surveys: Data Modernization Efforts on NCSES's Workforce Surveys

Gigi Jones, *National Center for Science and Engineering Statistics*

Lynn Milan, *National Center for Science and Engineering Statistics*

Lisa Mirel, *National Center for Science and Engineering Statistics*

Shelley Feuer, *National Center for Science and Engineering Statistics*

Like many federal agencies, the National Center for Science and Engineering Statistics (NCSES) is facing growing challenges in administering its workforce surveys, including declining response rates, rising costs, and reduced operational resources. In response, NCSES and agencies across the federal government are reexamining their survey portfolios to identify opportunities for improving efficiency and effectiveness through innovation. This could be accomplished by shifting from traditional, survey-only data collection methods toward hybrid approaches that incorporate alternative data sources that meet data quality standards while aiming to reduce respondent burden.

Federal initiatives are reinforcing this shift. The Evidence-Based Policymaking Act of 2018 (Public Law 115-435) promotes data sharing for evidence building, while ongoing modernization efforts across agencies emphasize improving survey efficiency and facilitating greater use of alternative data. Researchers, too, have long advocated for supplementing traditional survey data with alternative sources to improve efficiency.

With support from America's Data Consortium, NCSES is conducting a feasibility study as a National Secure Data Service Demonstration (NSDS-D) project. This study explores the integration of three flagship workforce surveys—the National Training, Education, and Workforce Survey (NTEWS), the National Survey of College Graduates (NSCG), and the Survey of Doctorate Recipients (SDR). The goal is to develop a single, integrated workforce survey that delivers high-quality data in a more cost-effective and resource-efficient manner.

In parallel, NCSES is assessing the potential of using alternative data sources to supplement or partially replace data currently collected through these surveys. This presentation will share findings and lessons learned from this initiative, offering insights that may serve as a model for other federal agencies considering similar approaches to survey modernization and data integration.

CTC and ACTC Participation Results and IRS-Census Match Methodology, Tax Years 2018-2020

Charles Hokayem, *U.S. Census Bureau*
Ciyata Coleman, *Internal Revenue Service*
Ashley Erceg, *U.S. Census Bureau*
Kirk Heffelmire, *Internal Revenue Service*
Sanghun (Eric) Kim, *Internal Revenue Service*
Ethan Krohn, *Internal Revenue Service*
Krishnan Patel, *Internal Revenue Service*
Dean Plueger, *Internal Revenue Service*

The Child Tax Credit (CTC) and Additional Child Tax Credit (ACTC) offer assistance to help ease the financial burden for families with children. This paper presents a new methodology for estimating participation (take-up) of the CTC and ACTC that relies on linking the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) to IRS administrative data. This approach, called the Exact Match, uses survey data to identify CTC/ACTC eligible taxpayers and IRS administrative data to indicate which eligible taxpayers claimed and received the credit. Overall, in tax years 2018-2020 eligible taxpayers participated in the CTC and ACTC program at a rate of 93 percent each year while dollar participation (take-up of eligible dollars) was about 91 percent each year. The presentation shows participation by various socioeconomic characteristics, adjusted gross income, eligible credit amount, and filing status. It also presents characteristics of eligible taxpayers who do not participate in the program. Future work will extend the methodology to also include the expanded Child Tax Credit of 2021 under the American Rescue Plan.

Business Formation Activity of College Graduates

Andrew Foote, *U.S. Census Bureau*
Emin Dinlersoz, *U.S. Census Bureau*
Cody Orr, *U.S. Census Bureau*

There is an extensive literature that has examined the economic returns to college for people who attend and complete their schooling. However, this research has focused almost exclusively on the wage and salary outcomes of graduates, because these outcomes are the easiest to measure, particularly with the availability of administrative records on earnings and graduates.

Measuring business formation captures economic outcomes not typically measured in the literature, which heavily relies on Unemployment Insurance wage record data, and the majority of business founders are not able to collect unemployment insurance. To the extent that these businesses become successful and hire employees, the social benefits of colleges are underestimated if we focus solely on wage and salary jobs.

In this paper, we link administrative records on college graduates from 13 states to the universe of applications for an Employer Identification Number (EIN) with the IRS (Form SS-4). This linkage allows us to identify business formation of post-secondary graduates at a number of degree levels. Furthermore, we link these EINs to the Longitudinal Business Database to identify which of these businesses became employer businesses within two years of the EIN application.

This novel linkage allows us to document a number of important facts about the business formation of graduates. First, we show that there is large variation in the rates of business formation by degree level and degree field, and we show that a number of these businesses are high quality. We document the extent to which graduates form businesses in similar industries as their employment. Importantly, we also show that graduates are more likely to form businesses in the state of their institution. Finally, we show that most businesses are formed in mid-career.

Employment and Earnings Trajectories of HUD Program Participants

Kyle Raze, *U.S. Census Bureau*
Rachel Shattuck, *U.S. Census Bureau*
David Pritchard, *U.S. Census Bureau*
Thomas B. Foster, *U.S. Census Bureau*
Sonya R. Porter, *U.S. Census Bureau*
Denise Flanagan-Doyle, *U.S. Census Bureau*

Veronica Garrison, *U.S. Department of Housing and Urban Development*
Jacqueline Bachand, *U.S. Department of Housing and Urban Development*
Ethan Krohn, *U.S. Census Bureau*

Federal housing assistance programs, such as those run by the U.S. Department of Housing and Urban Development (HUD), have been shown to reduce rent burden and improve housing stability for program participants, which may in turn have downstream impacts on their labor market attachment and career trajectories. However, existing studies from individual cities or states provide mixed evidence on the association of housing assistance with labor market outcomes. By linking HUD administrative records to matched employee-employer earnings records from the Longitudinal Employer-Household Dynamics (LEHD) program, we document how the labor market trajectories of program participants change as they enter and exit federal housing assistance programs, examining outcomes over a 14-year window surrounding entry or exit. In our analysis of entry, we find that the employment rates and earnings of first-time HUD program participants begin to increase upon entering a HUD program, which represents a reversal of prior declining trends in these outcomes. Suggestive of a positive association, these increases in employment and earnings trends exceed those of low-income non-participants from the American Community Survey (ACS). In our analysis of exits, we find that program participants who eventually leave a HUD program have increasing pre-exit trends in employment and earnings that then flatten upon exiting. Comparing these negative changes in trend to the relatively stable trajectories of those who remain in HUD programs throughout the analysis suggests that exits are associated with diminished employment and earnings trajectories.

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Session B-4: Mode Matters: How Design Choices Shape Survey Results

Organizer: Tywanquila Walker, *Bureau of Labor Statistics*
Chair: Tywanquila Walker, *Bureau of Labor Statistics*
Census Conference Rooms 1-2

Differences between Web and Mail Respondents of the 2012-2022 National Post-acute and Long-term Care Study

Samantha Spiers, *National Center for Health Statistics*
Manisha Sengupta, *National Center for Health Statistics*

The National Post-acute and Long-term Care Study (NPALS) provides nationally representative estimates about long-term care services providers in the United States through administrative data and a sample survey of residential care communities (RCCs) and adult day services centers (ADSCs). Beginning with the first wave in 2012, NPALS uses a multi-mode survey protocol with mail and web self-response, followed by computer-assisted telephone interviewing (CATI) nonresponse follow-up for ADSCs and RCCs. Web is the preferable response mode due to increased data quality compared to mail and decreased costs compared to mail and CATI. Thus, we examined differences between mail and web respondents to understand the mode preference of facilities who self-respond.

We examined differences in facility characteristics separately within ADSCs and RCCs in the 2012, 2014, 2016, 2018, 2020, and 2022 waves. Several characteristics were different between mail and web respondents across multiple waves. A higher percentage of RCCs that were web respondents used electronic health records and were chain affiliated compared to RCCs that responded by mail. Response by mail was more common than response by web among RCCs and ADSCs with smaller capacity. Identifying provider characteristics that are associated with mode of response may facilitate targeting providers with specific response mode options. These results will inform future survey operations, improving recruitment efforts and decreasing cost.

From Uploads to Integration: The Evolution of Spreadsheet Reporting in Establishment Surveys

Temika Holland, *U.S. Census Bureau*

With growing challenges in survey data collection, alternative strategies have been warranted to allow greater flexibility in survey response while maintaining data quality and efficiency. The U.S. Census Bureau has made strides in allowing greater flexibility in how respondents provide more detailed survey data with the use of spreadsheet reporting, where respondents are able to report survey data in a customized template. In the past this template would require respondents to download a preformatted spreadsheet using an external program (e.g., Excel, Numbers), enter response data offline, then upload the template back into the survey instrument for submission. In recent years, the Census Bureau has allowed respondents the ability to report via an online spreadsheet template option, where the spreadsheet is embedded in the online survey instrument.

Incorporating such flexibility offers a multitude of benefits to respondents and statistical agencies, like the U.S. Census Bureau. With spreadsheet reporting in general, data are customizable, filtered and can meet the demanding needs of respondents while being a cost-effective solution for survey administration. With the added utilization of online spreadsheet reporting, respondents are able to input response data directly into the survey site without having to use any external programs. Additionally, interactions with spreadsheet templates can be examined for further improvements to ensure they are meeting the growing needs of respondents.

This presentation seeks to highlight the expanded use of spreadsheet reporting as an alternative response method in online establishment surveys. Citing examples from the U.S. Census Bureau, this presentation will provide recommendations for spreadsheet template design, cite its prevalence and relevance in survey administration, and discuss the challenges that remain for establishment surveys. Furthermore, a discussion of the future implications for spreadsheet reporting will be discussed as we continue to evolve and expand upon traditional reporting methods.

A Tale of Two Surveys: How Differing Methodologies Impact Key Estimates

Herschel Sanders, *U.S. Census Bureau*

Renee Reeves, *U.S. Census Bureau*

Kayla Varela, *U.S. Census Bureau*

Jenny Hunter Childs, *U.S. Census Bureau*

Rachel Horwitz, *U.S. Census Bureau*

The Household Pulse Survey (HPS), launched in 2020 as an emergency data collection tool to capture household experiences during the coronavirus pandemic, had evolved into a broader instrument for tracking emergent social and economic issues. As the pandemic subsided, the Census Bureau transitioned HPS content into a new panel-based infrastructure: the Household Trends and Outlook Pulse Survey (HTOPS). This shift represents a fundamental methodological change from a cross-sectional survey using SMS and email recruitment to a longitudinal panel using mail recruitment and offering monetary incentives. This study evaluates how this transition affects estimates, comparing the HPS' non-incentivized cross-sectional approach to HTOPS' incentivized panel design.

To measure the impact of this change, we analyze estimates across overlapping survey items from each survey, focusing on basic demographics as well as critical domains including employment status, income distribution, housing conditions, and mental health indicators. Our analysis assesses whether the shift in recruitment strategy leads to systematic differences in respondent characteristics or substantive findings.

By documenting these methodological differences, this research provides guidance for researchers and data users seeking to understand how operational survey changes may influence estimates and trend analyses.

Do we get the same respondents after changing dietary data collection mode? Evidence from the National Health and Nutrition Examination Survey (NHANES)

Te-Ching Chen, *National Center for Health Statistics*

Matt Jans, *National Center for Health Statistics*

Anne Williams, *National Center for Health Statistics*

Lara Akinbami, *National Center for Health Statistics*

Cynthia Ogden, *National Center for Health Statistics*

Changing interview mode can create differential nonresponse among subgroups, possibly increasing nonresponse bias. We evaluated whether changing dietary recall interview mode in the National Health and Nutrition Examination Survey (NHANES) impacted respondent demographics, health characteristics, and operational factors. NHANES involves home-based in-person interviews, health examinations in a mobile examination center (MEC), and two 24-hour dietary interviews. The first dietary interview (Day 1) was historically collected in-person at the MEC. The second (Day 2) was collected by phone 3-10 days later. In August 2021, Day 1 became a phone interview conducted after the MEC exam. To assess the possible impact, we used chi-square tests to compare the association between phone Day 1 respondents and in-person Day 1 respondents from previous cycles (2015-2016, 2017-2018, 2017-March 2020). Demographics included sex, race and ethnicity, age, education, family income-to-poverty ratio, US birth. Health outcomes included self-reported general health, and diagnosed diabetes, along with measured diabetes based on A1c level, obesity and high total cholesterol. Operational factors included day of the week of the dietary recall, and season of the year of participation. No attempt was made to disentangle the effect of mode from the effect of time. Results show similar demographic characteristics and health conditions for the August 2021 - August 2023 Day 1 (phone) dietary interview and MEC (in-person) results from past cycles, suggesting phone respondents have similar characteristics and outcomes to in-person respondents. Operationally, more Day 1 phone interviews were on weekdays than weekends, different from past cycles. This could arise from having more weekend than weekday MEC exams but may also suggest that reaching people by phone on weekdays may be easier than weekends. Overall, we found little mode effect risk for demographic and health characteristics, but insights for mode change logistics.

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Session B-6: Show Me the Monetary: New Metrics for Measuring Value

Organizer: Matthew Chambers, *Bureau of Economic Analysis*

Chair: Matthew Chambers, *Bureau of Economic Analysis*

BLS Auditorium A

Pilot Physical and Monetary Energy Flows Account for the U.S.

Matthew Chambers, *Bureau of Economic Analysis*

Tina Highfill, *Bureau of Economic Analysis*

Dirk van Duym, *Bureau of Economic Analysis*

Energy—in all its many forms—is the lifeblood of an economy. All sectors and industries in the economy consume energy in one or more forms, and the extraction and generation of energy is a major point of interaction between the economy and the environment. Good decision-making hinges on both public and private stakeholders understanding how energy flows through the economy and how its use supports industries and households in their economic activity. Environmental accounting offers a framework to support this understanding, by placing economic activities in the context of their underlying physical processes, such as energy use. Measuring the flows of energy into, through, and out of the economy is a central component of environmental-economic accounting as outlined in the System of Environmental-Economic Accounting (SEEA), the UN standard for environmental accounting, a companion to the System of National Accounts (SNA).

In this paper we present an early-stage combined physical energy flow account and monetary energy flow account for the United States, covering the years 2017–2022. The proof-of-concept energy accounts include tabulated supply and use of energy by industries and institutional sectors. Additionally, we present an experimental state-level version of the monetary energy flow account that attempts to estimate the domestic supply of energy by state. Data on energy extraction, generation, and transformation primarily come from the U.S. Energy Information Administration. Other data sources include the Bureau of Economic Analysis's published supply and use tables and detailed stock estimates of fixed assets and consumer durable goods, the Bureau of Transportation Statistics's Transportation Satellite Accounts and National Transportation Statistics

tables, and academic studies. We discuss challenges encountered and solutions identified in compiling the accounts and briefly demonstrate analytical applications.

Comparing Early Estimates of U.S. Manufacturing Output

Peter B Meyer, *Bureau of Labor Statistics*

Labor productivity measures can be constructed using different measures of output. A challenge in using value-added output measures for quarterly US productivity has been that the official estimates are not available quickly after each quarter. In this paper, I construct preliminary estimates of quarterly durable and nondurable manufacturing output using data available within a month after each quarter. Output measures come from BEA. Predictors come from several sources: the Census Bureau Economic Indicators data on exports, housing starts, inventories, and business applications; the Federal Reserve's Industrial Production Indexes, and price indexes from BLS. I evaluate the accuracy of the preliminary estimates over the 2005–2024 period. I compare value-added productivity measures to other manufacturing productivity measures which account in different ways for outsourcing and transactions within manufacturing.

Satellite Account for Human Capital

Justine Mallatt, *Bureau of Economic Analysis*

This paper addresses a significant gap in U.S. economic accounting: the absence of human capital measurement in national accounts. While economists recognize human capital as a key driver of economic growth, current practice treats education spending as consumption rather than productive investment, unlike other capital assets. This omission risks biasing policy decisions toward physical capital over human capital investment.

The paper develops a Satellite Account for Human Capital using two complementary approaches. The cost-based method sums direct education spending plus indirect costs like student time. The income-based method values human capital through the present value of future earnings from education. Both methods substantially increase measured GDP, with cost-based adding 2.6% and income-based adding nearly 15%.

Key findings reveal human capital investment's enormous scale: cost-based investment averages 10% of GDP while income-based ranges from 20-30% of GDP—comparable to all other capital investment combined. Educational investment shows strong but declining returns, ranging from 100-230% over the study period, falling from peaks of 2.3 in 1997 to near 1.0 in 2021.

The account reveals important cyclical patterns. During the Great Recession (2007-2012), human capital investment was counter-cyclical as rising enrollment cushioned GDP declines. Conversely, during COVID-19 (2020-2023), it was pro-cyclical as falling enrollment amplified economic contraction. These contrasting patterns demonstrate how enrollment responses critically determine whether human capital investment stabilizes or amplifies economic fluctuations.

The policy implications are substantial. A simulation showing that enrolling just 5% of unenrolled high school males in 2-year programs would generate significant net GDP gains when accounting for lifetime earnings, though traditional accounting would show net costs due to foregone wages. This illustrates how failing to treat human capital as an asset systematically undervalues education and training policies, potentially leading to underinvestment in this crucial economic driver.

Valuation of Subsoil Energy Assets: Pilot Estimates for the U.S. National Balance Sheet

Matthew Chambers, *Bureau of Economic Analysis*

Steven Anderson, *US Geological Survey*

Julie L. Hass, *Bureau of Economic Analysis*

Melissa Lynes, *Energy Information Administration*

Ian Mead, *Energy Information Administration*

Aditi Mirani, *Bureau of Ocean Energy Management*

Scott Wentland, *Bureau of Economic Analysis*

Energy plays a critical role in the U.S. economy. While the U.S. Government produces high quality estimates of physical quantities of subsoil assets like oil and natural gas reserves, the national economic accounts do not currently report a detailed accounting of natural resources on the national balance sheet (or Integrated Macroeconomic Accounts) in monetary terms. This paper develops new estimates of the value of U.S. subsoil

energy assets, focusing on oil and natural gas as test cases for improving the national accounts.

We estimate the value of these assets as the net present value of natural resource rents for the oil and gas industries. These natural resource rents are commonly estimated as a residual, after subtracting intermediate input, labor, and estimated capital costs from total revenue. We develop a refined version of the residual method that uses the return on corporate bonds as a proxy for the rate of return to capital and adjusts depreciation costs using the corporate tax rate. This method adjusts inconsistencies that arise when using other formulations of the residual method, such as those that assume a constant rate of return to capital and is one of this paper's primary contributions.

The estimated lifespan of U.S. oil and gas natural resource assets depends to a large extent on the assumed paths of extraction and discovery. We discuss four possible sets of assumptions based on Energy Information Administration projections and establish lower (\$1 trillion) and upper bounds (\$15.7 trillion) for the total value of U.S. oil and gas resource assets. Under the most reasonable assumptions, we estimate the total value of U.S. oil and gas subsoil resource assets to be \$2.3 trillion–\$7 trillion. These estimates may serve as a first step toward including our subsoil energy assets on the national balance sheet.

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***Session B-7: Small Area Estimation (SAE) at the U.S. Census Bureau:
Bridging Data Gaps with Flagship Models and New Innovations***

Organizer: Heather King, *U.S. Census Bureau*

Chair: Bethany DeSalvo, *U.S. Census Bureau*

Discussant: Heather King, *U.S. Census Bureau*

BLS CTC 2A

Methodological Overview of the Small Area Income and Poverty Estimates (SAIPE) and Small Area Health Insurance (SAHIE) Programs

Siqi Liu, *U.S. Census Bureau*

This presentation provides a methodological overview of the Census Bureau's flagship Small Area Income and Poverty Estimates (SAIPE) and Small Area Health Insurance Estimates (SAHIE) programs. Both programs show how small area estimation delivers reliable statistics for program administration and fund allocation while reducing respondent burden through innovative data integration. We demonstrate how shared Bayesian hierarchical modeling frameworks combine American Community Survey data with administrative records to produce timely, reliable estimates that would be impossible through direct survey methods alone. The presentation illustrates how Fay-Herriot area-level models serve as the foundation for most Census Bureau SAE applications, enabling the production of high-quality statistics for geographies with insufficient sample sizes. These flagship programs showcase the Census Bureau's commitment to leveraging multiple data sources to meet federal statistical needs while minimizing collection costs and respondent burden, directly supporting the allocation of billions in federal program funds. This presentation aims to solicit feedback from the data user community regarding the utility of the data and their potential interest in expanded estimates. This same interest served as the impetus for the subsequent presentation in this session.

Expanding Poverty Small Area Estimation: SAIPE Extensions and Innovations

Jim Scurry, *U.S. Census Bureau*

This presentation demonstrates how the Census Bureau innovates within established small area estimation (SAE) frameworks to meet emerging federal data needs while maximizing the utility of existing data assets. We focus on a key extension of the Small Area Income and Poverty Estimates (SAIPE) program: the Modeled Tract Area Poverty Estimates (M-TAPE). Developed in response to the Economic Development Administration's need

for tract-level poverty estimates, M-TAPE integrates SAIPE county estimates with decennial census data, American Community Survey microdata, and tax return records from the Internal Revenue Service. This approach addresses geocoding and granularity challenges to provide detailed poverty statistics at the tract level, data that were previously unavailable. M-TAPE illustrates how SAE methodology enables the Census Bureau to generate timely, policy-relevant estimates without new data collection, enhancing responsiveness to stakeholder requirements. We invite discussion on scaling these methods and overcoming implementation challenges in varied data environments.

Estimating Community Resilience: The CRE and CRE for Heat

Isabelle Notter, *U.S. Census Bureau*

This presentation showcases how small area estimation enables the Census Bureau to produce timely, actionable statistics for emergency management and public health applications. The Community Resilience Estimates (CRE) program, developed in response to COVID-19, demonstrates SAE's ability to rapidly deliver policy-relevant data during national emergencies. We detail the methodology behind CRE and its extension for extreme heat events, which combines American Community Survey (ACS) data with decennial census counts and Population Estimates Program data to measure community vulnerability at local levels. These models produce tract-level estimates of social and economic characteristics that affect disaster preparedness and recovery capacity, filling critical data gaps for emergency planners and public health officials. This work illustrates how SAE methodology allows the Census Bureau to quickly adapt existing data infrastructure to address urgent national priorities, providing federal, state, and local agencies with granular vulnerability assessments that support evidence-based disaster planning and resource allocation while leveraging existing data collections without increasing respondent burden. We encourage feedback on extending these resilience modeling frameworks to additional hazards and community characteristics.

Modeled County-level Estimates from the American Housing Survey: Two Feasibility Studies

Jiwon Lee, *U.S. Census Bureau*

This presentation focuses on the methodological challenges of applying small area estimation (SAE) in surveys with a large proportion of unsampled counties, using the American Housing Survey (AHS) as a case study. The AHS, conducted biennially for the Department of Housing and Urban Development, provides detailed housing data nationally and for select metropolitan areas but directly samples fewer than 30% of U.S. counties, limiting its ability to produce reliable county-level estimates through direct methods alone. We compare two modeling frameworks: (1) a two-level frequentist Fay-Herriot (FH) model with Empirical Best Linear Unbiased Prediction (EBLUP) for county-level rodent sightings, incorporating ensemble-selected auxiliary variables and survey weight adjustments; and (2) a three-level Bayesian hierarchical extension of the FH model for estimating air conditioning prevalence, addressing missing data via imputation and including random effects for climate-urban zones and counties to capture spatial heterogeneity and contextual dependencies. These studies demonstrate how SAE methods must rely heavily on auxiliary data, imputation, and strong model assumptions when unsampled areas predominate, which limits precision, complicates validation, and increases risk of bias. Despite these challenges, model-based estimation remains a practical approach to extending survey utility without new data collection. We welcome discussion on strategies to enhance model robustness and usability in sparse-data contexts.

Expanding Survey Utility with Machine Learning, Cross-Survey Modeling, and Small Area Estimation

Chase Sawyer, *U.S. Census Bureau*

This presentation demonstrates ongoing research at the Census Bureau to integrate machine learning with small area estimation (SAE) to create innovative blended data products that enhance federal statistical capabilities. We explore combining cross-survey modeling, a technique developed to support the Community Resilience Estimates (CRE) for Heat program, with SAE unit-level methods to develop a flexible, bottom-up estimation approach. Cross-survey modeling trains predictive models on content-rich specialized surveys like the American Housing Survey (AHS) to impute responses in broader surveys such as the American Community

Survey (ACS), effectively expanding data coverage without additional collection burden. In the CRE for Heat application of cross-survey modeling, we use machine learning models trained on AHS data to predict air conditioning presence in ACS households. Building on this success, this study investigates methodological approaches for integrating these household-level predictions with SAE techniques to produce statistically valid county-level estimates. We examine strategies for combining machine learning's predictive power with SAE's statistical rigor while addressing challenges such as survey design differences and uncertainty quantification. This work represents the Census Bureau's exploration of advanced analytical techniques to maximize existing data assets and produce comprehensive statistics that meet evolving federal information needs. Feedback is welcomed on integration strategies, benchmarking approaches, survey weight handling, and methodological refinements from the federal statistical community.

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Session B-8: Methodological Advancements in Health Measurement

Organizer: Julie Weeks, *National Center for Health Statistics*

Chair: Julie Weeks, *National Center for Health Statistics*

BLS Auditorium C

Comparing Medicaid Coverage Between Administrative Data and Three Federal Surveys

Rachel Lindstrom, *U.S. Census Bureau*

Holly Fee, *U.S. Census Bureau*

The American Community Survey (ACS), Current Population Survey Annual Social and Economic Supplement (CPS ASEC), and the Survey of Income and Program Participation (SIPP) are widely used data sources for estimates of health insurance coverage, including Medicaid coverage. Previous research has shown that federal surveys underreport Medicaid coverage compared to administrative records. Undercounting Medicaid enrollees may result from a variety of factors, including program confusion on the part of survey respondents. The ACS, CPS ASEC, and SIPP all measure health insurance in different ways. The current study examines the extent of Medicaid reporting error in these three federal surveys. It also examines what respondent characteristics are associated with reporting Medicaid coverage accurately.

To answer our research questions, we conduct descriptive and multivariate analyses. We compare the 2022 ACS Content Test (Control and Roster/Test 2 samples; hereafter referred to as the ACSCT-Control and ACSCT-Test, respectively), 2023 CPS ASEC, and 2023 SIPP survey data (reference period is calendar year 2022) to administrative data from the 2022 Transformed Medicaid Statistical Information System (T-MSIS) database. Protective Identification Keys (PIKs) were used to link the T-MSIS administrative data with the respective survey datasets. PIK rates across the samples were: 86.3% (CPS ASEC), 93.8% (ACSCT-Test), 93.9% (ACSCT-Control), and 95.6% (SIPP).

Descriptive results will be presented for Medicaid coverage rates for survey and administrative data sources, as well as reported Medicaid coverage agreement rates between survey and administrative data. In the multivariate analysis, we will estimate a logistic regression model predicting agreement between the survey and administrative data using respondent characteristics. We will perform all analyses separately for the 2022 ACS Content Test, 2023 CPS ASEC, and 2023 SIPP.

This research is unique in that it compares Medicaid coverage agreement between survey and administrative data across three large federal surveys for the same reference year.

Comparison of Estimates From the 2020 National Hospital Care Survey to the 2020 Healthcare Cost and Utilization Project: National Inpatient Sample (NIS) and Nationwide Emergency Department Sample (NEDS)

Adaeze O’Jiaku-Okorie, *National Center for Health Statistics*

Donielle G. White, *National Center for Health Statistics*

Catherine Rappole, *National Center for Health Statistics*

Kai Kang, *National Center for Health Statistics*

Lello Guluma, *National Center for Health Statistics*

Christopher Cairns, *National Center for Health Statistics*

Susan M. Schappert, *National Center for Health Statistics*

Brian W. Ward, *National Center for Health Statistics*

Sonja N. Williams, *National Center for Health Statistics*

Starting with the 2020 data year, the National Hospital Care Survey (NHCS) (2011–present), which succeeded the National Hospital Discharge Survey (1965–2010) and the National Hospital Ambulatory Medical Care Survey (NHAMCS) (1992–2022), was first able to provide nationally representative data on healthcare utilization. NHCS utilizes administrative claims and electronic health records to generate reliable data from hospital settings. To benchmark these national estimates, selected outcome measures from hospitalizations and emergency department (ED) visits were compared using 2020 data from the Agency for Healthcare Research and Quality’s Healthcare Cost and Utilization Project: National Inpatient Sample (NIS) and Nationwide Emergency Department Sample (NEDS), and 2020 NHAMCS.

The analysis revealed that many count estimates from NHCS and NIS were similar; out of 61 count estimates examined, 54 were statistically similar, though statistically significant differences were identified in 11.5% of the estimates. For example, NHCS reported higher hospitalization estimates for patients ages 1-17 (1.8 million) and 18-44 (9.3 million) compared to NIS (933,000 and 7.9 million, respectively), and fewer self-pay/no charge hospitalizations (498,000) than NIS (1.5 million).

More statistically significant differences were noted between NHCS and NEDS, and NHCS and NHAMCS, likely due to variations in the methodologies of the data systems. For example, for the diagnosis “external cause: intent of injury, accidental/unintentional,” NHCS reported a lower ED visit rate compared to NEDS (4,039 vs. 5,899 per 100,000 people) but a higher rate compared to NHAMCS (455.3 per 100,000 people). NHCS also showed a higher percentage of routine discharges among all ED visits compared to NHAMCS (88.7% vs. 76.8%) but a lower percentage compared to NEDS (93.1%).

Overall, NHCS is generally comparable to NHAMCS, NEDS, and NIS; however, users should consider the differences in methodologies of each source when using or comparing NHCS and NIS data.

Utilizing Exploratory Data Analysis and Feature Engineering

Sarah Pfeiff, *U.S. Census Bureau*

Pedro Vasquez Perez, *U.S. Census Bureau*

Ivan Cardona Tomassini, *U.S. Census Bureau*

Matt Dearstyne, *U.S. Census Bureau*

An efficient machine learning model requires selecting the features most relevant to the problem. In this research, the principal goal was to implement exploratory data analysis (EDA) and feature selection methods to identify the optimal features for a model to predict if a person will enroll in Medicaid. Secondary goal was to compare five feature selection processes to determine which yielded strong, effective results in a timely and efficient manner. Data was gathered from the American Community Survey 1-year estimates in 2023, accessed through the IPUMS USA database. During EDA, correlation tests were run, missingness patterns were observed, and multiple variables were combined to begin the initial trimming of variables. Next, feature selection methods such as, Random Forest Classifier, Mutual Information, SelectKBest, SelectPercentile, and SelectFromModel were applied using Python to determine the best variables for a prediction model. We present a comparison between feature selection methods and their results. Additionally, we discuss some challenges associated with feature selection methods and potential solutions.

Applying New Causal Analytics to Patient Records Assessing Patient Safety, Cost-Efficiency, and Value at Federal Healthcare Facilities

T. Michael Kashner, *U.S. Department of Veterans Affairs*

Steven S. Henley, *U.S. Department of Veterans Affairs (Contractor)*

Catherine P. Kaminetzky, *U.S. Department of Veterans Affairs*

John M. Byrne, *U.S. Department of Veterans Affairs*

Andrea D. Birnbaum, *U.S. Department of Veterans Affairs*

Cristian H. Peralta, *U.S. Department of Veterans Affairs*

The Department of Veterans Affairs [VA] demonstrates how peer-reviewed, Best Approximating Modeling [BAM] technology improves on inverse propensity-score weighted doubly robust regressions to predict how outcomes change when a given ‘causal’ factor of interest is changed by direct intervention. Going beyond merely predicting outcomes, ‘causal’ analytics were applied to Federal electronic health records to estimate clinical workload production models used to evaluate cost efficiency, productivity, and future staffing of VA’s 120,000 health professions trainees who rotate annually through its medical centers. ‘Causal’ analytics are important to VA, Defense, HHS, Homeland Security, and Commerce who are creating a single health record system for 18 million Federal patients (Federal Electronic Health Record Modernization). We report how BAMs improve doubly robust regressions performance. First, unlike stepwise or heuristic approaches, BAM outcomes models were selected following stochastic and exhaustive searches that increases the chances selected models represent the true Data Generating Process. Second, candidate models were evaluated for misspecification using White’s generalized information matrix test (Hessian and Outer-Product Gradient covariance estimators equality) simplified into a battery of smaller dimension tests, both generalized and parameter focused, with ‘causal’ factors tested for conditional exogeneity. Third, all observationally equivalent (Occam’s Window) ‘best’ (GBIC) qualifying models were averaged, tested for robustness, with standard errors accounting for sampling and model uncertainty. Our results show BAMs are robust to predict how policy changes in how VA oversaw physician residents and nursing students (engaged in supervised primary and mental health care) impacted their contribution to workload for cost-efficiency, supervised for patient safety, and progressed to become qualified VA employees. Our results demonstrated causal analytics applied to Federal health records replaced direct observations and employee surveys that were labor intensive, untimely, often unpersuasive at great expense to taxpayers, and interfered with patient care.

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Session B-X: Signal & Noise: AI in Federal Statistics

Organizer: Safaa Amer, *Social Security Administration*

Chair: Safaa Amer, *Social Security Administration*

BLS CTC 2B

In addition to this designated time slot, this room will be open throughout the conference to those who want to engage with this important topic.

As AI adoption expands across federal agencies, important questions arise: How can AI meet the standards of official statistics? How do we separate meaningful signal from noise? And how do we responsibly evaluate, approve, and acquire AI tools in the federal environment?

This interactive theme room invites participants to engage with focused prompts on representativeness, uncertainty, data quality, bias, confidentiality, validation, governance, and tool approval pathways. We will also discuss practical considerations such as security review, procurement, and piloting new tools responsibly.

Drop in to join the conversation and help shape how AI is thoughtfully integrated into federal statistical practice.

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Session C-1: National Secure Data Service Use Cases: Advancing Federal Mission-Critical Priorities Through Shared Services

Organizers: Sharon A. Boivin & John Finamore, *National Center for Science and Engineering Statistics*

Chair: Emilda Rivers, *National Center for Science and Engineering Statistics*

Discussant: Ed Strocko, *Bureau of Transportation Statistics*

Census Auditorium 1

Linking State and Federal Data to Strengthen Federal Program Evaluation

Rhea Acuña, *Department of Housing and Urban Development*

The ability to link administrative data across federal and state sources is critical to understanding results achieved with federal funding, yet access to these data is often limited and hindered by regulatory and interoperability challenges. This presentation will provide an overview of an NSDS demonstration project piloting a data linkage process between federal housing assistance data and state childcare voucher records to create data files that can be used for evaluation purposes. The project seeks to develop a scalable toolkit to support state-federal data integration for program evaluation.

Measuring Large Language Model Understanding of Federal Statistical Data to Develop an AI-Readiness Plan

Dominique Duval-Diop, *Department of Commerce*

To ensure federal statistical data remain relevant and valuable in this rapidly evolving environment, they must be optimized and enriched within the context needed to support reliable interpretation, machine readability, and adherence to open data principles. This presentation will provide an overview of an NSDS demonstration project developing an empirical evaluation of how large language models respond to questions requiring federal statistical open government data and metadata, with the goal of producing an AI-readiness plan with a framework for assessing data preparedness and a methodology for evaluating response quality.

Fraud in Public Healthcare Programs: Developing Fraud Detection Models with Linked Data

Ashley Hirashima, *Department of Justice*

Fraud detection in public healthcare programs is undermined by fragmented data sources, limited sharing, and lack of interoperability across oversight bodies. This presentation will provide an overview of an NSDS demonstration project that uses healthcare fraud as a use case, integrating federal, state, and open-source data to develop novel fraud detection frameworks and risk-scoring tools that provide a comprehensive view of fraud and vulnerabilities in federally funded healthcare programs. The project aims to strengthen fraud analysis by incorporating underutilized and unstructured data to uncover patterns that might otherwise remain hidden and by offering practical recommendations to address data access barriers in public healthcare programs.

Modeling the Criminal Case Process with State and Local Data

Erica Smith, *Bureau of Justice Statistics*

Producing a comprehensive overview of the criminal case process is hindered by fragmented data systems, limited interoperability, and barriers to accessing sensitive case- and person-level information across state and local jurisdictions. This presentation will provide an overview of an NSDS demonstration project piloting privacy-preserving record linkage tools and processes to integrate operational data across agencies and jurisdictions in order to increase understanding of how cases progress through the criminal justice system, with the ultimate goal of increasing efficiencies in the criminal justice system. It aims to provide new insights into case processing, enhance interoperability, and develop a scalable framework that supports future cross-jurisdictional data integration by addressing barriers to data acquisition and linkage.

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Session C-2: Making Federal Data AI-Ready

Organizer: Bob Sivinski, *Office of Homeland Security Statistics*

Chair: Joe Parsons, *National Agricultural Statistics Service*

Census Auditorium 2

Generative Artificial Intelligence and Open Data: Guidelines and Best Practices

Brock Webb, *U.S. Census Bureau*

The FCSM AI-ready data paper called for experimentation on AI and open data. This talk shares observations and lessons learned from working with AI technologies and building systems around static data, what worked, what didn't, and what practitioners need to know.

AI-Ready Federal Statistical Data: An Extension of Communicating Data Quality

Travis Hoppe, *National Center for Health Statistics*

This presentation highlights the FCSM report AI-Ready Federal Statistical Data (FCSM 25-03) and early work in this area. It explores how agencies can prepare federal statistical data using approaches such as Model Context Protocols to support accurate, authoritative integration with generative AI tools and agents.

Improving LLM Access to Federal Data

Mark Aronson, *General Services Administration*

Bella Mendoza, *General Services Administration*

Sam Levy, *General Services Administration*

Mohammad Arifur Rahman, *General Services Administration*

Haley Johnson, *General Services Administration*

Gregg Rich, *General Services Administration*

The increasing adoption of AI models is transforming how people use federal statistical data, leading to new requirements for data accessibility. Federal statistical agencies must now develop machine-understandable documentation and interfaces AI agents can navigate and interpret. This need for AI-ready public data was articulated in a recent Federal Committee on Statistical Methodology (FCSM) paper, which called on agencies to modernize data access for machine interpretation.

Following the FCSM's recommendations, we began by conducting a baseline assessment of large language models' abilities to retrieve correct data from two federal datasets, CDC's PLACES and USAspending, by prompting a panel of questions for which the datasets contained specific numeric answers. We found that the models were largely incapable of extracting the correct data value. Engineering the prompts did not improve data recall nor were the models capable of forming coherent API queries for the desired data values. However, when given an API response of a relevant query for the question, the models identified the correct value. This capacity to effectively parse an API response indicated that developing Model Context Protocol (MCP) servers, which instruct models querying API endpoints, could be a powerful tool for federal statistical data retrieval. We then developed pilot MCP servers for each dataset, which led to near perfect data retrieval accuracy.

While our pilot servers were capable of simple data retrieval, these datasets, as with any federal statistical data, contain idiosyncrasies known deeply by the data stewards, pointing to the need for future federal data MCP servers to be developed in close collaboration with these stewards.

Our results indicate the enormous potential of MCP servers for advancing federal data accessibility to not only enable machine interpretation but to unlock more nuanced interactions with federal data for academic research, policy development, and evaluation.

From Statistical Data to Machine-Understandable Knowledge: Methodological Advances for AI Readiness in Federal Statistics

Irina Belyaeva, *U.S. Census Bureau*

Artificial intelligence (AI) is reshaping how users discover, interpret, and apply federal statistics. Yet its potential depends on whether statistical data and metadata are prepared in formats that AI systems can reliably consume. Much of today's statistical information is structured for human interpretation rather than automated reasoning, limiting its value for large language models (LLMs) and AI-driven, knowledge-guided analysis. This work introduces methodological advances for transforming statistical data and metadata into machine-understandable knowledge. Building on international standards such as the Generic Statistical Information Model (GSIM), the Data Documentation Initiative (DDI), and Statistical Data and Metadata eXchange (SDMX), the framework applies semantic modeling and knowledge graph techniques to represent programs, datasets, universes, concepts, and variables in structured formats optimized for LLMs and AI consumption. This approach enhances interoperability across federal data assets while ensuring that AI systems can interpret, retrieve, and integrate information in a trustworthy manner. The methodological contributions are threefold. First, semantic structuring: encoding statistical data and metadata as linked knowledge graphs that capture relationships enabling AI-reasoning and cross-survey integration. Second, evaluation frameworks: developing metrics to assess the performance of AI systems that consume statistical knowledge in AI-ready formats. Third, governance integration: embedding the Findable, Accessible, Interoperable, and Reusable (FAIR) data principles into design of AI-ready standards for statistical data. Results from pilot applications demonstrate that semantically structured data improves precision in AI-assisted reasoning and retrieval. Lessons learned highlight both technical and organizational challenges in transitioning to AI-ready data formats and representations. Preparing statistical data as machine-understandable knowledge ensures federal statistics remain AI-ready and accessible in an AI-driven future.

AI-Ready Agricultural Statistics: Deploying NASS's Quickstats with Natural Language Processing Advances

Justin Van Wart, *National Agricultural Statistical Service*

The National Agricultural Statistics Service (NASS) has made significant strides in enhancing the accessibility and usability of their Quick Stats database through the integration of advanced tools and technologies. This study focuses on NASS's experimentation with various platforms such as MCP servers, Google's Data Commons, Databricks, Azure copilot studio, and simple RAG with a particular emphasis on their development of a natural language processing (NLP) bot API for public data queries. The NLP bot api, allows users to interact with the Quick Stats database in a more intuitive and efficient manner, while maintaining security and guardrails that some other approaches have failed to accomplish. While MCP servers provide an excellent toolset for data access, and cloud platforms like Azure and AWS offer robust processing and point-and-click deployment capabilities, the NLP bot stands out by significantly improving user interaction and data accessibility while maintaining security. This was accomplished by crowd sourced training within the agency and simultaneous LLM-led parallel development of data ontology. This advancement underscores the importance of standardized data formats and comprehensive metadata in ensuring data consistency and usability, while also highlighting the critical role of security measures in protecting sensitive information.

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Session C-3: Next Steps in Record Linkage: Adding Value to Existing Data and Making Progress on Hard-To-Measure Concepts and Populations

Organizer: Keith Finlay, *U.S. Census Bureau*

Chair: Keith Finlay, *U.S. Census Bureau*

Census Auditorium 3

Open-source record linkage solutions to improve efficiency

Frances McCarty, *National Center for Health Statistics*

Orlando Davy, *National Center for Health Statistics*

Roshni Patel, *National Center for Health Statistics*

Cordell Golden, *National Center for Health Statistics*

The Data Linkage Program at the National Center for Health Statistics (NCHS) has a longstanding history of integrating survey data collected from national population and provider surveys with key sources of administrative data that expand the utility of the survey data. To conduct these linkages, innovative, customized algorithms that utilize both deterministic and probabilistic linkage approaches are implemented using SAS programs. In addition, open-source software, such as Python and R, include packages and applications specifically developed for record linkage. The use of established open-source packages and applications could improve efficiency by reducing the need for complex programming for each linkage that is conducted. This presentation will present results from linkages of the National Health Interview Survey (NHIS) linked with National Death Index (NDI) data conducted using selected open-source linkage packages and applications, such as the Python Record Linkage Toolkit and the R package fastLink. Results from the linkages conducted using these open-source solutions will be compared to those from NCHS' customized linkage previously implemented in SAS. For this evaluation, the SAS linkage results will serve as the benchmark and the following performance metrics are presented for each of the open-source solutions: recall, precision, and F1-score. The results of each linkage are provided along with lessons learned related to their implementation.

Comparing Ferry Statistics Using Results from the National Census of Ferry Operators and AIS Data

Carl Cloyed, *Bureau of Transportation Statistics*

April Gadsby, *Bureau of Transportation Statistics*

Clara Reschovsky, *Bureau of Transportation Statistics*

Kyle Titlow, *Bureau of Transportation Statistics*

Cha-Chi Fan, *Bureau of Transportation Statistics*

Surveys and censuses are the standard method to obtain data for official statistics. These datasets are frequently augmented by administrative data, which are collected for non-statistical purposes. Increasingly, large, alternative datasets, such as social media, mobile phone, and GPS data, are being utilized as administrative datasets and linked to survey data. However, little work has directly compared statistics generated between surveys and other sources. We measured uncertainty in ferry statistics—number of trips per year, segment duration and distance, and segment season length—as the difference between the National Census of Ferry Operators (NCFO) and Automatic Identification System (AIS) data estimates from 2020 and 2022. We modeled uncertainty by operator parameters (i.e., number of vessels, terminals, segments, operator passengers, and segment passengers, region, publicly funded, operator ID) to determine if operator characteristics and complexity drove patterns of uncertainty. Uncertainty was highest in number of trips per year in 2020 and segment duration in 2022. Uncertainty was frequently related to number of segment passengers, increasing for trips per year but decreasing for segment duration, distance, and season length. Uncertainty increased with the response magnitude and generally decreased when passengers were controlled by number of trips, segment duration, and segment distance, suggesting complex operators in terms of segment passengers had less uncertainty. We highlighted where and how errors can arise in survey data. Furthermore, incorporating aspects of these datasets into the survey can reduce burden in the NCFO. For example, we can

prefill surveys with segment durations and distances that were determined using measures of central tendency from distributions generated by each segment's set of trips, and operators can verify the information. Our study demonstrates that incorporating publicly available, alternative data can help verify survey data quality and increase data collection efficiency.

Using Linked Data to Measure Income for Tribal Areas

Carla Medalia, *U.S. Census Bureau*

Randy Akee, *Harvard University and NBER*

Carl Sanders, *U.S. Census Bureau*

Cass Dorius, *U.S. Census Bureau*

Sallie Keller, *U.S. Census Bureau*

Income measurement on American Indian Reservations is notoriously difficult, with small survey sample sizes and inconsistent administrative records population coverage. Recent research has established that linking household survey and administrative tax record data to produce individual income statistics can mitigate measurement error from both sources. This paper may be the first to apply this innovative approach to examining income for individuals living on federally-recognized tribal reservations, a unique geographic population that spans the United States and has different demographic and socioeconomic characteristics than the nation as a whole. Using the American Community Survey linked to Internal Revenue Service 1040 information returns and W-2 filings from 2021 to 2023, we examine discrepancies between survey and tax record reports of wages and income. Building on research developed by the Census Bureau's National Experimental Wellbeing Statistics (NEWS), we compare estimates for residents of tribal reservations to the nation by socioeconomic and demographic characteristics. Additionally, we analyze the correlation between individual characteristics and the size of the measurement error. Specifically, we look at differences in receipt of W-2s, filings of 1040s, individual-level differences in wages, and household-level differences total income. In addition to filling a crucial information gap in our understanding of income measurement, this paper provides a proof of concept for the Statistical Products First initiative and demonstrates the value of producing statistics useful to sovereign tribal nations.

National Survey of Children's Health Participant Location Efforts: Unique Challenges, Unique Solutions

Scott Albrecht, *U.S. Census Bureau*

Reem Ghandour, *Maternal and Child Health Bureau, HRSA*

Olivia Sappenfield, *Maternal and Child Health Bureau, HRSA*

Victoria Sheil, *U.S. Census Bureau*

John Voorheis, *U.S. Census Bureau*

Maria Perez-Patron, *U.S. Census Bureau*

The National Survey of Children's Health - Longitudinal Cohort (NSCH-LC) data collection was carried out between January and June 2024. It was designed to locate and re-interview participants (the selected child and reporting adult) of the National Survey of Children's Health (NSCH) annual data collections from 2018 and 2019. The NSCH-LC was a uniquely challenging data collection effort, because the original NSCH annual survey was not designed to facilitate location and re-interview of survey participants. It did not collect identifying or contact information – e.g. participants' full names, email or phone number. Also, to protect the confidentiality of past participants, contact materials for the NSCH-LC could not include personally identifiable information or reveal past participation for the target household. Therefore, to execute the NSCH-LC data collection, we used limited household information to locate likely movers. We verified the household and child selection based exclusively on information provided by NSCH-LC respondents. The U.S. Census Bureau's data linkage infrastructure allowed us to assign anonymous and unique individual and household linkage keys to the 2018/2019 annual surveys. We then linked the survey respondents to administrative records, which were used to construct address histories for the respondents and identify the most likely place of residence at the time of follow up. Finally, field representatives could be assigned for in-person follow-up when address information was unreliable or out of date, or if the address was unresponsive by mail. In this presentation, we will discuss the challenges we faced in executing the NSCH-LC data collection and the approaches we employed to overcome those challenges, including creative applications of administrative records and algorithmic within-instrument

matching. We will also discuss the effectiveness and limitations of these approaches, and lessons learned from this experience.

Like Needles in a Haystack: Identifying Long-Term Cohabitors Using a New Measure of Co-Residential Duration

Brian Knop, *U.S. Census Bureau*

Marta Murray-Close, *U.S. Census Bureau*

One of the most pronounced demographic changes in family and household composition over the last 50 years is the rise of non-marital cohabitation. These arrangements tend to be short-lived, with couples often breaking up or marrying within 2 years (Copen et al. 2013). Scholars have described these relationships as alternatives to singlehood rather than alternatives to marriage, given their short duration (Sassler 2004). But what about couples who don't marry or break up in the first few years? Qualitative studies show that long-term cohabitors exist and appear to differ from short-term cohabitors both demographically and in how they view their relationship (Sassler and Milller 2023). Quantifying long-term cohabitors requires information about the duration of these relationships, but questions about cohabiting duration are almost non-existent in nationally representative household surveys. To overcome this limitation and produce national estimates about the prevalence of long-term cohabitors, we construct a measure of cohabiting duration. We augment data from the American Community Survey (ACS) with relationship histories from several decades of Internal Revenue Service (IRS) 1040 tax records. We find that although most cohabiting couples have been together for a short duration, more than 1 in 5 cohabitors in the U.S. have been living together over 5 years. We show how these long-term cohabitors differ demographically from both short-term cohabitors and their married counterparts. Because of the lack of cohabiting duration data in surveys, researchers don't generally distinguish between long- and short-term cohabiting couples. And because they are a smaller share of cohabitors, long-term cohabitors have unique characteristics that are going unnoticed in analyses. Our research contributes to federal statistics by showing the usefulness of administrative records to fill in the gaps when survey data, alone, is not able to fully capture and distinguish subpopulations in the data.

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Session C-4: Modernizing the Survey of Income and Program Participation

Organizer: Neil Bennett, *U.S. Census Bureau*

Chair: Neil Bennett, *U.S. Census Bureau*

Census Conference Rooms 1-2

Modernizing the Survey of Income and Program Participation

Neil Bennett, *U.S. Census Bureau*

Sharon Stern, *U.S. Census Bureau*

Jason Fields, *U.S. Census Bureau (former)*

The Census Bureau is amid an agency-wide effort to innovate the enterprise approach to the data lifecycle with improvements in data collection, processing, imputation, and the use of administrative records. This period of transformation has created a prime opportunity to begin implementing the modernization of the Survey of Income and Program Participation (SIPP) that is part of SIPP's continuous improvement program. SIPP, one of Census' flagship surveys, is a nationally representative survey that measures monthly changes in employment, income, household dynamics, and government program participation. In this paper, we discuss the challenges that SIPP faces in the current survey operations landscape, from declining response rates to rising collection costs, and the opportunities for modernization. Our priorities in the new SIPP design are to reduce respondent burden, move to multimode data collection, and increase the use of administrative records. With these priorities in mind, we outline the roadmap for SIPP's modernization and the steps The Census Bureau is taking to improve upon the SIPP program.

Targeted CAPI Follow-up in Multimode Household Surveys: A Simulation-Based Evaluation

Kevin Tolliver, *U.S. Census Bureau*

Ashley Westra, *U.S. Census Bureau*

Shelley Irving, *U.S. Census Bureau*

Neil Bennett, *U.S. Census Bureau*

Survey non-response in web-push data collection is a perennial problem that leads to a biased sample. Survey programs using web-push data collections may mitigate some sample bias by prioritizing cases during in-person non-response follow-up. Despite the improvements that case prioritization techniques offer when it comes to developing a representative sample, rising survey costs still require that programs determine how to most efficiently follow up with high-priority households. The Survey of Income and Program Participation (SIPP) is amidst a modernization effort that will move the survey towards multimode data collection, reduce respondent burden, and incorporate administrative records across the survey lifecycle. Key priorities in SIPP's modernization effort are to develop a targeted non-response strategy that produces a representative sample while considering cost constraints. In this paper, we simulate a targeted in-person non-response follow-up and consider the tradeoff between data quality and cost savings when interviewing high-priority households.

Model-based or Hot deck? Imputing Item Non-response Data in the Survey of Income and Program Participation (SIPP)

Besufekad Alemu, *U.S. Census Bureau*

Angelica Phillips, *U.S. Census Bureau*

Ruth Sarafin, *U.S. Census Bureau, former*

Joey Marshall, *U.S. Census Bureau*

Sandy Dietrich, *U.S. Census Bureau*

Hot deck has been the default method of imputing missing data in the Survey of Income and Program Participation (SIPP) for many years. However, hot deck is not without its limitations which are even more pronounced in the face of increasing rates of survey non-response. In this research, we investigate the extent to which model-based imputation methods, relative to hot deck, enhance the quality and accuracy of imputed data for item-level nonresponse in the SIPP.

To address this topic, we used data from the 2022 SIPP and compared hot deck imputation against four model-based imputation methods: non-iterative Linear and Logistic regressions, Multiple Imputation by Chained Equations (MICE), and Extreme Gradient Boosted Trees (XGBoost). These methods were applied across different types of variables with various non-response rates. We randomly split the observed, non-hot decked, observations into training and test samples; the training sample was used to develop the imputation models, which were then applied to and evaluated on the test sample. The imputation evaluation metrics we used include accuracy, sensitivity, specificity, and F1 score for categorical variables; and mean error, correlation, and root mean squared error for continuous variables. We also assessed the processing time for these models to understand the implications from a production standpoint. Our results indicate that MICE and XGBoost significantly improved the quality of the imputations and reduced processing time relative to hot deck imputation. We discuss the implications of our work on the SIPP as well as broader considerations which can be applied to other surveys.

Looking for Work While Employed: Who, When, and How?

Matthew Pesner, *U.S. Census Bureau*

Alexandra Opanasets, *U.S. Census Bureau*

Research highlights that job search while employed is a critical determinant of labor market competitiveness and earnings growth. Although economic models have long incorporated employed job search – and longstanding evidence shows the employed search at much higher rates than the nonemployed – no federal survey consistently collects this information.

As part of ongoing research on modernizing the Survey of Income and Program Participation (SIPP) while

maintaining and strengthening the survey's innate longitudinal advantages for studying job mobility, we collected search information from both employed and nonemployed adults. Our setting was the Census Bureau's May 2025 Household Trends and Outlook Pulse Survey (HTOPS), an internet self-response survey designed for rapid evaluation. For all adults, we collected whether an individual had searched in the last 4 weeks, as well as the specific methods of search used, distinguishing between those methods that are active (e.g., interviewing) and passive (e.g., working on one's resume). Crucially, these questions were embedded within a broader sequence that collected job details and earnings information over a six-month reference period, allowing for statistics of on-the-job search by job type (e.g., employee or self-employed), multiple job holding, relative earnings, and other economic and demographic dimensions.

In this project, we provide one of the first large-scale, federal survey-based accounts of on-the-job search, documenting rates, patterns, and correlates of job search among the employed. We also seek to illustrate what kinds of novel measures could be constructed, were search information collected monthly as part of a redesigned SIPP. In particular, we explore how combining such information from both the employed and nonemployed with detailed job mobility information could yield a novel and timely labor market tightness index.

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Wednesday, March 11, 2026

1:15 PM

Session C-5: Evaluating the Unseen: Tackling Bias and Error in Federal Statistics

Organizer: Brian Ward, *National Center for Health Statistics*

Chair: Brian Ward, *National Center for Health Statistics*

Census Conference Rooms 3-4

High-Level Design of the 2030 Census Coverage Estimation Program for National and State Estimates

Scott Konicki, *U.S. Census Bureau*

The Census Bureau has historically estimated coverage of the decennial census using data collected from a post-enumeration survey. For 2030 Census coverage estimation, we plan to maximize our use of administrative records to produce dual-system estimates. Using administrative records offers substantial improvements to the historical survey-based design. Because the estimates are derived from national-level frames, they are not subject to sampling errors. Many of the costly and time-consuming data collection and matching activities of previous post-enumeration surveys can be eliminated or greatly simplified. However, administrative records have their own limitations and sources of nonsampling error that must be carefully addressed so that we can continue to produce high-quality estimates of census coverage. In this paper, we propose a high-level design of the 2030 Census coverage estimation to support the national and state-level estimates of net census coverage errors. This design leverages the Demographic Frame, a Census Bureau enterprise frame of people compiled from numerous administrative records sources. We will review the major activities for producing the dual-system estimates and how these differ from historical coverage estimation programs. Further, we will discuss challenges with this new approach and how we plan to address these challenges in future research. These strategies have been informed by research using the 2020 Census and Demographic Frame data.

Confounding Factors in Total Survey Error Measurement

Greg Erkens, *Bureau of Labor Statistics*

Surveys typically publish estimates of sampling error to help users gauge the accuracy of the survey's estimates, but surveys also experience the effects of nonsampling errors – errors which often remain unpublished. The Total Survey Error (TSE) framework (Groves and Lyberg, 2010) catalogues some errors that may influence

survey estimates' overall quality. Some surveys use Monte Carlo methods to develop measures of nonsampling errors (Elam-Evans, Hill, et al., 2023), and other authors use administrative data (Meyer and Mittag, 2021). Groves and Lyberg's TSE framework notes the impact of other error types such as processing errors, but those dimensions can also impact the calculation of and inject some unreliability into the survey error components derived using administrative data.

The Current Employment Statistics (CES) survey publishes estimates of monthly employment changes in US business establishments. Each year CES benchmarks its employment totals to values derived from the Quarterly Census of Employment and Wages (QCEW) – with the revision considered an approximation of total survey error.. Huff and Gershunskaya (2009) developed a methodology to decompose the benchmark revisions into multiple TSE components using administrative data from the QCEW. This paper illustrates the impact of several additional error sources – such as administrative data and microdata processing errors – on the CES surveys' TSE measurements, and how those error sources may confound analyses of total survey error.

Addressing Selection Effects from Price Relative Formation in Scanner Data Price Indexes with Inverse Probability Weights

Brendan Williams, *Bureau of Labor Statistics*

Conventional matched-model, scanner data price indexes are susceptible to a selection bias related to the frequency at which a product is observed. Lower volume products are proportionally less likely to be observed in adjacent time periods compared to high volume products and thus lower volume products are underrepresented as observed price changes in matched-model price indexes. Aggregated measures of price change may then be biased when low-frequency products have pricing behavior distinct from high-frequency products. We document the issue empirically in several transaction datasets and introduce an inverse probability weight correction as a general means of addressing this issue. As sales data are increasingly used in price index estimates, this frequency bias may be widespread but unnoticed.

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Session C-6: Innovations in State-Level Statistics at BEA: New Approaches to Trade, Government, and Gross Output Measurement

Organizer: Jennifer Bruner, *Bureau of Economic Analysis*

Chair: Jennifer Bruner, *Bureau of Economic Analysis*

BLS Auditorium A

Developing Federal Government Consumption Expenditures and Gross Investment Statistics by State

Kun Li, *Bureau of Economic Analysis*

Christian Awuku-Budu, *Bureau of Economic Analysis*

Kyle Hood, *Bureau of Economic Analysis*

This paper details BEA's ongoing efforts to develop experimental statistics on federal government consumption expenditures and gross investment by state. These new statistics would provide valuable information for policymaking, research, and business decision-making at the state level, complementing BEA's national data on federal spending. The methodology used to prepare the experimental estimates relies on a wide variety of source data and it is consistent with the methodology used for the corresponding national statistics. When state-level detail is available in the source data used for the national statistics, it is directly applied; when such detail is lacking, alternative sources and a modified methodology are used. In general, the statistics are derived from allocators developed for each detailed component of federal government expenditures, which are used to allocate national component totals to states. Preliminary estimates for 2012–2023 show considerable variation in the economic impact of federal government spending across states. For example, on average from 2012 to 2023, federal spending as a percentage of state gross domestic product ranged from 2.5 percent in Delaware to 27.6 percent in Virginia, with the District of Columbia averaging 51.9 percent. As these estimates are preliminary, BEA invites feedback from stakeholders and data users to improve the methodology and accuracy of these statistics.

A Step Toward State-Level Supply-Use Tables: A Strategic Roadmap and Early Preview of Gross Output and Merchandise Trade by State

Gabriel Medeiros, *Bureau of Economic Analysis*

Ledia Guci, *Bureau of Economic Analysis*

Dirk van Duym, *Bureau of Economic Analysis*

Kyle Hood, *Bureau of Economic Analysis*

Mahsa Gholizadeh, *Bureau of Economic Analysis*

Supply and Use Tables (SUTs) are a foundational tool in national economic accounting, illustrating economic interdependencies by tracking the flow of goods and services across sectors. They also serve as a valuable framework for reconciling various economic statistics—such as gross domestic product (GDP), value added, and gross output—by aligning production with its uses.

The Regional Economic Accounts Directorate at the Bureau of Economic Analysis (BEA) has set a long-term goal to develop SUTs at the state level. These state-level SUTs aim to integrate diverse economic statistics by state and offer insights into supply chains across industries and geographic regions. This talk outlines a strategic roadmap for developing state-level SUTs, emphasizing their utility and identifying important data and methodological challenges.

This roadmap entails not only incorporating existing BEA statistics on value added and personal consumption expenditures (PCE) by state into the SUT framework but also requires the development of new statistics that are precursors towards state SUTs. This presentation will preview two such statistics under development - gross output by state and merchandise trade by state - detailing their data sources, methodologies, and presenting experimental results for the 2017 calendar year.

Estimating Services Exports by State: A Linked Microdata Approach

Jessica McCloskey, *Bureau of Economic Analysis*

BEA conducts enterprise-level surveys of international trade in services, covering various business services, transactions in intellectual property, financial services, and insurance services. This presentation will describe a project to develop estimates of services exports at the state level based on BEA's survey data and Census establishment-level data. This work uses a BEA-Census microdata link that was developed to enable new research, which is available to staff of both agencies and users of the Federal Statistical Research Data Centers. Match rates were very high, covering most of the value of U.S. exports. Detailed establishment-level information on activity by location from the Economic Census is used to apportion enterprise-level service exports for 2007, 2012, and 2017. The project developed an allocation procedure that strives to identify the establishments that are likely to be involved in exporting, beginning with the most likely traders with similar characteristics to the BEA exporter and proceeding to alternative methods based on the data available. This presentation will describe the data and allocation methods as well as present exploratory state-level estimates of services exports derived using this approach.

Toward U.S. Trade in Services by State: Exploratory Results of a Survey-Based Approach for Selected Services and Intellectual Property

Scott Wingo, *Bureau of Economic Analysis*

BEA produces statistics on U.S. international trade in services using a combination of surveys and non-survey-based methods. These statistics are currently available at the national level, and BEA is working to produce state-level estimates. As part of this process, BEA added questions on the composition of trade in services by state to the 2022 Benchmark Survey of Transactions in Selected Services and Intellectual Property with Foreign Persons (BE-120). Respondents were asked to list the top three states or territories for their sales to (purchases from) foreign persons as well as the associated shares of their total sales to (purchases from) foreign persons. The resulting firm-level data is the first step in developing estimates covering all service types. Parallel questions were included on the 2023 Insurance Transactions by U.S. Insurance Companies with Foreign Persons (BE-140) and the 2024 Financial Services Transactions Between U.S. Financial Services Providers and Foreign Persons (BE-180). BEA is also developing methods to estimate trade by state for service types estimated using non-survey data and methods. The resulting statistics can inform policymakers and stakeholders about the geographic distribution of international trade in services.

This paper presents information about response rates across different parts of the BE-120 sample before introducing preliminary results of the survey. These results are an interim measure pending ongoing work to estimate values for firms that did not respond or whose reported shares sum to less than 100 percent of trade. This paper will then evaluate the reported data through comparison to other state-level economic data. The paper will conclude with lessons and plans for future work.

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Session C-7: Innovative Solutions for Small Area Problems: From Machine Learning to Novel Modeling

Organizer: Yan Liu, *Internal Revenue Service*

Chair: Yan Liu, *Internal Revenue Service*

BLS CTC 2A

Integrating Spatial and “Nearest Neighbors” Effects to Estimate Diabetes Prevalence for Small Areas in Puerto Rico

Hui Xie, *Centers for Disease Control and Prevention*

Kai McKeever Bullard, *Centers for Disease Control and Prevention*

Li Rui, *Centers for Disease Control and Prevention*

Holly Du, *Centers for Disease Control and Prevention*

Diabetes remains a persistent public health concern in Puerto Rico, with high prevalence rates posing substantial challenges to the health system. Accurate small area estimates might be essential for developing targeted prevention and optimizing health care delivery. However, insufficient sample sizes at the municipio (county equivalent) level could hinder reliable estimates. Spatial small area estimation (SAE) techniques are commonly applied, borrowing “strength” from neighboring municipios based on spatial autocorrelation. Despite their utility, the key assumption of spatial SAE—that geographically close areas are more likely to exhibit similar characteristics due to spatial autocorrelation—is often violated. Adjacent municipios frequently differ in population demographics and socioeconomic characteristics, which can weaken or negate spatial dependencies and compromise the accuracy of estimates. To address this limitation, our study proposes a machine learning-based approach to identify “nearest neighbors” for each municipio based on demographic and socioeconomic factors rather than geographical proximity. This novel method allows SAE models to borrow “strength” from both spatial and sociodemographic “Nearest neighbors,” enhancing the robustness of estimates. The proposed approach was validated with empirical and simulation data. The results showed a 39.6% reduction in mean squared error and a 25.1% reduction in mean bias error compared to traditional spatial SAE models. This innovation could provide a robust framework for improving diabetes prevalence estimation at the municipio level, even those with limited or no sample size, and reliable insights for pinpointing areas for public health interventions.

Improving Estimates of the Civilian Non-Institutional Population for Counties and County Equivalents in the U.S. and Puerto Rico

Andrew C. Forrester, *Bureau of Labor Statistics*

The Local Area Unemployment Statistics (LAUS) program at the U.S. Bureau of Labor Statistics produces estimates of employment and unemployment for states, counties, and cities with a population of at least 25,000. Previous research by the LAUS program developed experimental estimates of the civilian non-institutional population (CNP) for counties at a monthly frequency. Since their first iteration described in Forrester (2024), the LAUS program further refined the county CNP estimates to incorporate additional input data and reconcile geographic boundary over time. Notable improvements make use of special tabulations of the Privacy Protected Microdata Files from the 2010 and 2020 enumerations, incorporate newer postcensal population data from the Vintage 2024 estimates, and apply new statewide intercensal population controls introduced in March 2025 (Forrester and Wilson, 2025). The improved county-level data are presented in comparison with the previous methodology and in relation to statewide benchmarks. Further LAUS research efforts to develop experimental labor force rates and ratios for substate geographies will be discussed.

Response Model Selection in Small Area Estimation in Case of Not Missing at Random Nonresponse and Only Population Total Available for Auxiliary Variables

Michael Sverchkov, *Bureau of Labor Statistics*

Sverchkov and Pfeiffermann (S-P, 2023) suggested Information Criteria, similar to AIC and BIC for response model selection for Small Area Estimation under informative probability sampling of areas and within the sampled areas, and not missing at random (NMAR) nonresponse. The approach considered by S-P requires knowledge of auxiliary data for complete sample before response occurs. In this paper we generalize this approach to the case where auxiliaries are not known for the non-respondents, but population (or sample) totals for the auxiliaries are known.

Minimizing Seasonal Adjustment Suppressions due to MSA Redefinitions

Paige Schroeder, *Bureau of Labor Statistics*

The Current Employment Statistics State and Area (CESSA) program seasonally adjusts their employment estimates for metropolitan statistical areas (MSAs) at the total nonfarm level. This process uses a two-step method that requires both the benchmarked data along with sample-based estimates. To set the seasonal model for the coming year, at least three years of sample-based estimates are needed. However, every ten years, revised delineations to the metropolitan statistical areas are issued. The re-delineations may cause large compositional changes within existing MSAs or produce new MSAs altogether. The previous policy was to suppress seasonally adjusted sample-based estimates for new and substantially changed areas until there were adequate sample-based histories available. During the last re-delineations in 2014, this resulted in CESSA suppressing seasonally adjusted estimates for 91 new or largely changing areas for three years.

As an attempt to suppress fewer seasonally adjusted series, a new option was explored this year that wasn't available in the previous re-delineations – instead of using actual histories, a method was developed to simulate the sample-based estimates for use in the creation of seasonal models. To test this, sample-based simulations were generated going back to benchmark year 2017. This paper elaborates on the method and analyzes the results.

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Session C-8: Connecting the Dots: Linking Data for Smarter Social and Economic Measures

Organizer: Tywanquila Walker, *Bureau of Labor Statistics*

Chair: Tywanquila Walker, *Bureau of Labor Statistics*

BLS Auditorium C

Effect of Evictions on Children's Later Life Wellbeing

Ashley Erceg, *U.S. Census Bureau*

Carl Gershenson, *Princeton Eviction Lab*

Nick Graetz, *University of Minnesota, Princeton Eviction Lab*

Nicole Perales, *U.S. Census Bureau*

Experiencing eviction is common among children; one in seven will experience eviction by age 15 and more will experience an eviction. Eviction in childhood is associated with poor health in the short-term, but little is known about the effect of evictions on medium- and long-term health and non-health outcomes for children. Further, most studies rely on comparisons between experiencing an eviction court filing versus a court-ordered eviction, but little is known about the effect of the initial filing.

We construct a novel dataset which links nationwide eviction court records from 2008-2018 to the Decennial Census (DC), American Community Survey (ACS), and Census Household Composition Key (CHCK), creating the

largest national samples of children at risk of eviction in the US, including in-utero exposure. To identify children living in households who experienced an eviction filing, we start by following the existing literature and use household rosters observed the year prior to an eviction filing. To address the undercounting of children under age two and exclusion of in-utero exposure in this approach, we supplement these rosters with births observed in the CHCK. The CHCK uses administrative records to link children, born starting in 1997, to their parents. We use matched control design to compare children who never experienced an eviction filing to estimate the effect of eviction.

For each child in the samples, we link to survey (ACS) and administrative records to observe later life outcomes, including education, fertility, housing stability, and living arrangements. We estimate the effects of eviction filings and evictions by comparing outcomes between children (including in-utero exposure) who did not experience an eviction filing, experienced an eviction filing but were not evicted, and experienced an eviction. We explore heterogeneous effects by geographic and sociodemographic characteristics, as well as features of the local housing market and legal protections.

Improving Measurement of People, Businesses and the Environment: New Advances from the Environmental Impacts Frame

John Voorheis, *U.S. Census Bureau*

Understanding relationships between people, businesses and the environment provides crucial context for the economic and social measurement mission of the US Statistical System. The Census Bureau's Environmental Impacts Frame (EIF) project provides one such framework and infrastructure for advancing our knowledge on these questions. The EIF is a modular microdata infrastructure, providing access to researcher to harmonized address history information from 1970 to the present, harmonized geospatial data on environmental hazards and amenities, and high quality information on demographic, economic and housing characteristics. The EIF infrastructure has been used to produce high quality research on the distribution of flood risk, the impacts of exposure to wildfires, and the long run impact of exposure to environmental hazards such as particulate matter and toxic chemicals such as PFAS.

To provide insights from this important infrastructure to the public, the EIF project has been developing public facing data products. The most important of these is the Gridded Environmental Impacts Frame, which provides access to a privacy protected statistics on population and housing units on a fixed 0.01 degree grid, along with harmonized environmental hazards and amenities in the same grid topology. One important use of this gridded data is to provide near-real time analysis of the populations and housing stock exposed to environmental hazards (such as the wildfire smoke-related air quality events, hurricanes and wildfires). This gridded data also underlies in progress work to develop interactive tools that allow users to explore the distribution of environmental hazards such as air pollution and flood risk over the last two decades.

This presentation, if accepted, would summarize these strands of work and describe future efforts to expand the EIF infrastructure.

Developing a Measure of Monthly Comprehensive Well-being in the United States

Brian Glassman, *U.S. Census Bureau*

Adrienne DiTommaso, *U.S. Census Bureau*

Christina Bijou, *U.S. Census Bureau*

Marissa Davila, *U.S. Census Bureau*

There is growing interest in developing comprehensive measures of well-being to provide a more holistic understanding of social and economic conditions in the United States. While many agencies and organizations release indices of economic well-being in the U.S., few are based on a comprehensive conception of well-being that moves beyond income to include both objective and subjective components of well-being. Fewer are available at a frequency that allows observation of to-the-moment changes in well-being.

To address this gap, this paper outlines the development of a monthly indicator intended to capture changes across various domains of well-being. Specifically, the proposed indicator incorporates measures across five domains: financial hardship, resources, housing, health, and perceived life situation. The financial hardship domain includes economic measures from the Bureau of Labor Statistics (BLS) and the Federal Reserve Bank of New York to assess unemployment and 90+ day delinquency rates on revolving credit cards. The resources domain utilizes data from the U.S. Bureau of Economic Analysis (BEA) and the BLS to capture real weekly

earnings and consumption spending. Within the housing domain, we incorporate data from Zillow to include measures of housing affordability. The health domain relies on mental health measures from the Household Trends and Outlook Pulse Survey (HTOPS). Finally, the perceived life situation domain uses data from HTOPS and the University of Michigan to measure difficulty paying expenses, life satisfaction, and consumer sentiment. Each measure is standardized and weighted by domain to calculate a monthly composite. The composite well-being indicator value is not intended to be interpreted as a single meaningful value, but rather a month-to-month gauge of changing social and economic conditions in the U.S. The indicator's monthly periodicity will enable a better understanding of how policy changes and broader events coincide with dimensions of well-being.

National Coral Reef Monitoring Program: Results from the Field and Future Developments

Sarah Siegel, *National Oceanic Atmospheric Administration*

Amanda Alva, *National Oceanic Atmospheric Administration*

Mary Allen, *Lynker, Inc.*

Aleia Clark Fobia, *National Oceanic Atmospheric Administration*

Erica Towle, *National Oceanic Atmospheric Administration*

The socioeconomic component of the National Oceanic and Atmospheric Administration (NOAA)'s National Coral Reef Monitoring Program (NCRMP) collects socioeconomic data, alongside biological and environmental data across U.S. coral reef states and territories. These data inform human dimensions indicators that broadly cover population demographics, human use of coral reef resources, and knowledge, attitudes, and perceptions of coral reef management. The program aims to track relevant information about communities and their relationship to coral reef ecosystems. The data is intended to inform local, regional, and national programs designed to protect coral reefs as well as to inform continuing research and communication projects.

Every year, NCRMP targets one of seven populated coral reef states and territories for primary data collection by conducting a representative survey of the population. Using address-based sampling, surveys use either a mail push-to-web design or face-to-face interviewing. The program then updates remaining indicators with available secondary data. Secondary data sources for indicators include data sets such as Census American Community Survey data, Environmental Protection Agency data, and NOAA economic data.

The presentation will share results from the most recent primary data collection in the U.S. Virgin Islands (USVI). The data collection effort occurred in the spring of 2025 and constituted the second round of data for the territory. The presentation will share high-level results comparing differences between the three islands, St. John, St. Thomas, and St. Croix, as well as different marine resource user groups. Furthermore, the presentation will explore future avenues for integrating primary and secondary data into indicators. As survey costs continue to rise, this work presents an example of integrating secondary data into indicator development for federal statistical programs.

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Session C-X: Meet a Member of the Interagency Council on Statistical Policy (ICSP)

Organizer: Carla Medalia and Jenny Hunter Childs, *U.S. Census Bureau*, John Finamore, *National Center for Science and Engineering Statistics*

Chair: Carla Medalia and Jenny Hunter Childs, *U.S. Census Bureau*, John Finamore, *National Center for Science and Engineering Statistics*

BLS Board Room

The Interagency Council on Statistical Policy (ICSP) plays a central role in coordinating the Federal Statistical System—connecting the heads of the principal statistical agencies, Statistical Officials across cabinet

departments, the U.S. Chief Statistician, and the Office of Management and Budget to align priorities, standards, and strategy.

Please join this breakout session for a unique opportunity to engage directly with ICSP members in an open, participant-driven conversation. Bring your questions about cross-agency coordination, statistical policy, emerging priorities, or the future direction of federal statistics.

Wednesday, March 11, 2026

3:00 PM

Session D-1: Open for Collaboration and Innovation: Partnerships Across the Federal Statistical System

Organizers: Jenna Truman, *Bureau of Justic Statistics*; Jennifer Hunter Childs, *U.S. Census Bureau*; Carla Medalia, *U.S. Census Bureau*

Chair: Kevin Scott, *Bureau of Justic Statistics*

Census Auditorium 1

Lessons Learned from 10 Years of the Interagency Council on Statistical Policy Mentoring Program

Kelly L. Myrick, *National Center for Health Statistics*

Bob Sivinski, *Office of Homeland Security Statistics*

Venicia Nichols, *Bureau of Economic Analysis*

Laniera Jones, *Bureau of Economic Analysis*

The main goals of the Interagency Council on Statistical Policy Mentoring Program (ICSP MP) are to help participants grow in their careers; and to build connections between ICSP federal statistical member agencies and units. The ICSP MP celebrated its 10-year anniversary in 2025. Over the years, ICSP MP navigated the coronavirus pandemic, changes in the federal statistical system, and changes in federal government priorities. This presentation will discuss lessons learned that support the federal statistical system.

Yearly since 2015, between 7 to 14 federal statistical agencies/units participate. The agencies/units provide mentors, mentees, and working group members (WG). The WG coordinates recruitment, selection of participants, and plans program events. The program has added new agencies/units, increased cohort size, and served over 200 participants since 2015. Several ICSP MP mentees have returned as mentors, several mentors volunteer more than once, and mentees and mentors have joined the WG. The program began as an in-person program for staff of the 13 Principal Statistical Agencies in the national capital region. The first cohort had 14 mentee-mentor pairs. Following the passage of the Evidence Act and the expansion of the ICSP in 2019, the ICSP MP also expanded to include a broader set of agencies across the federal government. To adapt to the coronavirus pandemic, the program became virtual and recruited participants from across the country. The ICSP MP remains primarily virtual, although optional in-person events were added starting in 2022, e.g., Federal Committee on Statistical Methodology Research and Policy Conference.

We use participant feedback from evaluations to improve our programs and better serve our participants. The 2025 cohort had 22 mentee-mentor pairs; our largest cohort. Despite many changes, the ICSP MP continues to serve our federal statistical system by providing a community for past, present, and future federal statistical leaders.

Building a Baseline for Disaster Recovery Measurement: The Census Bureau's CAREs Dataset and Hurricane Helene

R. Chase Sawyer, *U.S. Census Bureau*

Hurricane Helene was one of the most catastrophic hurricanes in recent U.S. history, responsible for at least 249 fatalities and widespread disruption. Measuring its impact requires both timely data and innovative approaches to integrate disparate federal sources. This presentation introduces the Census Bureau's Assessment, Recovery, and Evaluations Datasets (CAREs), a new resource designed to provide a standardized,

accessible framework for disaster-related analysis.

CAREs combines data from the U.S. Census Bureau, Federal Emergency Management Agency (FEMA), Department of Energy (DOE), National Oceanic and Atmospheric Administration (NOAA), and Bureau of Labor Statistics (BLS). By curating information on populations, disaster exposure, and recovery outcomes, CAREs enables researchers and policymakers to evaluate recovery needs comprehensively in a single dataset.

We present several case studies that demonstrate the value of CAREs in understanding recovery dynamics. These include examining relationships between disaster exposure, such as power outages, high winds, and heavy rainfall and subsequent recovery outcomes. The results show clear associations between the scale of exposure and indicators of community recovery, including assistance registrations. By linking these elements, CAREs illustrates how exposure influences recovery needs and the interaction with the population in the area. This work highlights both the analytic utility of standardized cross-agency datasets and the importance of establishing baselines for disaster recovery measurement. CAREs provides a model for how federal data integration can support evidence-based decision-making and strengthen resilience in future disaster response and recovery efforts.

A Strategic Vision for Innovating Federal Statistics

Carla Medalia, *U.S. Census Bureau*

Jenny Hunter Childs, *U.S. Census Bureau*

Tim Gilbert, *U.S. Census Bureau*

Declining survey response rates, growing demands for near real-time and granular data, and increasing mandates to improve efficiency have accelerated ongoing efforts to improve the way the Census Bureau produces statistics. This presentation lays out a strategic vision for innovating federal statistics driven by the Statistical Products First (SPF) framework, which reorients the process around the end goal of delivering the right statistics to the right users at the right time. The SPF approach includes six dimensions. First, we begin by asking which statistics stakeholders need to make evidence-based decisions and which statistics are statutorily required. Second, we determine how we can use the Census Bureau's Data Linkage Infrastructure, which includes surveys, censuses, administrative records, third-party data, and other sources, to address these questions. Third, we identify data gaps by examining where we lack coverage and what we need to ask respondents directly to fill in these gaps. Fourth, we explore ways to use an innovative survey frame, abbreviated survey, internet-first mode, and a targeted field operation to fill in the gaps. Fifth, we identify a roadmap to disseminate cutting-edge, blended, model-based statistics in a way the public can understand and use. And sixth, we evaluate these innovations in a small-scale pilot and then apply the lessons-learned to other programs. To illustrate this approach, we will outline a potential survey design that applies each step of the framework to a specific topic, showing how it could guide decision-making from concept to implementation.

FCSM AI-Ready Data Working Group: A focus on data quality

Brandon Kopp, *Bureau of Labor Statistics*

Justin Cole, *Department of Treasury*

Matt Jans, *National Center for Health Statistics*

Tiffany Julian, *National Center for Science and Engineering Statistics*

Luke Keller, *U.S. Census Bureau*

David Laird, *Office of Homeland Security Statistics*

Caiyi Lang, *Statistics of Income*

Janice Lent, *Energy Information Administration*

Clay Moulton, *Statistics of Income*

Tina Norris, *Centers for Disease Control and Prevention*

Ben Rogers, *Centers for Disease Control and Prevention*

Organizations across the federal government have recognized the importance of AI-Ready Data; making data products and associated metadata easily searchable and consumable by AI services to improve the quality of responses when those services are queried about related federal data. In June 2025, the Federal Committee on Statistical Methodology (FCSM) chartered a cross-agency working group to research the amount and types of metadata most useful in the production of high-quality AI responses to queries about federal statistics. The working group focused on several data products across agencies and assembled relevant metadata. Those

metadata were provided as context along with varying queries to an array of AI services and the quality of the responses were assessed for their accuracy, relevance, and readability. This presentation will cover the work of the FCSM AI-Ready Data Working Group and its findings including the methods used to assemble metadata and evaluate AI responses. It will also cover lessons learned and recommendations for future research.

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Wednesday, March 11, 2026

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Session D-2: Keeping secrets while telling the truth: Synthetic data, methods, and applications

Organizer: Melissa Cidade, *Bureau of Justice Statistics*

Chair: Melissa Cidade, *Bureau of Justice Statistics*

Census Auditorium 2

Privacy Amplification for Model-based Synthetic Data

Terrance Savitsky, *Bureau of Labor Statistics*

Monika Hu, *Binghamton University*

Matthew Williams, *RTI International*

We introduce a new class of range restricted formal data privacy standards that condition on owner beliefs about sensitive data ranges. The range restricted formal privacy standards protect only a subset (or ball) of data values that exclude ranges (or balls) believed to be already publicly known. The privacy standards are designed for the risk-weighted pseudo posterior (model) mechanism (PPM) used to generate synthetic data under an asymptotic Differential (aDP) privacy guarantee. The PPM downweights the likelihood contribution for each record proportionally to its disclosure risk. The PPM is adapted under inclusion of beliefs by adjusting the risk-weighted pseudo likelihood. We introduce two alternative adjustments. The first expresses data owner knowledge of the sensitive range as a probability, λ , that a datum value drawn from the underlying generating distribution lies *outside* the ball or subspace of values that are sensitive. The portion of each datum likelihood contribution deemed sensitive is then $(1-\lambda) \leq 1$ and is the only portion of the likelihood subject to risk downweighting. The second adjustment encodes knowledge as the difference in probability masses $P(R) \leq 1$ between the edges of the sensitive range, R . We use the resulting *conditional* (pseudo) likelihood for a sensitive record, which boosts its worst case tail values away from 0. We compare privacy and utility properties for the PPM under the aDP and range restricted privacy standards.

Evaluation of Alternate Sampling and Weighting Methods Using Synthetic Current Population Survey Data with a Parametric Nonignorable Nonresponse Model

Tim Trudell, *U.S. Census Bureau*

Redouane Betrouni, *U.S. Census Bureau*

Declining response rates and the threat of nonresponse bias potentially erode confidence in the data quality of surveys such as the Current Population Survey (CPS). By developing a realistic but synthetic population, we can evaluate, via Monte Carlo simulation, the potential impact of alternative household sampling designs and estimation methods. We create a synthetic population based on public data from the CPS and American Community Survey (ACS), along with a parametric model for nonignorable nonresponse, in order to provide the evaluation framework for these proposed alternative methods. We consider Principle Components Analysis (PCA) systematic sampling as an alternative to the current CPS second stage sampling design and Hainmueller's Entropy Balanced Weighting as an alternative to iterative proportional fitting. The parametric model for nonignorable nonresponse takes a baseline missing at random (MAR) model then introduces parametric parameters to allow for varying degrees of nonignorable nonresponse. Finally, we compare the choice of

sampling design and weighting procedure for bias and variance of unemployment and labor force participation as a function of the nonignorable nonresponse parameters.

Empirical Assessment of Privacy for Estimates Produced with Synthetic Microdata

Luca Sartore, *National Institute of Statistical Sciences*

Habtamu Benecha, *National Agricultural Statistics Service*

Valbona Bejleri, *National Agricultural Statistics Service*

Synthetic data generation serves as a tool to preserve the privacy of data providers while maintaining key statistical properties at the population level. After processing original data sets, the United States Department of Agriculture's National Agricultural Statistics Service (USDA's NASS) and other statistical agencies must apply disclosure-control methods to privatize summaries that can be safely disseminated in official publications without compromising confidentiality. These official summaries are often generated using model-based adjustments to account for potential issues due to undercoverage and nonresponse. These adjustments can be produced through dual-system estimation (DSE) or capture-recapture models. Successively, calibration methods also adjust record-level weights to ensure that the estimates meet known population benchmarks. Although the properties of these standard statistical processes are well-known, the effects of privacy-preserving mechanisms and related concerns over disclosure risk are not well studied yet. In this paper, the properties of synthetic microdata used in privacy-preserving pipelines are assessed under a generalized definition of privacy. The utility and disclosure risk of final statistical summaries are empirically evaluated through a simulation study.

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Wednesday, March 11, 2026

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Session D-3: Assembling the Mosaic: Linking Data for Coverage, Poverty, and Consent

Organizer: Te-Ching Chen, *National Center for Health Statistics*

Chair: Matt Jans, *National Center for Health Statistics*

Census Auditorium 3

Record Linkage with Justice Data to Improve Coverage in the Decennial Census

Michaellyn Garcia, *U.S. Census Bureau*

This presentation summarizes work to develop a database, the Justice Facility Frame (JFF), that integrates facility records from the Bureau of Justice Statistics and the Office of Juvenile Justice and Delinquency Prevention with group quarters records from the Census Bureau's Master Address File. The JFF has been used to measure the coverage of prisons and detention facilities in the 2020 Census and updates from linked records in the JFF have been used to update the Master Address File. Staff use a JFF web application to increase validation efficiency, algorithmically identify records that could best benefit from review, and ultimately support continuous improvement and integration of new data sources.

Parents' Informed Consent, Data Privacy, and Data Linkage in the National Longitudinal Survey of Youth

Tywanquila Walker, *Bureau of Labor Statistics*

Robin Kaplan, *Bureau of Labor Statistics*

Rebecca L. Morrison, *Bureau of Labor Statistics*

Safia Abdirizak, *Bureau of Labor Statistics*

Linking survey and administrative data reduces respondent burden, supplements data, and enhances data

quality. The National Longitudinal Survey of Youth (NLSY) collects extensive information on American respondents' labor market behavior and educational experiences over time. During preliminary testing for a new NLSY cohort, we evaluated parents' willingness to consent to link their NLSY caregiver survey data to other administrative data for research purposes. We conducted online surveys (N=793) and moderated cognitive interviews (N=10) with parents of youth aged 11 to 17. We tested the parent consent form language and examined the effects of framing linking as a benefit to make their data more useful to researchers. Studies of consent form wording show inconsistent effects, or no effects at all. Instead, parents' decisions to link their data tend to be tied to their attitudes toward data privacy, government studies, and data use. We discuss parent expectations of the informed consent process, comprehension of consent form language, and willingness to participate and link survey responses to other data sources.

An Overview of the Evaluations of the Census 2030 Coverage Estimation Program

Michael Beaghen, *U.S. Census Bureau*

The U.S. Census Bureau has conducted a census every decade since 1790. Starting with the 1950 Census, the Census Bureau has also implemented programs to measure the census coverage, that is, the overcounts or undercounts of geographies and demographic groups. The coverage measurement programs from 1980 through 2020 included an independent sample of the population, which was matched to the Census, allowing for the production of dual-system estimates (DSE) of the population. For the 2030 Census, the Coverage Estimation program plans to create DSEs using administrative records. The Census Bureau will also conduct a series of evaluations, which will focus on assessing the quality of the DSEs. Further, the evaluations will produce an estimate of total survey error of the DSEs, which may provide the option of correcting the DSEs if substantial errors are detected in the DSEs.

This paper will provide an overview of the goals and methods of these evaluations. It includes evaluations of the following components of error in the 2030 DSEs: overcoverage, correlation bias, matching error, error in the identification of duplication, geographic misclassification, demographic misclassification, ratio estimator error, and synthetic error.

Estimating Income and Poverty with Administrative Data

Jonathan Rothbaum, *U.S. Census Bureau*

Adam Bee, *U.S. Census Bureau*

John Creamer, *U.S. Census Bureau*

Joshua Mitchell, *U.S. Census Bureau*

Nikolas Mittag, *CERGE-EI, Charles University*

Elizabeth Pelletier, *U.S. Census Bureau*

Carl Sanders, *U.S. Census Bureau*

Lawrence Schmidt, *MIT Sloan School of Management*

Matthew Unrath, *University of Southern California*

The National Experimental Well-being Statistics (NEWS) project aims to produce the best possible estimates of income and poverty given all available survey, decennial census, administrative, and third-party data. We estimate improved income and poverty statistics by addressing bias from unit non-response, missing information and measurement error. We find that income and poverty estimates are biased, and the biases vary by group and over time. This can cause estimates to miss important facts about how economic well-being is changing over time and why. For example, the NEWS and survey estimate of child poverty differ over time due to changes in survey nonresponse bias and sources of income, particularly with the large expansion of unemployment insurance during the pandemic. Child poverty was 1.5pp lower in NEWS than the survey in 2018, they were not statistically different in 2019, and the NEWS estimate was 3.8pp lower in 2020 (according to the Supplemental Poverty Measure).

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Session D-4: Building the Future of Juvenile Facility Data: Modernizing the Census Program

Organizer: Liz Willhide, *U.S. Census Bureau*

Chair: Erica Marquette, *U.S. Census Bureau*

Discussant: Ben Adams, *National Institute of Justice*

Census Conference Rooms 1-2

Setting the Stage for Transformation: Evaluation and Redesign of the Juvenile Censuses

Kaitlyn Sill, *National Institute of Justice*

Ben Adams, *National Institute of Justice*

This presentation will describe an external evaluation conducted by RTI International of the Census of Juveniles in Residential Placement (CJRP) and the Juvenile Residential Facility Census (JRFC), and how its recommendations shaped the foundation of the Juvenile Facility Census Program (JFCP). The evaluation examined content, methodology, and operational practices, and identified opportunities to streamline instruments, eliminate redundancies, and strengthen analytic value.

In response, the Census Bureau and the National Institute of Justice (NIJ) worked closely to develop a plan for redesign. This included revising and consolidating content, conducting cognitive testing with facility staff to ensure clarity and feasibility, and implementing targeted changes that balanced methodological efficiency with subject-matter needs.

By highlighting both the evaluation findings and the collaborative redesign process, this presentation shows how the recommendations became the blueprint for the larger transformation of the program.

From Two Censuses to One: Building the Juvenile Facility Census Program

Megan Minnich, *U.S. Census Bureau*

Jon Albers, *U.S. Census Bureau*

This presentation describes the consolidation of two long-running juvenile justice surveys—the Census of Juveniles in Residential Placement (CJRP) and the Juvenile Residential Facility Census (JRFC)—into a single, streamlined program: the Juvenile Facility Census Program (JFCP). Conducted in close partnership with the National Institute of Justice (NIJ) and the Office of Juvenile Justice and Delinquency Prevention (OJJDP), the integration offered opportunities to harmonize survey instruments, align workflows, and reduce duplication in operations.

A key innovation of the new program is its two-year modular structure, which alternates between the Youth Population Module and the Facility Operations Module, each anchored by a shared set of core questions essential for maintaining a current and accurate survey frame. This approach streamlines management, reduces respondent burden, and supports long-term sustainability, all while preserving the unique strengths and insights the two original surveys contributed.

The presentation will outline the transformation process, including the decision-making steps, design challenges, and the importance of agency collaboration. Collaboration was essential in balancing methodological priorities with subject-matter needs, so that the combined program would continue to serve a wide range of data users. The result is a more efficient and sustainable census program that meets both operational and analytic goals.

Reimagining Imputation: Transitioning from Legacy Code to Modern Methods in the Juvenile Facility Census Program

Samuel Soares Walton, *U.S. Census Bureau*

Nora Tran, *U.S. Census Bureau*

To sustain its long-term value, the Juvenile Facility Census Program (JFCP) recently underwent a major redesign of its imputation system. The legacy process relied on hundreds of SAS programs that were difficult

to maintain and limited in flexibility. The new system, developed in Python and supported by structured parameter files, is more efficient, transparent, and sustainable while reducing reliance on proprietary software. Beyond technical modernization, the redesign emphasized methodological enhancements. New approaches included hierarchical hot deck methods tailored to facility-level data and machine learning models trained on historical facility counts, both of which improved accuracy while maintaining adaptability to changes in survey content. These advances ensure that missing data are addressed in ways that reflect the evolving realities of the juvenile justice field.

Equally important, the project was carried out in close collaboration with subject-matter experts at the National Institute of Justice (NIJ). Collaboration was essential in balancing methodological priorities with subject-matter needs, so that the combined program would preserve the integrity of historical time-series data and continue to serve a wide range of data users. This partnership allowed business requirements to be translated into statistical best practices, ensuring that both policy-relevant relationships and methodological rigor were preserved.

The presentation will highlight the development timeline, key technical and methodological enhancements, and lessons learned in creating a framework that supports both current production and future innovation.

Leveraging Annual Juvenile Facility Data to Strengthen Decennial Coverage

Liz Willhide, *U.S. Census Bureau*

Juvenile facilities experience significant churn over a decade—closures, openings, and changes in status—that can challenge the accuracy of decennial group quarters updates. Historically, the Decennial Census relied on updates tied to the ten-year cycle, leaving gaps in coverage of juvenile facilities between censuses. Meanwhile, the Juvenile Facility Census Program (JFCP) maintains an annual inventory, providing a more current picture of the nation’s juvenile facilities. This presentation explores how that ongoing work was leveraged to support Decennial operations while continuing to meet JFCP’s operational needs.

Achieving this required close collaboration between the Census Bureau, the National Institute of Justice (NIJ), and the Office of Juvenile Justice and Delinquency Prevention (OJJDP). Establishing a secure and operationally effective data sharing arrangement involved navigating policy constraints, aligning interagency processes, and building consensus among multiple teams. The process itself—negotiating access, defining responsibilities, and formalizing agreements—was the key milestone, demonstrating the steps necessary to bridge program-specific and national-scale statistical operations.

The resulting framework creates a sustainable approach that benefits both programs. Decennial gains more current coverage of juvenile group quarters, while JFCP operations are reinforced by improved data alignment and efficiency. This initiative is a compelling example of how multiple agencies, working from the same frame, can achieve mutual benefits through a collaborative maintenance effort, reducing duplication and strengthening the foundation for ongoing and future data collection.

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Session D-5: Defining, Measuring, and Monitoring Data Quality

Organizer: Nancy Early, *Social Security Administration*

Chair: Otto-Obong Essen, *Social Security Administration*

Census Conference Rooms 3-4

Disaggregating and Harmonizing the Quasi-Legal Alien Population in DHS Administrative Data

Erin Babich, *Office of Homeland Security Statistics*

The quasi-legal population is comprised of individuals physically present in the United States with active and traceable immigration records who fall between lawful and unauthorized immigrant classifications. This group

includes visa overstayers, asylum applicants with pending cases, and individuals under temporary humanitarian protections such as parole or Temporary Protected Status (TPS). Reliable measurement of the quasi-legal population is essential for correcting administrative records, improving reliability in internal and external data reporting, and supporting data-driven decisions for immigration law and policy, resource allocation, and enforcement strategies. However, repeated inclusion of individuals in population counts, fragmented systems, and inconsistent collection practices within the Department of Homeland Security (DHS) have hindered accurate population estimation. This paper proposes a methodological framework to disaggregate and harmonize DHS administrative data and identify unique individuals within the quasi-legal population. The approach integrates enforcement and benefit data from DHS components—U.S. Customs and Border Protection (CBP), U.S. Immigration and Customs Enforcement (ICE), and U.S. Citizenship and Immigration Services (USCIS)—to construct a deduplicated dataset. A hierarchical entity resolution algorithm is applied, leveraging biographic, biometric, and administrative identifiers to resolve duplicate records across disparate datasets. Fuzzy matching techniques address discrepancies in text fields, while a decision framework reconciles overlapping statuses to determine the primary immigration status. This framework is expected to improve the accuracy of quasi-legal population estimation and address discrepancies in prior counts. The methodological contributions include enhanced data harmonization and deduplication processes, which have implications for federal statistics, particularly in refining quasi-legal and unauthorized estimates. Anticipated results include insights into demographic characteristics, immigration history, and the effects of policy changes, offering a robust foundation for evidence-based policymaking and advancing the reliability of DHS administrative data.

Comprehensive Data Quality Monitoring: Upgrades to the Census Bureau's Field Quality Monitoring Program

Rachel Huang, *U.S. Census Bureau*

Elizabeth Mahoney, *U.S. Census Bureau*

Laura Hergert, *U.S. Census Bureau*

Anthony Chiado, *U.S. Census Bureau*

Natalie Belford, *U.S. Census Bureau*

As the nation's premier data collector, the Census Bureau bears the crucial responsibility of ensuring the accuracy and quality of its data. This intricate task entails monitoring the work of thousands of interviewers across numerous surveys, meticulously verifying adherence to established procedures. To address this challenge, the Field Quality Monitoring (FQM) Program was conceived in 2021, implementing a near real-time monitoring system utilizing paradata to detect, mitigate, and rectify potential data quality issues.

Since its inception, the FQM program has expanded to monitor data quality cross-survey in an effort to monitor the entire volume of an interviewer's work in near real-time. This update has improved the ability to identify deviations from procedures and potential data quality problems earlier, rather than looking at individual surveys. Data-driven analysis is used to investigate and correct quality problems before they impact survey data. Using innovative outlier detection methodology, several core metrics were developed including some to identify systemic problems with data quality such as speeding during interviews, high item non-response, and under-representation on household rostering. Other metrics are targeted to identify circumstantial issues such as incorrect classification of sample addresses, consecutive interviews in a short period, and overnight work. This presentation focuses on the capability of cross-survey monitoring to identify large-scale data quality issues based on a suite of metrics, tools, and outlier detection methodology.

In the future the FQM program aims to utilize advanced predictive models to proactively anticipate and avert data quality issues. Additionally, the program aims to integrate machine learning algorithms to automate outlier detection and streamline the investigative process. By embracing these advancements, the FQM program will continue to safeguard the integrity of the Census Bureau's data, ensuring its reliability and trustworthiness for generations to come.

The Educational Requirements of Jobs: A Comparison of Worker and Employer Ratings

Maury Gittleman, *Bureau of Labor Statistics*

Michael Handel, *George Washington Institute of Public Policy*

Job skill requirements, including job educational requirements, have been a longstanding concern of researchers, policy makers, and the public. Federal statistics have addressed this interest in several projects. Nevertheless, the measurement of job educational requirements and other job skill requirements have raised conceptual and methodological questions. Researchers have expressed concerns regarding possible upward biases in worker self-reports, arguing employer surveys and ratings by job analysts are likely to be more objective. There are also concerns that responses to survey questions regarding required education may be ambiguous or vary depending on whether they refer to the education needed for minimally acceptable job performance, average performance, or some kind of optimal performance. Despite these concerns or objections, there has been little research on the extent to which survey reports of job required education are sensitive to the source of data or to variations in question wording.

Many of these concerns can be addressed now using a relatively new and underutilized database, the Occupational Requirements Survey (ORS), which collects job ratings from employers, in conjunction with the better-known Occupational Information Network (O*NET) database, which collects information from workers. In addition, the employer survey asks for the minimum acceptable education level for a given occupation, while the worker survey item leaves job performance level open.

Analyses of intra-occupational agreement within both worker and employer samples show high and similar levels of internal consensus, indicating both groups have relatively well-defined views of the education required by jobs. Comparisons of average occupational ratings between workers and employers show very strong consistency in occupational rankings and only very slightly higher absolute levels among workers, as expected based on respondent identity and question wording. Overall, ratings of job educational requirements are quite robust to choice of respondent and survey question wording, suggesting high levels of reliability and validity.

The Clash: Identifying New Standards for Automated Editing Systems in Agricultural Surveys

Denise A. Abreu, *National Agricultural Statistics Service*

Megan Lipke, *National Agricultural Statistics Service*

Sean Rhodes, *National Agricultural Statistics Service*

Darcy Miller, *National Agricultural Statistics Service*

Historically, the United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS) has been reviewing and editing agricultural survey data through a costly and time-consuming manual process to produce official statistics. The process has greatly benefited the Agency's mission throughout the years. Current resource limitations necessitate a shift towards more efficient editing methods and reprioritization of staff time. In recent years, NASS has been developing and testing a generalized system to automate the imputation and error correction occurring after survey data collection but before the outlier review process. However, the use of automation and cutting-edge technology require realigning current review standards to maximize allocation of resources. Evaluation of this new process of data editing and error correction has evolved over the development of the system, balancing competing desired outcomes. This presentation discusses the establishment and application of metrics used over time in decision-making about the implementation of new automated editing processes in production. Successes and challenges of these new standards are also addressed.

A Data Quality Plan for Commodity Flow Estimates with an Air Cargo Case Study

Monique Stinson, *Bureau of Transportation Statistics*

Praveen Hettige, *Bureau of Transportation Statistics*

Jingsi Shaw, *Bureau of Transportation Statistics*

Stephanie Lawrence, *Bureau of Transportation Statistics*

This study aims to quantify the accuracy and reliability of an integrated data product, the Freight Analysis

Framework (FAF), which provides estimates of cargo volumes throughout the U.S. by origin, destination, commodity type, and mode (truck, rail, air, etc.). Since the US Department of Transportation (USDOT) Bureau of Transportation Statistics (BTS) develops FAF using survey data plus other statistical, administrative, and transactional data, FAF is an integrated data product (FCSM 2020). While FCSM (2020) offers a framework to evaluate such products, Schmitt (2021) found “additional tools to measure quality in blended data sets” as a gap. Some studies (e.g., Bieler 2021) have made progress in addressing this gap, but challenges remain, especially in evaluating the quality of blended data that comprise origin-to-destination flow estimates. Challenges include finding comparable data and devising methods to make available benchmark data comparable to the estimated source. Our case study addresses these challenges, comparing estimated FAF airborne commodity flows to BTS’ benchmark T-100 flight-level cargo volumes data. We find that FAF aligns reasonably well with T-100 for import and export cargo due to data collection techniques of each source. However, T-100 shows much higher domestic flows than FAF. This is partly because T-100 data largely reflect airport-to-airport trips, while FAF represents the end-to-end trip: cargo that transfers planes will have two records in T-100 and one in FAF. To make the data more comparable, we plan to use Origin-Destination Matrix Estimation or ODME. We expect the ODME exercise to permit identification of hub-to-hub flows, which can then be removed to estimate benchmark end-to-end flows. We will also examine the impacts of definitional differences in air cargo. Our presentation will discuss our methodology, as well as its advantages and disadvantages, as one approach to evaluating the quality of an integrated commodity flow data product.

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Wednesday, March 11, 2026

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Session D-6: Where in the World? Defining and Applying Measures of Geography

Organizer: Brandon Kopp, *Bureau of Labor Statistics*

Chair: Elina T. Page, *Economic Research Service*

BLS Auditorium A

K-Means Clustering for Geographic Stratification in Census Bureau Fieldwork

Elizabeth Mahoney, *U.S. Census Bureau*

The Census Bureau uses a clustering system to classify geographies based on their relative difficulty of interviewing. These clusters are used to determine interviewer’s performance benchmarks, with higher-difficulty clusters having lower response rate targets. Given the changing demographics and geography of the United States, a redesign of these clusters was necessary.

To develop the new clusters, the Census Bureau’s Planning Database, which contains housing, demographic, and socio-economic statistics was leveraged. An initial regression analysis assessed the relevance of variables used in a previous model. K-means clustering, an unsupervised learning algorithm, was used to partition block groups based on these variables, with refined scaling for better representation of interview difficulty factors. The resulting clusters were validated using both Decennial and current survey response rates.

Analysis revealed a shift in the distribution of block groups across clusters. Notably, the number of block groups classified as “difficult” increased, suggesting that the challenges associated with conducting interviews have evolved, necessitating a revised clustering approach to reflect the current landscape.

Innovative Approaches to Building Food Access Metrics: Google Places API, NLP, and Open-Source Routing in FoodAPS

Patrick Marino, *U.S. Census Bureau*

This presentation highlights the innovative methodology used by the National Household Food Acquisition and Purchase Survey (FoodAPS) analysts to construct geography-based metrics for access to SNAP-authorized retailers and food acquisition locations to meet the survey sponsors post-processing requirements. A central

feature of the FoodAPS self-response instrument was the integration of the Google Places API, which allowed respondents to search for food acquisition locations through an intuitive search bar. API calls returned both address and geographic coordinates. A manual entry option was provided when the API could not resolve a location. In manual entry, respondents entered the location name (required), along with optional street address, city, state, and ZIP code.

The creation of the geography metrics involved five core processes: (1) obtaining location coordinates through Google Places API calls; (2) normalizing manual entries using a natural language processing (NLP) approach, then geocoding them with an internal Census Bureau geocoder; (3) normalizing and geocoding manual sample address updates made by respondents; (4) applying the Census Bureau's in-house, open-source routing engine, mini-WARIO, to estimate driving and walking time and distance; and (5) using the Python package GeoPandas to map SNAP retailers within a 50-mile radius of sample households and generate variables assessing household access to different retailer types.

Using the API integration during data collection produced standardized address information to streamline the analysis and the team leveraged open-source software to create the required metrics for the survey sponsor.

Finding Suburbia in the Census

Todd Gardner, *U.S. Census Bureau*

This paper presents a methodology for classifying U.S. settlement patterns that extends beyond the conventional urban–rural dichotomy by distinguishing urban cores, suburbs, exurbs, outlying towns, and rural areas. The approach relies on the Historical Housing Unit and Urbanization Database (HHUUD10), which provides consistent housing unit estimates for 2010 census tract boundaries for all decennial census years from 1940 through 2020. By harmonizing tract-level data across eight decades, the method enables longitudinal analyses of settlement change while preserving geographic comparability.

A key methodological contribution is the use of fixed tract boundaries to address the problem of changing census geography over time. Tracts are classified according to both their era of urban development and their proximity to metropolitan centers, allowing for a standardized typology of settlement types that can be replicated nationwide. This framework facilitates small-area estimation of urbanization trends, measurement of suburban and exurban growth, and assessment of long-term demographic shifts.

Application of this typology demonstrates its ability to capture the timing and magnitude of postwar suburbanization, the emergence of exurban development, and the persistence of rural and small-town settlement structures. Beyond substantive findings, the approach offers a broadly applicable tool for federal statistical agencies. It enhances the capacity to link historical and contemporary datasets, to generate AI-ready features for modeling, and to produce spatially consistent measures of settlement change.

By addressing longstanding challenges of geographic comparability and classification, this methodology advances best practices for producing high-quality, relevant, and timely federal statistics. It offers lessons for data dissemination, data linkage, and longitudinal analysis that are directly relevant to agencies engaged in measuring demographic, housing, and economic change.

A Blended Data Approach to Measuring Monthly Housing Starts: Satellite Imagery, Survey Data and More!

Nicole Czaplicki, *U.S. Census Bureau*

Colin Shevlin, *Reveal Global Consulting*

Aidan Smith, *U.S. Census Bureau*

Hector Ferronato, *Reveal Global Consulting*

Dwarakh Nayam, *Reveal Global Consulting*

Lei Peng, *Reveal Global Consulting*

Scott Springer, *U.S. Census Bureau*

Yasmeen Varma, *Reveal Global Consulting*

Doren Walker, *Reveal Global Consulting*

As part of the comprehensive Construction Re-engineering Initiative at the U.S. Census Bureau, alternative data sources are being considered to supplement or replace current data collection methods. For the Survey of Construction (SOC), which measures new residential construction, this includes observing housing starts from satellite imagery in place of the current interviews for housing starts conducted by field representatives.

Satellite images are obtained monthly for a subset of places in the SOC sample. Convolutional Neural Network (CNN) models are then applied to images to predict likely new residential construction projects, with the current focus being single-family housing starts. Several post prediction processing steps are applied including exclusions based on intersections with known buildings or roads, treatments for missing data due to cloud cover, and adjustments for the length of time between consecutive images, to ultimately produce place level estimates of housing starts. These place level estimates are then combined with the existing building permit level survey data to produce estimates of division level housing starts, a new experimental data product from the Census Bureau.

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Session D-7: Precision at Scale: Modern Approaches to Small Area Estimation

Organizer: Elan Segarra, *Bureau of Labor Statistics*

Chair: Elan Segarra, *Bureau of Labor Statistics*

BLS CTC 2A

Small area estimation: a Bayesian approach to benchmarking and ranking

Gauri S. Datta, *U.S. Census Bureau*

Abhyuday Mandal, *University of Georgia*

Jerry Maples, *U.S. Census Bureau*

Model-based inference plays an integral role in small area estimation (SAE). The hierarchical Bayesian modeling provides a logical and easily implementable approach in SAE. This approach facilitates borrowing of information via non-trivial modeling of data from various sources. Model-based estimates of small area means are often required to be compatible with more precise data at higher levels of aggregation. Many existing benchmarking methods adjust the regular Bayesian estimates of the small area means to comply with the aggregation requirement, but do not properly account for the variances of the benchmarked estimates.

Sugasawa et al. (2024) argued that by incorporating the benchmark constraints the regular posterior distribution of the small area means can be perturbed to a new posterior distribution that automatically produces benchmarked SAE and their appropriate associated uncertainty. We use this principled approach to make point and interval estimation of small means satisfying the constraints.

While inference for the small means is very crucial in making important policy decisions by national statistical agencies, reliable ranking of the small areas based on their true means is also a useful policy tool. The uncertainty in the ranking of the estimated small area means is often ignored. We propose a Bayesian solution to this problem using the above-mentioned perturbed posterior to develop a joint distribution of the rank vector. We demonstrate the usefulness of our method to estimate median incomes of three demographic groups from 72 PUMAs in Georgia.

Version 2.0 of the Enhanced Modified Kalman Filter for Small Domain Estimation: Accounting for Breaks in the Underlying Trend Data

Lauren Rossen, *National Center for Health Statistics*

Makram Talih, *National Center for Health Statistics*

Measuring health outcomes for small subpopulations can be challenging because direct estimates for small subgroups may be unstable or unreliable due to small sample sizes or small numbers of events. As a result, estimates for small subgroups may not meet data presentation standards, and analysts may need to aggregate data over larger areas, time periods, or groups to obtain reliable estimates. The enhanced Modified Kalman Filter (MKF) is a statistical modeling tool developed for small domain estimation, borrowing strength over time and across groups to produce more reliable model-based estimates for small subgroups. In this updated version 2.0 of the eMKF, the tool was expanded to account for potential breaks in the underlying trend data, such as

following survey redesigns or other interruptions or changes. This new feature allows analysts to make better use of available data and to account for changes in survey design or other point-in-time changes. Using simulated data, model-based estimates using the eMKF tool resulted in improvements in root mean squared error relative to direct estimates, across a wide array of analytic scenarios. The latest eMKF version will be especially useful in cases where there may not yet be enough data points post-trend break to conduct a standalone trend analysis, allowing users to produce more timely model-based estimates of health outcomes for small subpopulations, which will improve the availability of data for the assessment and monitoring of health outcomes in small groups.

Applying the Enhanced Modified Kalman Filter for Small Domain Estimation of Blood Pressure Patterns Over the Lifespan — United States, August 2021–August 2023

Nimit N. Shah, *National Center for Health Statistics*

Lauren M. Rossen, *National Center for Health Statistics*

Brian K. Kit, *National Center for Health Statistics*

Cheryl Fryar, *National Center for Health Statistics*

Makram Talih, *National Center for Health Statistics (Contractor)*

Lara A. Akinbami, *National Center for Health Statistics*

High blood pressure (BP), a major risk factor for heart disease, increases with age and varies by sex and race/ethnicity. Elevated childhood BP is associated with high BP and heart disease in adulthood. However, ages at which BP differences by sex and race/ethnicity first emerge remain unclear. Population-level life course analyses on the emergence and patterns of BP differences by sex and race/ethnicity across age are limited and challenging because subgrouping data leads to small sample sizes resulting in imprecise estimates. We assessed BP differences by sex and race/ethnicity across age by applying the Enhanced Modified Kalman Filter (eMKF), a statistical modeling tool designed for small domain estimation. We used BP examination data from the National Health and Nutrition Examination Survey (1999–2000 through August 2021–August 2023 cycles) to estimate mean systolic and diastolic BP variations by sex and race/ethnicity, across age for participants aged 8–77 years during the August 2021–August 2023 cycle. We applied the eMKF, to generate reliable model-based estimates for 2-year age subgroups, leveraging data across cycles, age, and demographic subgroups. We compared eMKF and direct estimates using relative root mean squared error (RRMSE), and visualized eMKF estimates by sex and race/ethnicity across age for the August 2021–August 2023 cycle. Model-based eMKF estimates had lower RMSE compared with direct estimates for both systolic and diastolic BP. For some subgroups, relative gains in equivalent sample size reached 500%. Visualization of mean systolic and diastolic BP across age showed that males had different BP patterns than females. For both sexes, non-Hispanic Black individuals showed different mean BP patterns than their non-Hispanic White and Hispanic counterparts. By enabling reliable model-based estimates for small populations, the eMKF enhances the availability and precision of data needed to track and address BP differences and variations across subgroups by age.

Small Area Estimation for the Annual Integrated Economic Survey

Stephen J. Kaputa, *U.S. Census Bureau*

The Annual Integrated Economic Survey (AIES) collects annual establishment data on key economic statistics such as employment, payroll, first quarter payroll and receipts. AIES will produce estimates of totals for the core items (employment, payroll, first quarter payroll and receipts) within industries detailed at the 4-digit North American Industry Classification System (NAICS) level for all fifty U.S. states and the District of Columbia. For select sectors, additional break outs include type of operation and tax status. Direct survey estimates at this granular level are imprecise because of limited sample size to reduce respondent burden and survey cost. This research explores the use Bayesian Fay-Herriot (F-H) area-level models from small area estimation methodology, which are a composite of the direct survey estimates and indirect model-based estimates, aimed at improving precision. We discuss our unique application to skewed economic survey data and the model selection process which draws from covariates created from historic administrative records and Economic Census data.

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Abstract Listings for Thursday, March 12

- | | |
|-------------------------|---------------------|
| • Concurrent Sessions E | 8:30 am – 10:00 am |
| • Concurrent Sessions F | 10:15 am – 11:45 am |
| • Concurrent Sessions G | 1:15 pm – 2:45 pm |
| • Concurrent Sessions H | 3:00 pm – 4:30 pm |

Session E-1: Updates on the 2030 Census Disclosure Avoidance Research Program

Organizer: Michael Hawes, *U.S. Census Bureau*
Chair: Michael Hawes, *U.S. Census Bureau*
Discussant: Sallie Ann Keller, *U.S. Census Bureau*
Census Auditorium 1

Algorithmic Improvements to the TopDown Algorithm

Ryan Cumings-Menon, *U.S. Census Bureau*

The 2020 TopDown Algorithm (TDA) is a disclosure avoidance technique based on formally private mechanisms that was used to protect the confidentiality of 2020 Census respondents. This algorithm provides a transparent approach to balance the triple tradeoff of privacy, accuracy, and data availability by allocating a privacy loss budget (PLB) between the many tabulations produced for the decennial census. For example, the accuracy of the total population counts for census blocks can be increased, at the expense of reducing the privacy of respondents, by increasing the PLB allocated to these total population tabulations. In preparation for the 2030 Census, the Census Bureau's Disclosure Avoidance System team is in the process of making improvements to the TDA, which, broadly speaking, fit into one of three categories. First, some of these improvements are aimed at using our PLB more efficiently when producing tabulations for towns and cities. Second, other changes allow TDA to "reuse" PLB spent on one set of tabulations to increase accuracy on others. Third, a final set of changes will decrease the prevalence of consistency issues when comparing certain tabulations of households and tabulations of people.

Statistical Post-Processing Methods for Differentially Private Supplemental Demographic and Housing Characteristics Tables in the Decennial Census

Ryan Janicki, *U.S. Census Bureau*
Mikaela Meyer, *MITRE Corporation*
Adam Hall, *U.S. Census Bureau*
Scott H. Holan, *U.S. Census Bureau & University of Missouri*
Kyle M. Irimata, *U.S. Census Bureau*
James Livsey, *U.S. Census Bureau*
Monica-Ann Mendoza, *MITRE Corporation*
Andrew Raim, *U.S. Census Bureau*

For the 2020 decennial census, the U.S. Census Bureau adopted differential privacy (DP) as its primary disclosure avoidance method to protect respondent data. Like other methods for statistical disclosure limitation, the use of DP can lower the accuracy and utility of the data. Statistical models have been proposed as a post processing method to improve the precision of protected data while preserving DP guarantees. We demonstrate the proposed models on county-level estimates for the decennial census Supplemental Demographic and Housing Characteristics tables and show that the approach can produce estimates that are more precise than the original DP estimates, especially for less populous counties and race groups.

Bridging Race and Ethnicity to Support a Revised SPD 15

Matt Spence, *U.S. Census Bureau*

On March 28, 2024, the U.S. Office of Management and Budget (OMB) finalized revisions to Statistical Policy Directive No. 15: Standards for Maintaining, Collecting, and Presenting Federal Data on Race and Ethnicity (SPD 15). Since 1977, SPD 15 has provided a common language to promote uniformity and comparability for data on race and ethnicity, but it has undergone two revisions in response to new research and to better measure and present data on race and ethnicity. Among other improvements, the revised SPD 15 provides for a

combined question to gather data on race and ethnicity, whereas the prior version required separate Hispanic Origin and race questions.

When OMB publishes new race and ethnicity data standards, federal agencies require data bridging guidance to compare race and ethnicity data collected under the revised standards with data collected under the old standards. This bridging will permit estimation and comparison of race-specific statistics at a point in time or across time. Effective bridging is also necessary to inform and evaluate research to improve the 2030 Census Disclosure Avoidance System, as the research requires having specifications and input data that match the new reporting categories.

This presentation will discuss the methodology that has been developed to bridge between these two versions, as well as practical ways to apply bridging factors to federal, state, or local data. It will also discuss ongoing research and future plans for dissemination.

Supporting Transparency with Synthetic Data

Rolando Rodríguez, *U.S. Census Bureau*

The 2020 Census Disclosure Avoidance System (DAS) brought unprecedented transparency into the U.S. Census Bureau's process of providing confidentiality to its respondents. While public stakeholders were able to review 2020 DAS demonstration data products and the code used to generate them, they could not replicate the entire workflow themselves, as they lacked access to the legally protected microdata that served as input. We present research to address this opacity via synthetic data: statistical microdata created from models fit to the underlying, protected microdata. We discuss modeling strategies for realistic synthetic data generation, methods for assessing confidentiality risk from synthetic data release, the use of synthetic data to stress test the nascent 2030 Census DAS, and end-user considerations for performing statistical analysis on synthetic data.

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8:30 AM

Session E-2: Estimating in an Imperfect World: Understanding and Addressing Error in Federal Statistical Estimates

Organizer: Jennifer Pace, *U.S. Census Bureau*

Chair: Jennifer Pace, *U.S. Census Bureau*

Census Auditorium 2

Monthly Revisions in Current Employment Statistics

Steve Mance, *Bureau of Labor Statistics*

The Current Employment Statistics (CES) program publishes timely, closely watched estimates of national nonfarm payroll employment shortly after the end of the reference month. Initial estimates are revised in the next two subsequent monthly releases, with receipt of additional survey reports and recalculation of seasonal adjustment factors, and again in an annual benchmark revision. We explore the distribution and predictability of monthly payroll employment revisions and examine assumptions CES makes regarding late response.

It's about Time (Series): A Simple Correction for Difference-in-Differences Estimators

Gary Cornwall, *Bureau of Economic Analysis*

Scott Wentland, *Bureau of Economic Analysis*

This paper reconsiders the difference-in-differences (DiD) research design when the outcome of interest follows an autoregressive process and serial correlation stems from first-order (rather than second-order) model misspecification. When time-series issues like this are overlooked, we show that the traditional parallel trends assumption is insufficient and DiD designs will exhibit an autoregressive bias. For most panel applications ($T > 25$ periods), explicitly accounting for this process in the design can correct this bias and

increase precision of standard two-way fixed effects (TWFE) and dynamic Callaway-Sant'Anna (CS) estimators. We evaluate this modification through Monte Carlo simulations and explore its implications with an empirical example using a balanced panel. The example leverages a policy shock used in recent literature to reevaluate the impact of household credit constraints on outcomes like state-level GDP growth and labor market participation. When we implement the proposed correction, which can be as simple as incorporating a lagged outcome and group interaction into the model, the results confirm a reduction in bias predicted by theory, yielding a more generalizable and more precise estimator.

On Two Errors Sometimes Made in Statistics

William R. Bell, *U.S. Census Bureau*

I consider two errors I have sometimes seen made by statisticians. (1) Expressing uncertainty about the estimate of a stochastic quantity using the variance of the estimator instead of the variance of the prediction error. This error seems to arise from people assuming that quantifying uncertainty when estimating a stochastic quantity is the same problem as quantifying uncertainty about statistical model parameters such as means and regression coefficients. This is not the case. I will illustrate this error and its consequences for the simple case of a Fay-Herriot small area model with known parameters. (2) Using survey estimates with nonnegligible and heteroskedastic sampling error as regression variables in models. This is a particular case of the well-known “errors in variables” problem in regression that leads to biased estimates of regression parameters. I will show that in prediction problems this error leads to inefficient predictors and invalid prediction error variances. I will then note how this problem can be addressed by using either of two measurement error models – the functional measurement error model or the structural measurement error model – and point out why the latter has advantages for small area estimation.

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Session E-3: The 2030 Administrative Data Frame and Use Case

Organizer: Meagan Tydings, *U.S. Census Bureau*

Chair: TBD

Census Auditorium 3

Putting People in [the right] Place: Assigning Residential Location in the Census Bureau’s Demographic Frame

Jennifer Ortman, *U.S. Census Bureau*

The Demographic Frame is a secure, comprehensive database of person-level characteristics drawn from administrative data, census, and survey data. It employs unique identifiers to link individual data over time, enabling reliable longitudinal analyses. A critical feature is accurately assigning residential locations to individuals, essential for producing high-quality data that reflect U.S. community dynamics.

This presentation outlines the motivation and methodology behind modeling residential location within the Demographic Frame, highlighting how the Person-Place Model assigns probable residential addresses using established and emerging techniques adaptable to various applications. We will discuss current capabilities, innovations in model development, and accuracy evaluations, illustrating how this approach can advance the Census Bureau’s mission and foster data-driven insights into population dynamics.

2030 Census Person Characteristic Frame

Jessica Rader, *U.S. Census Bureau*

Decennial census efforts to count the population rely primarily on methods to encourage people to self-respond, followed by in-person interviewing for nonresponding households. To increase efficiency and reduce

costs, the 2020 Census leveraged administrative data to enhance respondent-provided information, validate respondent submissions, and to reduce in-person field follow-up visits. The 2030 Census will continue these 2020 Census innovations while also exploring additional administrative data sources and uses to further streamline census data collection and minimize respondent burden. The Demographic Frame is expected to play a significant role in 2030 Census's expanded administrative data efforts and will likely serve as a major source of administrative data for the 2030 Decennial Census Program.

This presentation will detail 2030 Census research into its plans to use the Demographic Frame as the basis for the Person Characteristic Frame, the source of person-level administrative data for the 2030 Census. We will highlight initial plans regarding 2030 Census administrative data uses, describe how the Demographic Frame may be used to create the Person Characteristic Frame, and discuss the 2026 Census Test plans for building and testing the Person Characteristic Frame.

Demographic Frame Enhancements

Marta Murray-Close, *U.S. Census Bureau*

The Demographic Frame combines data from numerous administrative, census, and survey sources on people associated with U.S. addresses and their demographic characteristics. The Demographic Frame is used in research and production activities across the agency, so monitoring and improving its quality is essential. This presentation describes several enhancements researchers are developing for the Demographic Frame. They include measures of data quality for person-level demographic characteristics, person-level residential location, and household-level residential stability. They also include efforts to better capture population inflows and outflows by incorporating additional data on young children and emigration.

Incorporating New Administrative Data Sources

Andres Mira, *U.S. Census Bureau*

Nikolas Pharris-Ciurej, *U.S. Census Bureau*

The quality of the Census Bureau's administrative data composites depends critically on the quality of the administrative data sources used as inputs. To accurately represent the U.S. population, a composite should include sources that, when combined, cover every U.S. resident. It should also include sources with accurate data on key characteristics like residential locations and demographics. This presentation will show how quality assessments of new data sources and existing composites have identified gaps in coverage and possible inaccuracies in the data—with corresponding opportunities for improvement through new data acquisitions. We will discuss desirable attributes of potential acquisitions in light of these findings and will suggest datasets with large potential benefits for the Census Bureau's administrative data-based products.

In-Office Enumeration Decision Rules

Stephanie Coffey, *U.S. Census Bureau*

The In-Field Enumeration (IFE) operation in the decennial census is costly but necessary for fully enumerating the US population. Each decade tens of thousands of enumerators are hired to knock on doors and collect information about housing units that did not self-respond to the decennial census questionnaire, either by internet, mail, or inbound telephone call. A data- and quality-driven approach to using administrative data to enumerate IFE-eligible housing units could reduce the overall cost of the IFE operation, while allowing enumerators to focus on cases for which available administrative data is either unavailable or of poor quality. Building upon administrative data usage for enumeration in the 2020 Census, the 2026 Census Test will implement decision rules work to determine which housing units could accurately and sufficiently be enumerated using administrative data, reducing the need for IFE contact attempts on these cases. This talk will include a discussion of the overall decision rules framework, the component models, and select simulation results.

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Session E-4: Modernizing a Longstanding Federal Survey: Overview of Updates to the Current Population Survey

Organizer: Jessica Holzberg, *U.S. Census Bureau*

Chair: Rachel Horwitz, *U.S. Census Bureau*

Census Conference Rooms 1-2

Design Considerations when Adding a New Mode: Steps in the Current Population Survey Modernization

Kyra Linse, *U.S. Census Bureau*

Introducing a new mode of data collection involves far more than simply replicating an existing paper or interviewer-based survey to create an online instrument. Each design decision—from wording changes and login procedures to survey invitation strategies—can affect response, measurement, and overall data quality. This presentation will review key considerations and tradeoffs that arise when incorporating an internet self-response option into an established survey, using the Current Population Survey (CPS) as a case study.

Introduction of the Internet Self-Response Mode in a Longitudinal Panel – Does it Work?

Herschel Sanders, *U.S. Census Bureau*

Beth Satsky, *U.S. Census Bureau*

Allison Zotti, *U.S. Census Bureau*

The Current Population Survey (CPS) is a unique survey with a longitudinal design but a short interviewing window. Introducing internet self-response (ISR) may boost participation, as it could be a preferred mode for many respondents. However, timeliness is critical; responses must be submitted early in the data collection window to allow time for nonresponse follow-up by interviewers. Ideally, respondents will continue to use ISR each month to maximize cost savings. In this presentation, we will examine how many respondents used ISR, how quickly they responded, and the characteristics of those who did. We will also assess whether respondents continued to use ISR over time or whether follow-up through field interviewing was still required.

Findings from Cognitive and Usability Testing of the CPS Internet Instrument

Jonathan Katz, *U.S. Census Bureau*

Renee Stepler, *U.S. Census Bureau*

To evaluate how the Current Population Survey (CPS) questionnaire would perform in an Internet self-response (ISR) mode, we conducted cognitive and usability testing of the entire questionnaire. Two rounds of cognitive testing were conducted focusing on how participants comprehended and answered questions without an interviewer present. To ensure that each pathway of the CPS instrument was thoroughly tested, we focused on recruiting people from a variety of labor force participation categories (e.g., those with more than one job, those not working but actively looking for work). Probing in the cognitive interviews focused on how participants responded to and interpreted some of the complex terminology featured in the CPS questionnaire, such as reporting their usual hours worked versus their actual hours worked, how they described activities they did to look for work, and how self-employed people interpreted a question on whether they did any work for pay.

In usability testing, the goal was to evaluate participants completing the CPS in the Census data collection enterprise instrument to be fielded; we used a longitudinal testing design to mimic dependent interviewing where we interviewed the same participants across two different rounds. There were specific question features (e.g., drop down, questions with prefilled data, household roster dashboard) in the instrument that we wanted to test to see how participants interacted with these features. Therefore, we developed probing questions to capture participant's experiences interacting with these online features.

In combination, cognitive and usability testing informed the CPS ISR questionnaire in data collection. This

presentation will cover the main findings of cognitive and usability testing and general takeaways in conducting qualitative testing for surveys that are considering adding an ISR mode.

Optimizing Short Field Periods: A Test of an Early Bird Digital Incentive in a Sequential Mixed-Mode Design

Renee Stepler, *U.S. Census Bureau*

Jessica Holzberg, *U.S. Census Bureau*

In the Current Population Survey 2025 Field Test, a sequential mixed-mode design was tested in the second interview and higher. Most respondents were asked to respond online first, with the goal of minimizing overall data collection costs. Obtaining web self-response early in CPS data collection is particularly important due to the short 10-day field period. Prior research has shown that “early bird” incentives offered to respondents only if they respond by a certain date have increased response rates during critical periods of data collection (Ward et al, 2014; Coopersmith et al., 2016; Calderwood et al., 2022). In 2025, CPS conducted a pilot test to evaluate the integration of the web survey mode and to determine how to best optimize early self-response. As such, an experimental \$15 early bird digital incentive was offered conditionally to those who completed the survey online by the deadline in the survey invitation (either day 4 or 5 of data collection, depending on their assignment to another experiment). About half of respondents were eligible for the incentive. The other half served as a control group and did not receive any incentive. In our evaluation of the early bird digital incentive, we compare the response rates, timing of response, and the level of effort in collecting responses between the incentive and non-incentive group. As this is among the first federal surveys to include the use of digital incentives, we will also present analyses on the redemption of the digital incentives. These findings will enhance the field’s knowledge on the impacts of early bird digital incentives, especially for surveys with short field periods.

When time is short: Testing contact strategies to increase and speed up web response

Douglas Williams, *Bureau of Labor Statistics*

Robin Kaplan, *Bureau of Labor Statistics*

Erica Yu, *Bureau of Labor Statistics*

To inform the move to mixed-mode administration, BLS and Census jointly conducted experiments to test contact strategies that optimize the tradeoff between cost and response. Contacts were made by email and postal mail to push sample members to respond online within a constrained four-day period, after which an interviewer would visit their home.

In one experiment, approximately half of all households were randomly selected to be sent a postal reminder, with the other households being sent no postal reminder. In the context of a short data collection period where reminders were also sent via email, the value of an additional postal mailing is unknown. Printing and mailing letters is costly, and given the short fielding period of the CPS, the postal reminder must be sent before data collection opens and must be sent to the full sample. Data were collected for three months of pilot test (July to September of 2025). We will share findings from analyses of the effect of the postal reminder on overall response rate, daily response rate, break-off rates, and cost per complete.

In a second experiment, approximately half of all households were randomly selected to be sent emails on each of the first five days of the data collection period (five total emails), with the other households being sent emails on only alternate days (three total emails). While methods for pushing people to web have been well established, traditional protocols require several weeks and even months to complete; there is little literature informing contact procedures over a short collection period – a requirement of the CPS. Data were collected for three months of pilot data collection (July to September of 2025). We will share findings from analyses of the effect of the increased contacts on overall CPS response across various dimensions.

Multi-Method Blitz: Developing the Internet Self-Response (ISR) Questionnaire for a Modernized Annual Social and Economic Supplement (ASEC)

Lisa Cheok, *U.S. Census Bureau*

The Current Population Survey (CPS) Annual Social and Economic Supplement (ASEC) is developing an internet self-response (ISR) mode as part of a broader effort to introduce ISR on CPS. The ASEC is a rich source of data

for evaluating economic trends and understanding their impact on households, families, and individuals. Since ASEC is the source for the Official Poverty Measure, the Supplemental Poverty Measure, and widely used estimates of health insurance coverage, a smooth transition to a multi-mode collection approach requires multiple rounds of testing and evaluation.

Historically, the ASEC has been administered by trained interviewers who guide respondents through the complex questionnaire. Therefore, initial modernization efforts focused on how well self-respondents understand and navigate the survey content. Findings from this phase will be incorporated into the questionnaire ahead of upcoming usability and field tests.

There is a tight window of opportunity for questionnaire development before the planned ASEC field test. To maximize the value of the feedback collected during this phase, questions were tested using a variety of methods, with some test running concurrently. Given the supplement's extensive content--covering work experience, income from all sources, receipt of noncash benefits, health insurance coverage, and geographic mobility--we also strategically divided the content to accommodate the time constraints of various testing opportunities. This approach allowed for testing of multiple iterations of the questionnaire within this development window.

This presentation will provide an overview of the testing methods used and how their results are integrated into a complete questionnaire for the next phase of testing.

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Session E-6: Identifying and Overcoming Hurdles in Consumer Data: Recent Research on Nonresponse and Stratification for the Consumer Expenditure Surveys and Consumer Price Index

Organizer: Stephen Ash, *Bureau of Labor Statistics*

Chair: Michael White, *U.S. Census Bureau*

BLS Auditorium A

Assessing Nonresponse Bias in the CPI's Commodities and Services and Housing Surveys

Jenny Fitzgerald, *Bureau of Labor Statistics*

The Consumer Price Index (CPI) estimates the change in prices over time of the goods and services U.S. consumers buy for day-to-day living. The CPI collects its data in two surveys: a "Commodity and Services" (C&S) survey, and a "Housing" survey. Prior to the COVID-19 pandemic, the C&S survey's outlet and quote repricing response rates were about 90 percent and 80 percent, respectively, and the unit repricing response rate for the Housing Survey was above 70 percent. As COVID-19 caused brick-and-mortar stores to shut their doors, the response rates for both surveys steadily declined. In June 2025, the outlet and quote repricing response rates for the C&S Survey were still only 76 percent and 68 percent, respectively, and the unit response rate for the Housing Survey was about 65 percent. To meet the Office of Management and Budget (OMB) standards and guidelines for statistical surveys regarding response rates, research was conducted using a three-phase approach to assess the effect of nonresponse bias on the C&S and Housing Indexes. In the first phase of the study, sampling unit characteristics contributing to nonresponse at sample initiation were identified using Random Forest for the C&S Survey, and nonresponse adjustment cells were created using decision trees. In the second phase of the study, sampling unit characteristics contributing to nonresponse at repricing were investigated using Random Forests, Random Survival Forests, and Generalized Linear Mixed Model Trees for the Housing Survey. In the final phase the study, counterfactual indexes were calculated to determine whether the level of nonresponse error in the price change estimates was significant by imputing price relatives for nonresponding units using models developed based on the results of the previous two stages.

Model Validation and Benchmarking for CPI Sample Redesign

Onimissi Sheidu, *Bureau of Labor Statistics*

The Consumer Price Index (CPI) updates its primary sampling units (PSUs) after every decennial census using stratification variables that reflect the latest geographic and demographic information available. These decennial updates cost money, and their increasing costs have led some people to recommend developing a reliable permanent stratification model to curb the rising cost. The task of the study is to assess the validity and evaluate six stratification models derived from 12 publicly available variables primarily sourced from the American Community Survey (ACS). The study uses Cross-validation (CV) procedure in conjunction with confidence interval (CI) on monthly and yearly CPI data to validate the models' predictions and to pinpoint the most accurate model for implementation. The result shows that the models could be valid for predicting the yearly data, but are not suited for predicting the monthly data, and that one particular stratification model performs better than the other five models. The study demonstrates a more simplified way of model validation and layouts the framework on how to assess models' accuracy in the context of their validity.

The Past, Present, and Future of Response Rates' Impact on Cell Collapsing Procedures for the Consumer Expenditure Surveys

Lauren Vermeer, *Bureau of Labor Statistics*

Response rates for most household surveys are decreasing, and the Consumer Expenditure Surveys (CE) are no exception. BLS uses cell-collapsing methodology as part of their nonresponse adjustment process for the CE, which assigns both interview units and noninterview units to cells based on certain demographic variables. In cases where there are too few units, or the nonresponse adjustment factors exceed a certain threshold, cells are then collapsed until the cell reaches an acceptable number of units and the factor is less than the threshold. Cell collapsing is sensitive to fluctuation in response rates, and as response rates decrease, the number of collapsed cells increases. One example of this sensitivity occurred in 2020, when survey response rates temporarily and quickly reached a historic low due to the COVID-19 pandemic. To address this unexpected situation, the maximum factor thresholds were subsequently increased to prevent all cells from collapsing together and still exceeding the factor threshold. Since 2020, the response rates have recovered but are still decreasing overall. So, the maximum nonresponse adjustment factors were increased again. In this study, we examine the timeline of response rates' impact on the nonresponse adjustment factor threshold, and how to best adapt cell collapsing methodology in a world of decreasing response rates.

A Comparison of Nonresponse Adjustment Methods with an Application to the Consumer Expenditures Survey

Stephen Ash, *Bureau of Labor Statistics*

An effective nonresponse adjustment is a critical component of producing survey estimates. This paper considers the problem of choosing the nonresponse method given the available methods that include the weighting cell method, calibration, calibration with instrumental variables, maximum likelihood, constrained maximum likelihood, or weighting cell methods that employ a mix of predicted mean stratification and predicted response stratification. The choice of the nonresponse method is vitally important given the low response rates of surveys, the decreasing nature of response rates over time, and the many types of auxiliary variables available that can be used with a nonresponse adjustment. Our research will compare the bias and variance of the alternative nonresponse adjustment methods with a small simulation and an application to the Consumer Expenditure's Interview survey estimates. We will also discuss the advantages and disadvantages of the methods in terms of flexibility of the methods and their use of auxiliary and instrumental variables.

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Session E-7: Using advanced computational methods for complex statistical problems

Organizer: Ben Franco, *Social Security Administration*

Chair: Ben Franco, *Social Security Administration*

BLS CTC 2A

Heterogeneity and Measurement Error for the Expectation-Maximization Algorithm for Log-Linear Models in Capture-Recapture Estimation

Thomas Mule (Vincent Thomas Mule Jr.), *U.S. Census Bureau*

Scott Konicki, *U.S. Census Bureau*

Timothy Kennel, *U.S. Census Bureau*

Dirk Bullock, *U.S. Census Bureau*

Statistics Netherlands developed an expectation-maximization algorithm for log-linear models to use in capture-recapture estimation. This method has a novel application that allows a characteristic to be measured differently in multiple lists. Statistics New Zealand has researched how to use this approach to estimate their Māori population when using multiple lists. The U.S. Census Bureau has been examining this method as our agency is researching capture-recapture approaches that can utilize this novel application with census enumerations and administrative data. Due to changes in question wording, response options and processing, the census can collect race and Hispanic origin information differently than what may have been previously available. Our presentation gives an overview of this method when using two systems. We document the bias implications in certain instances when heterogeneity and/or measurement error in the different systems are present. We conclude by introducing an adjustment that can be used in certain instances to remove or minimize the bias when you can assume that one source has none or negligible measurement error.

Using Conditional Linear Regression Trees to Identify Potential Construct Validity and Measurement Error Differences in Health Outcomes from a U.S. National Survey

Morgan Earp, *National Center for Health Statistics*

Lauren M. Rossen, *National Center for Health Statistics*

Sarah Forrest, *National Center for Health Statistics*

Trent D. Buskirk, *Old Dominion University*

Accurate measurement is critical for the assessment of health outcomes and behaviors across population subgroups. Estimated differences in the prevalence of a given health outcome or risk factor can be affected by whether the outcomes are measured or self-reported. Using data from the National Health and Nutrition Examination Survey (NHANES) which contains both measured and self-reported health information, we used conditional linear regression trees to assess differential measurement error across several chronic health conditions and risk factors, where measurement error is defined as differences between self-reported and measured (e.g., gold standard or objective) health outcomes or behaviors.

Using data on four health conditions/behaviors (e.g., diabetes, hypertension, high cholesterol, and current smoking) from NHANES (2015 through March 2020), we assessed differences between self-reported and measured outcomes by various sociodemographic characteristics (e.g., age, sex, race/ethnicity, education, marital status, health insurance, and poverty). We used conditional linear regression trees via the 'rpms' package in R to identify demographic subgroups with larger differences between self-reported and measured health conditions/behaviors.

Identifying subgroups where measurement error may be larger or smaller could help inform future work on improving the estimation of the prevalence of chronic conditions and health behaviors.

Multiple Imputation Variance Estimator (MIVE) for the Occupational Requirements Survey (ORS) small domain estimation

Xingyou Zhang, *Bureau of Labor Statistics*

Michael Hudak, *Bureau of Labor Statistics*

Michael Lettau, *Bureau of Labor Statistics*

Erin McNulty, *Bureau of Labor Statistics*

Richard Uhrig, *Bureau of Labor Statistics*

Yu(Cosmo) Zhang, *Bureau of Labor Statistics*

Small domain estimation, by nature, is a massive statistical data imputation process. For unit-level small domain models, the imputation (prediction) is applied to every unit from a large survey sample or the entire population frame to produce reliable estimates for small domains that could not be achieved via original survey estimation. If a large survey sample is used for prediction, how do we simultaneously account for the uncertainties associated with the original survey, model prediction, and the large survey? This remains a challenging task but also a controversial topic in practice. We have developed a Multiple Imputation Variance Estimator (MIVE), which is based on the law of total variance, to account for all possible statistical uncertainties for small domain estimation via Multilevel Regression and Poststratification (MRP) for the Occupational Requirements Survey (ORS), an establishment survey conducted by the Bureau of Labor Statistics (BLS).

ORS small domain estimation via MRP involves two basic steps: 1) construct and fit multilevel regression models for job requirements using ORS; 2) apply the fitted regression models to predict job requirements for each observation in the Occupational Employment and Wage Statistics (OEWS) and produce occupation-specific (small domain) estimates of job requirements. OEWS is the largest BLS establishment survey that produces employment and wage estimates annually for approximately 830 occupations. During this regression model-based imputation process, we use ORS replicate weight-based model predictions to capture the uncertainty from ORS sample design effect; we apply parametric bootstrapping to generate multiple imputed values to capture the uncertainty from model prediction; and we employ a weighted-sample variance estimator to capture the uncertainty associated with OEWS sample design effect. MIVE integrates all these three variance components to generate total variance estimates for model-based small domain estimates. MIVE provides a flexible and consistent statistical framework for variance estimation with model-based small domain estimation.

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Thursday, March 12, 2026

8:30 AM

Session E-8: Engaging every voice: Strategies for understanding and reaching all respondents

Organizer: Julie Weeks, *National Center for Health Statistics*

Chair: Julie Weeks, *National Center for Health Statistics*

BLS Auditorium C

Developing a respondent persona for mobile diary survey

Lin Wang, *U.S. Census Bureau*

Alda Rivas, *U.S. Census Bureau*

Maxwell Hope, *U.S. Census Bureau*

Diary surveys refer to those surveys that respondents take at regular intervals, e.g., daily, to record behaviours like sleep, purchases, etc. While responses to the diary surveys could provide more comprehensive information on a topic of interest than that of cross-sectional surveys, diary surveys often pose greater burden on the respondents. When designing a diary survey, survey designers have to balance the burden and the amount of information to be collected. In order to develop a diary survey that could engage the respondents with

acceptable burden, it is essential to understand the relevant characteristics of the target respondent population. Developing personas that characterize potential “typical” users of a new product in design is a widely-practiced technique for designing a human-centred product. Such approach could have an implication to survey design as well – using personas to guide the design of a survey that suites the target respondent population who are characterized by the persona. We developed a method of developing personas characterizing potential mobile diary survey respondents using data collected from usability evaluations of diary survey instruments. The method involves qualitative and quantitative analyses of observed and self-reported data collected during usability testing. Personas were constructed in six dimensions: biography, willingness to respond the survey, motivation for responding the survey, mental capacity, physical capacity, and lifestyle. We use a food acquisition diary survey as a case study to demonstrate our methodologies.

The Promise and Perils of Panels: Understanding Participation, Attrition, and Panel Sustainability in the Census Household Panel

Casey Eggleston, *U.S. Census Bureau*

To address declining participation rates in federal surveys, the Federal Statistical System (FSS) is increasingly turning to national panels—both privately maintained and government-operated—as an alternative data collection strategy. These panels, when carefully curated to maintain representativeness and supported by incentives for high-quality, complete responses, offer a promising approach to mitigating nonresponse bias in voluntary surveys. However, leveraging panel data effectively requires a nuanced understanding of respondent engagement, attrition patterns, and the application of advanced weighting techniques to correct for evolving gaps and biases over time. In this presentation, I analyze respondent behavior in the Census Household Panel (CHP) over a full year of monthly data collections to identify patterns of engagement, as well as characteristics associated with inconsistent participation and attrition. Drawing on this evidence, I explore the implications for representativeness and potential bias, discuss strategies for reengaging lapsed respondents, and consider how methodological changes may affect panel stability over time. Findings highlight both the potential and the limitations of national panels as a response to declining participation in federal surveys, offering practical insights for sustaining panel representativeness in a changing data collection landscape.

Honey, I Forgot the Kids: Underreporting of Children in Probability-Based Web Panels

Jennifer Rammon, *National Center for Health Statistics*

Katherine Irimata, *National Center for Health Statistics*

Jim Dahlhamer, *National Center for Health Statistics*

Jess Jones, *National Center for Health Statistics*

Traditional household-based surveys experience a systemic undercounting of children, particularly young children under 5 years. The magnitude to which children are missed in household surveys is known to vary substantially by demographic characteristics (e.g., age, race, ethnicity, income, household structure). This undercounting is consequential for the measurement of children’s health and healthcare needs. Research on undercounting has not traditionally focused on the effects of rostering and reporting errors in probability-based, web panel surveys. Additionally, little is known about how the magnitude and patterns of undercounting for surveys influence estimates of key public health indicators. The National Center for Health Statistics Rapid Surveys System (RSS) utilizes two commercial, probability-based web panels to produce timely, nationally- representative, and policy-relevant data. Surveys are conducted several times a year, and content varies with each fielding. While RSS was designed to produce estimates of adults, Round 5 was innovated to utilize parental reporting to produce estimates on the health and well-being of children ages 0 to 17. Panel members provide profile data including the number and ages of children in the household which can be used to guide sample selection for surveys targeted to parents. However, to evaluate potential underreporting of children due to missing or inaccurate profile data, two samples were drawn from distinct groups and screened for eligibility: main (profile data indicating presence of children) and supplemental (profile data indicating no children, or missing information). This presentation discusses the characteristics of children identified with the supplemental sample and describes the potential impact on health estimates if these children had been excluded from the study. Results elucidate the potential underreporting of children in adult panelist profile data and the impact of this underreporting on estimates of health indicators for children.

Latest Updates to the Low Response Score Model

Luke J Larsen, *U.S. Census Bureau*

Ana I Sanchez Rivera, *U.S. Census Bureau*

Ralph Culver III, *U.S. Census Bureau*

Maxwell Hope, *U.S. Census Bureau*

Ian Le, *U.S. Census Bureau*

Karlesha LeGrier, *U.S. Census Bureau*

Steven Scheid, *U.S. Census Bureau*

Faraz Shaikh, *U.S. Census Bureau*

Anthony Knapp, *U.S. Census Bureau*

Mark Frame, *U.S. Census Bureau*

Nathan Morse, *U.S. Census Bureau*

The Census Bureau's Low Response Score (LRS) is a regression-based metric that utilizes demographic and socioeconomic indicators derived from Census data to identify areas in the United States that are likely to incur low degrees of self-response to the decennial census. Published at the tract and block-group levels on the Census Planning Database (PDB), the LRS has many uses among not only internal Census analysts, but also data scientists, policymakers, and the general public. Due to changes in the data collection landscape in recent years both planned and unplanned, researchers in the Census Bureau determined a growing need to update the LRS model and improve the accuracy of its predictions. Over the past few years, an internal workgroup was established to research, develop, and deploy this new metric onto a forthcoming release of the PDB.

In this talk, the authors discuss structural revisions to the regression model, which was used to generate LRS predictions that were published on the 2024 vintage of the PDB and are the first to be derived from 2020 Census data. This model includes several new features such as state-level interactions and information about 2020 Census contact strategies assigned to each tract and block group. These additions yielded substantial improvement in model fit over the 2010-era LRS model structure and are expected to better reflect collective self-response behavior in a post-2020 environment. The authors provide some analysis of the published LRS predictions from the 2024 PDB and update the audience on a roadmap of future research and development. Through continued improvement of this predictive metric, this research improves efficiency by accurately anticipating resource needs for data collection at local geographic levels.

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Thursday, March 12, 2026

10:15 AM

Session F-1: How Safe Is Safe Enough? Disclosure Avoidance and Differential Privacy

Organizer: Jennifer Rhorer, *National Agricultural Statistics Service*

Chair: Jennifer Rhorer, *National Agricultural Statistics Service*

Census Auditorium 1

TaCo: A Python Tool for Auditing and Augmenting Suppression Methods

Elan Segarra, *Bureau of Labor Statistics*

Suppression is one of the most common statistical disclosure control (SDC) methods at BLS and many other statistical agencies. The general approach involves suppressing sensitive cells (primary suppressions) and then suppressing additional cells (complementary suppressions) to provide a level of protection to all suppressed cells. When algorithms with formal protection guarantees cannot be applied (perhaps due to computational or feasibility constraints), it can be difficult to assess how well ad-hoc protection mechanisms are functioning. The python tool Table Completer, or TaCo, has been developed to help audit suppression methods and improve the information that could be provided to data users. This tool computes the tightest implicit bounds on suppressed cells given all other published statistics and the relationships among them. Calculating the bounds on an

omitted cell can often be written down as a linear programming problem, and TaCo supplies methods and tools to both construct implied constraints and efficiently solve this problem. When there are hundreds of thousands of suppressions, and therefore many LP problems to solve, the tool also includes a constraint propagation algorithm to quickly tackle these situations and provide bounds for a relaxed definition of implicit bounds. Additionally, this tool can be used to publish more information for data users by publishing these implicit ranges rather than fully suppressing cells. TaCo has recently been leveraged in the production of the 2024 Census of Fatal Occupational Injuries (CFOI) where millions of interrelated statistics are produced (subject to thousands of constraints) and hundreds of thousands of suppressed cells must be checked. The package has helped identify where additional suppressions are required and quantify disclosure risk among competing suppression algorithms.

Multinomial Randomized Response for Data Sharing

Caleb Floyd, *U.S. Census Bureau*

Justin Doty, *U.S. Census Bureau*

Rolando Rodriguez, *U.S. Census Bureau*

Inter-agency data sharing strengthens evidence-based policymaking by enabling agencies to leverage cross-agency data resources to evaluate policy effectiveness. The U.S. Census Bureau collects demographic data that is linkable to other agencies' data; however, the Census Bureau is bound by law to release this data for statistical purposes only, and only after sufficient protections for respondent confidentiality are in place. Since other federal agencies might have use cases outside of purely statistical purposes and are not bound to the same legal confidentiality requirements, the Census Bureau must develop frameworks that allow for the balancing of the accuracy, availability, and confidentiality of linkable data releases.

Formally private frameworks, such as local differential privacy (local DP), provide quantifiable bounds on the statistical inferences that can be made when data is released. In the case of categorical demographic data, this can be achieved by a randomized response algorithm that perturbs each response according to some probability as a function of a privacy parameter. In this paper, we propose using multiple imputation methods to correct statistical inference for randomized attributes.

We consider the use case of sharing data on race and ethnicity (RE) data with the Internal Revenue Service (IRS). First, we discuss how to perturb each RE response in the microdata. We then present the results of an R package we developed to correct for statistical inferences after randomization. We show that this step is crucial for maintaining a balance between confidentiality and accuracy, as opposed to using the randomized responses alone. We conclude with directions for future research and other issues of privacy budget allocation.

Differential Privacy for County Business Patterns

Anthony Caruso, *U.S. Census Bureau*

As early as 1840, Census officials recognized the distinct importance of ensuring the confidentiality of economic data provided by the public. At the time, this simply meant that "on statistical tables no name is inserted" and "the figures stand opposite no man's name." Fortunately, as economic statistics evolved, so did the techniques employed to safeguard the survey and administrative data used in their production. Differential privacy is one of the most recent privacy-preserving techniques to be developed, and careful implementation results in a mathematically provable privacy guarantee. While differential privacy (or DP) was adopted by the 2020 Census, a different approach was needed for use with highly skewed economic data. A team of researchers from Census, MITRE, and Tumult Labs worked to develop Slowly Scaling Per-Record Differential Privacy (SSPRDP), a variant of DP suitable for use with economic data. The first implementation of this method will be for the 2024 County Business Patterns (CBP) publication to be released in 2026. A brief overview of DP, SSPRDP, and the proposed implementation for CBP will be discussed.

Applying the Boltzmann Machine Privacy Framework to a Randomized Response Setting

Mark Fleischer, *U.S. Census Bureau*

Pavel Zhuravlev, *U.S. Census Bureau*

Randomized Response (RR) is a method of disclosure avoidance (DA) described well before the invention of

differential privacy (DP). In RR, answers to true/false survey questions are randomly modified and thereby provide a level of privacy yet still provide useful aggregate information. It later turned out to be a form of differential privacy (DP) in that it fits within the framework and definition of DP.

We apply the Boltzmann Machine Privacy Framework (BMPF), a meta-mechanism for DP, to an RR problem domain. The BMPF relies on the simulated annealing (SA) algorithm, often referred to as a meta-heuristic, to "unsolve" an optimization problem. By keeping the temperature parameter in SA at a positive value, the SA algorithm tends to avoid the global optimum, which corresponds to the true dataset, and establishes a stationary distribution vector that increases the probability of sub-optimal solutions that do not correspond to the true dataset. Sub-optimal solutions are therefore randomized versions of the true dataset. We show that this framework satisfies the definition of DP. In the RR setting, the SA algorithm perturbs a vector of binary values subject to an objective function. We show proofs demonstrating how the framework using a binary based configuration space satisfies the definition of DP and its equivalence to RR. Experimental results are also provided illustrating application of the theories presented.

Cell suppression and noise injection in the context of causal inference

Nicholas Potter, *Economic Research Service*

Federal statistical agency tabulations of survey and census data are invaluable tool for social science researchers investigating causal relationships in economics, education, health, and other fields. Preventing precise inference of individual responses from these data products is a paramount concern, but methods of doing so have implications for the usefulness of these tabulations for causal inference.

Cell suppression and noise injection are two methods of protecting individual responses. In cell suppression, summary statistics are suppressed if a cell has too few observations or if the statistic is dominated by a few observations. Cells may also be suppressed if not doing so would allow precise inference primary suppressed cell values. Cell suppression reduces the usefulness of data products for causal inference in at least two ways: (1) by effectively reducing the sample size due to missing values, and (2) by potentially biasing parameter estimates when effects are heterogeneous with respect to characteristics that also affect whether a cell is suppressed.

Noise injection is an alternative to cell suppression in which individual responses are perturbed before summary statistics are calculated. Noise injection has the advantage of reducing the additional risk presented by special tabulations and new data releases while providing more information in data sets that would otherwise require substantial cell suppression. In addition, noise injection accomplishes this without biasing cell summary statistics. However, noise injection effectively introduces additional measurement error, making precise estimation of causal effects more difficult.

Relative to cell suppression, noise injection offers increased sample size and reduced potential bias in heterogeneous causal estimates at the cost of increased measurement error. I use this tradeoff to develop guidelines for choosing between cell suppression and noise injection using estimates of causal effects of canal lining on conveyance loss based on data from the USDA's 2019 Survey of Irrigation Organizations.

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Session F-2: Reading Between the Lines: Using Text Analysis in Bureau of Labor Statistics Surveys

Organizer: Victoria Narine, *Bureau of Labor Statistics*

Chair: Josh Langeland, *Bureau of Labor Statistics*

Census Auditorium 2

A model to predict NAICS code from business name in the Quarterly Census of Employment and Wages

Erin Boon, *Bureau of Labor Statistics*

The Quarterly Census of Employment and Wages (QCEW) program at the Bureau of Labor Statistics (BLS) publishes a quarterly count of employment and wages as reported by employers which covers more than 95 percent of U.S. jobs. The QCEW also serves as the frame for establishment surveys at BLS. The source for the QCEW is administrative data from state unemployment insurance programs, supplemented by data collected through two BLS surveys. A small percentage of QCEW records, however, do not include a valid industry code (NAICS). These unclassified records cannot be used for reports that are broken out by industry or samples that are stratified by industry. This presentation will introduce a prototype QCEW-NAICS AutoCoder, a random forest classifier trained on millions of QCEW records to predict 6-digit NAICS codes from words and phrases in a business name. The goal of this tool is to approximate a manual review by a human coder to assign a NAICS code to an unclassified record using only the information contained in the legal and trade name fields on the record. Limitations and challenges with this approach will be discussed, including how we constrained model features (i.e., vocabulary) to terms that are generalizable to business operations, and how we achieved a target model accuracy by limiting which records could be auto coded.

New and evolving occupations in the Occupational Employment and Wage Statistics Program

Sertan Akinci, *Bureau of Labor Statistics*

The Occupational Employment and Wage Statistics (OEWS) program collects job title and wage information annually, producing employment and wage estimates for 832 occupations. These job titles are coded to their appropriate Standard Occupational Classification (SOC) code to publish occupational data at the national, state, metropolitan area, and industry level. As the economy changes, and occupations shift, it is important to identify occupations that are new or evolving and how they are being reflected in OEWS data. Emerging occupations refer to new roles that arise in response to recent innovations, technological development, new organizational needs, or other external factors. Using the clustering techniques like K-means and DBSCAN, SOC codes are clustered to identify subsets or outliers within the codes, providing groupings based on job titles, some of which are potential new and emerging occupations.

An Autocoder tool to convert open-ended text into item codes in the Consumer Expenditures Diary Survey

Melissa Pollock, *Bureau of Labor Statistics*

The Consumer Expenditure Surveys (CE) program at the U.S. Bureau of Labor Statistics (BLS) provides data on expenditures, income, and demographic characteristics of consumers in the United States. Data for more minor or frequently purchased items are collected via a respondent self-administered Diary Survey. These expenditures are described by respondents through free-form text. The volume of Diary Survey expenditures collected each month is approximately 30,000 records. The CE Diary Autocoder, launched in 2024, applies a Natural Language Processing approach to classifying diary entries into item codes, significantly reducing processing time and costs. This presentation will show how free-form text in a respondent diary ladders up to item codes through illustrative examples, describe the methods which comprise the CE Diary Autocoder, explain how human intervention factors into the automated process, and discuss maintenance considerations.

Making sense of "other" disposition codes: A text analysis approach

Victoria Narine, *Bureau of Labor Statistics*

Josh Langeland, *Bureau of Labor Statistics*

The Occupational Employment and Wage Statistics (OEWS) program at the U.S. Bureau of Labor Statistics (BLS) uses an advance letter to inform sampled establishments of the upcoming survey request and collect contact information for the establishment's preferred respondent. In 2023, the authors reported results from an experiment conducted in a single state indicating that placing a telephone call to respondents after an advance letter is sent helps with collecting contact information. Two follow-up studies replicated the findings in a different state and took advantage of enhanced paradata collection that captured disposition information from the telephone calls (e.g., "left message," "spoke with business," "refusal," etc.). One category in the disposition information is "Other," which consists of open-ended text responses with descriptions of results from the telephone calls. These open-ended responses contain rich information about the business and the analysts' interaction with it; we suspected it could be leveraged to determine whether the standard disposition codes are sufficient or in need of improvement. Preliminary analysis of the open-ended responses indicated that analysts used the "Other" disposition code as a catch-all, but with limited variability. That is, analysts used the "Other" disposition code to describe similar situations. The current research extends the previous findings by using text clustering to explore the content of the open-ended fields on a bigger scale, focusing on whether a) cases could be coded into the standard categories, and b) the codes themselves might need future revision. In addition to sharing results from the text analysis, we will discuss how analysis of unstructured data can complement quantitative analyses in survey methodology research by providing insight into contextual factors that may not be captured in a categorical variable.

Filling in the blanks: Exploring an autocomplete question to collect occupation and industry from household participants

Robin L. Kaplan, *Bureau of Labor Statistics*

Erica Yu, *Bureau of Labor Statistics*

Doug Williams, *Bureau of Labor Statistics*

Autocomplete features of web surveys predict or complete responses as typed in real-time by drawing on lists of pre-existing answers that display suggested options. Autocomplete questions can save time, reduce errors, and improve the user experience. As survey programs look to add a web-mode to collections that are currently interviewer-administered, one challenge is how to administer open-ended questions, such as those collecting respondent occupation and industry. This is challenging because respondents may not provide the level of detail required to produce estimates without assistance from an interviewer to probe for more detail. One solution would be to provide a list of occupations and industries from which respondents can select that would map onto existing coding structures like the Standard Occupational Classification (SOC) and the North American Industry Classification Systems (NAICS). This approach could also save on survey post-processing time by having respondents self-code their own data. The current research explored the use of an autocomplete survey question to collect occupation and industry. A total of 1439 participants were recruited from a nonprobability web panel to participate in a self-administered survey. Participants first described in their own words their occupation and industry. On a separate page, they were then asked to describe their occupation and industry again, this time using an autocomplete feature, to select the option that best matched their occupation and industry. Participants also rated how well they felt their selections matched their actual occupation and industry. We present results assessing how well the participants' open-ended descriptions of their occupation and industry matched their autocomplete-assisted selections, and participants' self-reported ratings of how well their selection matched their occupation and industry. We discuss the implications for the use of autocomplete questions in web surveys.

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Session F-3: Addressing Addresses: Challenges and Solutions to Enterprise-Wide Address Management and Matching

Organizer: April Avnayim, *U.S. Census Bureau*

Chair: Lori Zehr, *U.S. Census Bureau*

Census Auditorium 3

Efficiencies and Challenges in Enterprise Address Matching

April Avnayim, *U.S. Census Bureau*

Underlying all Census Bureau data products are one or more sources of more detailed data, including responses to household and business censuses and surveys or records on government or private-sector databases. Since most Census Bureau datasets contain address information, addresses can be used to join datasets together using the shared characteristic of location. Locational information can also be used for joining data when the user is interested in analyzing more than one dataset at the same location.

The Master Address File/Topologically Integrated Geographic Encoding and Referencing (MAF/TIGER) System (MTS) is the Census Bureau's dynamic geospatial inventory of addresses, features, and boundaries. Within the MTS, unique persistent identifiers called MAFIDs are assigned to structures and their associated addresses. These MAFIDs are assigned to most Census Bureau datasets containing address information, which enables them to more efficiently serve as a key to link incoming addressed data to existing data within the Census Bureau, and to link Census Bureau datasets to each other. Addresses and MAFIDs are also an important component in accurately assigning the protected identification key (PIK), another foundational unique identifier for Census Bureau datasets.

While addresses and MAFIDs are powerful data elements for joining and linking datasets, the Census Bureau has encountered many challenges in using addresses to join disparate data within a large enterprise that uses different standardization methods, and different matching tools.

This presentation provides a brief overview of the methods of address matching used at the Census Bureau, highlights some of the challenges encountered, and provides some suggested solutions for implementing consistent address standardization and matching across a diverse enterprise.

Master Address File Address Standardization and Matching- Benefits of an Enterprise Solution

Taylor Harper, *U.S. Census Bureau*

The U.S. Census Bureau's Geography Division (GEO) and Decennial Information Technology Division (DITD) have maintained addresses within the Master Address File (MAF) and Topologically Integrated Geographic Encoding and Reference (TIGER) System (MAF/TIGER System; MTS) for over three decades. MAF maintenance today includes robust processes that evaluate and standardize incoming address data and match the standardized data against existing MTS addresses, address ranges, and supporting features (e.g., roads), with the overarching goal of identifying addresses that already exist in the MAF, versus those that may need to be added.

Other areas within the Census Bureau have also independently developed address standardization and matching methodologies, since addresses are a common and important data point on most Census Bureau datasets. Some of these processes are designed to match to an extract of the MAF to assign a MAF Identifier (MAFID) to addresses in the dataset. While equally robust and fit-for-purpose, these varying methods result in inconsistent standardization, matching, and MAFID assignment, which have impact on the ability to link or join records across datasets.

Since the 2020 Census, efforts to move foundational datasets and processes to cloud-based technologies are underway, including GEO and DITD's time-tested address maintenance processes- enabling more users to leverage GEO and DITD's standardization and matching process. This presentation provides insight into standardization and matching processes of GEO and DITD, and how adoption of a consistent methodology for matching can be beneficial for all parties within an organization.

Improving Address Location Accuracy Through Automation

Peter Lee, *U.S. Census Bureau*

The Census Bureau updates its Master Address File (MAF), twice annually. The United States Postal Service (USPS) Delivery Sequence File (DSF) is the primary source of address and ZIP Code information for the MAF; however, in order for these addresses to be included in the Decennial Census the Census Bureau must also associate MAF addresses with census blocks - a process called geocoding. Currently, the DSF does not include the specific location or geocode of its addresses. Nationwide parcel datasets are a valuable complement to the DSF because they enable DSF and MAF addresses to be more precisely associated with latitude and longitude coordinates that can improve geocoding. Parcel data sets also enable validation of existing MAF addresses and can serve as a source of new addresses.

Recently the Census Bureau developed a new automated process, known as the Parcel Address Matching (PAM) process, that uses commercial parcel datasets and building footprints to confirm, improve, or derive a precise location for addresses in the MAF. Through PAM, the Census Bureau has successfully automated activities that were previously completed manually by clerical staff, and in the process significantly increased both the efficiency of the operation and the number of accurately geocoded MAF addresses. The PAM process has improved the location for more than 50 million addresses and added coordinates for an additional 14 million addresses making them easier for field staff to locate for enumeration and improving the spatial accuracy of tabulated data. This presentation will provide an overview of the PAM process and inform address data users on the benefit of deriving locations for address data using automated methods and alternative data sets such as parcels and building footprints.

Tracking Change in the Census Bureau's Master Address File: A Framework for Longitudinal Consistency in Administrative Data

Paul Scholes, *U.S. Census Bureau*

Casey Blalock, *U.S. Census Bureau*

The U. S. Census Bureau maintains a database called the Master Address File (MAF), which contains residential, non-residential structures, and their associated addresses. The MAF is updated from various state, federal, and commercial sources. While the Census Bureau continuously maintains the data in the MAF, most users use a semi-annual extract, the MAFX.

Addresses on administrative records are probabilistically matched to addresses in the MAFX and assigned a MAFID. Over time, MAFX content changes accumulate so some MAFIDs assigned to addresses in older data are no longer valid, and in some cases, are not represented in current MAFXs.

The Demographic Frame, a comprehensive database of person-level data matched to addresses across time, is sourced primarily from administrative records. When producing extracts of historical data, users are unable to identify some MAFIDs in current MAF extracts. The Demographic Frame reconciles changes across MAF extracts and updates the MAFIDs in its own products, and indirectly, the MAFIDs in administrative records.

The Demographic Frame developed the MAF Master table to track MAF Units and their characteristics over time. It can be used by other projects that utilize administrative records or other data where MAFIDs are assigned across multiple time periods.

The MAF Master table enhances the utility of the MAF for projects that study housing, migration, and neighborhood change by permitting a longitudinal treatment of the data. It informs best practices for managing dynamic administrative data used in official statistics and illustrates the need for record linkage systems within federal agencies to be based on stable and durable identifiers.

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Thursday, March 12, 2026

10:15 AM

Session F-4: From interviewer to interface: experimental and qualitative research to inform web survey development

Organizer: Aleia Clark Fobia, *National Oceanic and Atmospheric Administration*

Chair: Aleia Clark Fobia, *National Oceanic and Atmospheric Administration*

Census Conference Rooms 1-2

Democratizing In-Depth Interviews: Involving All Team Members in the Process

Jean E. Fox, *Bureau of Labor Statistics*

Victoria R. Narine, *Bureau of Labor Statistics*

Guadalupe Cerritos, *Bureau of Labor Statistics*

Justin Holt, *Bureau of Labor Statistics*

Jennifer Price, *Bureau of Labor Statistics*

Nicholas Schaffer, *Bureau of Labor Statistics*

In-depth interviews are a useful method to gather input from users because the discussions can be thorough and lead to unexpected findings. Though interviews are typically led by researchers trained in survey methodology, user experience, or other qualitative methods to ensure that the method follows best practices, there are meaningful advantages to having other team members lead the interviews. For example, leading interviews gives team members an opportunity to connect with and develop empathy for the participants. This experience can align the team members' understandings of the users, thereby facilitating team discussions, streamlining decision-making, and helping the team create better content for users. However, some challenges with this approach include ensuring that the team members are sufficiently trained and comfortable leading interviews.

In this presentation, the authors will share their experience having team members conduct remote interviews ahead of the redesign of a government statistical agency's website for teachers and students. For this effort, two trained qualitative researchers guided other team members in a series of interviews to better understand the participants' jobs, their needs, and how they might use the site. With this approach, members of the team were better able to understand users' needs, and the team was able to conduct more interviews in the time available than would have been possible otherwise. The presentation will cover the benefits and drawbacks of this approach, how it helped the team, the training provided to the interviewers, options for analyzing the findings, as well as lessons learned. The presentation will share perspectives from both the researchers and the other team members.

Bring the Internet to the Interview: Applying Qualitative Mixed Methods Pretesting to Introduce a New Data Collection Mode

Jonathan Katz, *U.S. Census Bureau*

Eric Stone, *U.S. Census Bureau*

Kristen Kohm, *U.S. Census Bureau*

Jennifer Pace, *U.S. Census Bureau*

Ian Le, *U.S. Census Bureau*

Kathleen Kephart, *U.S. Census Bureau*

Jessica Jones, *National Center for Health Statistics*

Grace Medley, *National Center for Health Statistics*

Adena Galinsky, *National Center for Health Statistics*

Benjamin Zablotsky, *National Center for Health Statistics*

The National Health Interview Survey (NHIS), currently an in-person survey conducted by the National Center of Health Statistics (NCHS), is exploring the use of mixed-mode data collection, to include a web survey. To inform this exploration, the Census Bureau conducted focus groups with NHIS interviewers and supervisors and cognitive testing with general population participants. During questionnaire development, qualitative researchers often use methods like cognitive testing and focus groups to investigate potential survey questions

and response issues. In previous studies, focus groups with interviewers and supervisors of in-person surveys have provided important feedback on many aspects of the survey design and implementation process, particularly on respondents' reactions to the questionnaire. This research will focus on whether there is alignment in concerns between general population participants and interviewers and supervisors when developing a web mode.

Focus groups were conducted in 2024 with interviewers and supervisors who worked on the NHIS. These focus groups produced vital feedback on which features might be most challenging in a web survey. In 2025, the Census Bureau conducted cognitive testing on a web-based instrument, which included (1) items flagged by interviewers and supervisors as potentially difficult in a web survey and (2) items that NCHS and Census Bureau identified as priority and mode sensitive. Using the combination of the focus groups and cognitive interviews, we evaluate similarities and differences in the concerns that interviewers, supervisors, and respondents have with completing a survey on web. This presentation will help methodologists and researchers understand the value of using these qualitative methods to examine the perspective of different key stakeholders (e.g., survey respondents, interviewers).

Triangulating Migration: Leveraging Data from Qualitative and Quantitative Collections to Improve Survey Design

Owen Denoeux, *U.S. Census Bureau*

The Current Population Survey Annual Social and Economic Supplement (CPS ASEC) captures data on migration in the United States, including reasons for moving (RFM) to respondents' current residences. The RFM questions collect household members' main reason for moving. As part of an overall modernization of CPS ASEC, we are testing updates to the RFM questions. These updates include collecting a secondary reason for move, allowing household members who moved with the respondent to report their own reason(s) for move, and an updated set of response options. The response options are a set of categories, which aim to be clear (the category meanings are easily and widely understood by respondents), distinct (a given reason for move falls under only one category), and complete (any given reason for move falls under an existing category). Below each category listed, some examples of what sorts of answers would fall under that category are provided for added clarity and ease of association.

To assess the performance of our revamped RFM questions, we combine insights from the Census Household Pulse Survey (CHP), Household Trends and Outlook Pulse Survey (HTOPS), and two rounds of cognitive testing. The CHP data gave us an initial impression of the distribution of responses for a relatively large sample. The HTOPS data, which includes write-in responses for those who selected 'Other' reason for moving, supplied us with a sense of what categories respondents do (and don't) resonate with as well as what categories appear in write-in data when it is clerically coded. The cognitive interviews provided and continue to provide context as to why respondents select a specific category, select 'Other', or refuse to select a category at all. These insights can be taken together to improve the categories, examples, and even the overall format of our survey questions to the benefit of future data quality.

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Thursday, March 12, 2026

10:15 AM

Session F-5: Navigating Nonresponse: Modeling and Mitigation

Organizer: Jonathan Mendelson, *Bureau of Labor Statistics*

Chair: Erica Yu, *Bureau of Labor Statistics*

Census Conference Rooms 3-4

Edit and Imputation of Nested Data Using Multi-Level Latent Class Models

Emanuel Ben-David, *U.S. Census Bureau*

Tom Mule, *U.S. Census Bureau*

Joseph Schafer, *U.S. Census Bureau*

This research develops and evaluates model-based approaches for imputing missing data in nested data, such as individuals within households and households within census tracts. Traditional imputation methods often overlook dependencies across nesting levels, resulting in biased estimates and the loss of important structural relationships. To address this, we implement multi-level latent class models that perform data editing and imputation simultaneously, ensuring consistent treatment of, for example, both individual- and household-level variables.

To rigorously test these methods, we conduct simulation studies using the 2012 Public Use Microdata Sample (PUMS). Missing patterns are introduced at varying levels, including cases where entire households are missing. The proposed framework is then benchmarked against widely used hot-deck imputation techniques, with performance assessed across multiple criteria, indicating preservation of multivariate distributions.

Our findings demonstrate that multi-level latent class models provide a robust and flexible solution for enhancing imputations in nested data. These models can effectively recover relationships across hierarchical levels, even with high rates of missing data, and can enhance the accuracy and utility of large-scale survey data for analysis, providing valuable insights for researchers

Optimal Geographic Scale for measuring Community-Level Predictors of Decennial Census Self-Response

Richard Turner, *U.S. Census Bureau*

Ivan J. Cardona-Tomassini, *U.S. Census Bureau*

Statistical models predicting how likely a housing unit's (HU) occupants are to self-respond to the decennial census can be used to develop less costly data collection strategies, while still maintaining data quality. Realizing such benefits, however, depends on the models' predictive accuracy and, by extension, how well they account for address-level and community-level factors which influence likelihood of self-responding to the census. Drawing on 2020 Census data, this study aims to facilitate development of more precise census self-response models by analyzing the optimal geographic scale for measuring community-level predictors of self-response (e.g., percentage of addresses in the local area that are nondeliverable). Specifically, intraclass correlation coefficients (ICCs) are calculated for 2020 Census self-response status at the (1) county, (2) tract, and (3) block-group levels. Each ICC score captures the percentage of the overall variation in self-response status in the 2020 Census that existed between units of the respective geography (as opposed to within the geography). A higher ICC value would suggest there is more self-response variation to explain at the respective scale, potentially making it a better candidate for the level at which to construct community-level self-response predictors. Implications of the results for self-response modeling are discussed.

Assessing nonresponse mitigation strategies for national forest inventory data

Andrew J. Lister, *U.S. Forest Service*

James Westfall, *U.S. Forest Service*

Joseph McCollum, *U.S. Forest Service*

Jennifer Bakken, *U.S. Forest Service*

Paul Patterson, *U.S. Forest Service, retired*

Alana Bayless, *University of Nevada, Las Vegas*

The USDA Forest Service's Forest Inventory and Analysis Program (FIA) conducts the national forest inventory (NFI) of the United States. The NFI consists of a spatially-balanced sample of approximately 300,000 0.067-ha NFI plots that are distributed across all land and water area within the United States and its territories. Nonresponse, which mostly originates from a lack of access to plots due to denial of access by owners or safety issues, raises bias concerns.

To address these concerns, the FIA program conducted a study to assess the impacts of various candidate estimation strategies on bias using a simulation experiment in which a model of nonresponse probability was developed and applied to intact, regional NFI plot datasets to generate 5000 realizations of samples of 3000 plots with each of several levels of nonresponse – 0%, 1%, 5% 10% and 20%. Four estimation strategies were applied to each of the samples: use of response homogeneity groups (RHG), model assisted regression estimation (MAR), raking (RAK), and imputation (IMP). Where appropriate, methods included the use of stratified estimation. For each combination of method and nonresponse level, the mean of the estimate of

aboveground biomass of forest trees (an NFI variable computed from data collected on the plots) and its variance were calculated and stored to create distributions of these values. The mean and variance of each distribution were calculated. To assess bias mitigation potential, the means of the distributions were compared with the known means from the intact datasets, and the variances of the distributions of means were compared with the mean of the corresponding distributions of variances to assess performance of the variance estimators. Details of methods and results will be discussed, along with strategies for selecting a procedure to implement from among the candidate strategies.

Predicting refusal reasons in the National Health and Nutrition Examination Survey

Steven Fink, *National Center for Health Statistics*

Matt Jans, *National Center for Health Statistics*

Jill Fleming, *National Center for Health Statistics*

Minsun Riddles, *Westat, Inc.*

Understanding survey refusal reasons aids refusal avoidance and conversion. We explored which area, dwelling unit (DU), and person characteristics predict specific reasons, and whether DU and person-level results differ. The National Health and Nutrition Examination Survey (NHANES) represents the noninstitutionalized civilian U.S. population by screening DUs and interviewing people in-person. Up to three DU screener and survey participant (SP) interview refusal reasons were collected for each refusal. Using DU screener ($n = 8,027$) and SP interview refusals ($n = 14,736$, age 18+) between 2017 and 2023, we fit logistic regressions for five reasons (No Interest, Privacy, Distrust in Government or Surveys, Busy, and Personal or Health reasons) and refusals without a reason (six models each for DU and SR). DU predictors were ACS tract-level characteristics, other tract-level Census Bureau Planning Database information, percentage of registered voters who voted in 2020 as a civic engagement measure, and DU observations by NHANES interviewers. Person-level predictors added person-level information from the screener. Initial results (significant ($p < 0.05$)) show DU-level lack of interest was less likely in areas with more non-family households. Privacy reasons were more likely in areas with higher median age. Distrust was higher in areas with more males, older median age, and fewer non-family households. Busyness was not predicted by any area level characteristics. However, not providing a reason was more likely in areas with older median age, and fewer non-family households. Distrust and not providing a reason were also predicted by Census Division, with a possible contribution of urbanicity. Both person-level demographics and area demographics significantly predicted SP interview refusal reasons, showing the importance of both person and area level characteristics for understanding nonresponse. Interestingly, lack of interest was higher in areas with lower civic engagement (voting), but Busyness was higher in areas with more civic engagement.

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Thursday, March 12, 2026

10:15 AM

Session F-6: Economic Measurement in Practice: From Consumer Finance to Land Use

Organizer: Elan Segarra, *Bureau of Labor Statistics*

Chair: Brandon Kopp, *Bureau of Labor Statistics*

BLS Auditorium A

Comparing the Consumer Expenditure Surveys and Survey of Consumer Finances: Distributions of Income and Expenditures

Griffin St. Martin, *Bureau of Labor Statistics*

Robert S. Martin, *Arizona State University*

Accurate measurement of household economic well-being is critical for federal statistics and economic research. The Consumer Expenditure Surveys (CE) and the Survey of Consumer Finances (SCF) are nationally

representative household surveys that provide complementary information on income, expenditures, balance sheets, and demographics. The CE offers comprehensive data on consumer spending, while the SCF provides detailed measures of household finances and incomes. Comparing the two surveys highlights important issues of data quality, survey design, and measurement comparability that are central to producing high-quality federal statistics and to the use of these data in research and analysis.

This paper compares the CE and SCF on their measurement of household income, selected household characteristics, and overlapping expenditures. We construct comparable measures across the two surveys and analyze differences in population means as well as by quantile of before-tax income.

Our results show that the CE and SCF yield qualitatively similar estimates for housing tenure, age of the reference person, and food expenditures across the income distribution. However, the SCF better captures total income and wage and salary income at the upper end of the distribution, as well as property-related components of income across the entire distribution. By contrast, the CE performs better in collecting self-employment income. For wealth-related attributes and expenditures—such as the market value of the primary residence, mortgage payments, and property taxes—SCF averages are statistically significantly higher at the 0.05 level than CE estimates for the top two income quantiles, likely reflecting differences in the surveys' sample designs.

These findings underscore the importance of survey design and sampling for capturing different components of household economic activity. By documenting similarities and differences in key measures, this comparison provides guidance for researchers selecting data sources and supports survey linkage efforts by identifying where variables are comparable across the CE and SCF.

A Value Chain Approach to Measuring the U.S. Production Structure for Components of Final Demand

Gonca Senel, *Bureau of Economic Analysis*

Jon D. Samuels, *Bureau of Economic Analysis*

This paper develops a framework to examine the factor composition embedded in U.S. value-added across final demand components by integrating value-added data with Integrated Industry-level Production Account (KLEMS) statistics. This approach quantifies factor-specific value-added within U.S. global value chains (GVCs) using the U.S. Bureau of Economic Analysis' (BEA's) domestic requirements tables and the BEA-BLS Integrated Industry-level Production Account. The decomposition tracks both direct and indirect factor shares across upstream networks, revealing several patterns across final demand categories: exports are most intensive in research and development (R&D) capital, investment relies heavily on non-college labor, government spending has the highest share of college-educated labor, and consumption is least intensive in R&D and information technology (IT) capital. Sectoral comparisons highlight structural asymmetries: chemicals have the highest direct R&D intensity, financial services intensively use college-educated labor, and transportation and textiles primarily rely on non-college labor. These divergences highlight how upstream industries differ significantly from final-stage producers in factor intensity.

Economic Statistics Tool for Public Land Management

Hebin Lin, *U.S. Bureau of Land Management*

This presentation demonstrates the benefit of the Economic Statistics Tool developed by the Bureau of Land Management (BLM) in enhancing the efficiency, consistency, and quality of economic analyses while reducing the costs of project management. The Economic Statistics Tool in EXCEL format integrates (1) key economic indicators such as population, income, employment structure by economic sector, age structure, education level published by the U.S. Census Bureau, (2) macroeconomic indicators such as Gross Domestic Product (GDP) published by the U.S. Bureau of Economic Analysis, consumer price index published by the U.S. Bureau of Labor Statistics, and (3) fiscal indicators such as Payment-in-Lieu-of-Tax (PILT) published by the Department of the Interior. It provides users with ready-to-use data tables, column charts, pie charts, and summary descriptions showing (a) overall economic characteristics in 2023, (b) changing characteristics from 2016 to 2023, and (c) comparative economic characteristics in 2023. The Economic Statistics Tool has been applied to expediting National Environmental Policy Act (NEPA) projects, business plans, and land exchange projects, among others. The design and development of the Economic Statistics Tool provides insights for innovation and collaboration in integrating the production and the use of federal statistics across agencies.

The Composition and Re-employment Outcomes of Unemployment Insurance Claimants: Introducing Unemployment Insurance Economic Outcomes (UIEO)

Lawrence Warren, *U.S. Census Bureau*

David Wasser, *U.S. Census Bureau*

Caelan Wilkie-Rogers, *U.S. Census Bureau*

Unemployment Insurance (UI) programs play a central role in cushioning job loss and supporting labor market recovery, yet public administrative and survey data sources have limitations in measuring who collects UI and how recipients fare after joblessness. This project is a pilot for the planned Unemployment Insurance Employment Outcomes (UIEO) data product, which will provide new public-use statistics on the characteristics and labor market trajectories of UI claimants.

We assemble administrative data on the universe of weekly UI claims from six states, linked to longitudinal quarterly earnings records from the U.S. Census Bureau's Longitudinal Employer-Household Dynamics (LEHD) program. These linkages allow us to construct precise measures of unemployment spells, including timing, duration, and benefit receipt, and to track workers' pre- and post-spell employers, industries, and earnings.

Our findings document substantial heterogeneity in re-employment outcomes, shaped by demographics, prior earnings, and re-employment type. We highlight the distinct role of recall versus new matches, the earnings consequences of cross-industry and cross-state mobility, and the strong association between spell duration and subsequent earnings losses.

By combining claims and earnings data, this pilot demonstrates the feasibility and value of producing detailed, timely, and policy-relevant statistics on UI claimants. The UIEO data product will expand coverage to additional states and years, providing a new resource for researchers, policymakers, and the federal statistical system to better understand job loss, social insurance, and worker outcomes.

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Thursday, March 12, 2026

10:15 AM

Session F-7: Strengthening Federal Statistics Through Weighting Strategies and Adjustment Techniques

Organizer: Yan Liu, *Internal Revenue Service*

Chair: Yan Liu, *Internal Revenue Service*

BLS CTC 2A

Leveraging Augmented Inverse Propensity Weighting for Causal Inference: Insights from Confidential Versus Anonymous Survey Administrations in Federal Statistics

Zhiwei Zhang, *Department of War*

The Defense Organizational Climate Survey (DEOCS), administered by the Department of War (DoW), is a critical tool for assessing organizational climate across military units and DoW organizations. Recent methodological advancements evaluate the causal effects of survey administration conditions—confidential versus anonymous—using Augmented Inverse Propensity Weighting (AIPW). This paper presents a robust statistical method extending beyond traditional correlation-based analyses to enable causal inference in observational data, addressing inherent challenges. AIPW combines regression adjustment and inverse probability weighting to estimate the Average Treatment Effect of survey conditions while mitigating bias from confounding variables. The methodology was applied to DEOCS data from over 225,000 Service members with unit-level randomized survey registrations, leveraging demographic and organizational covariates such as pay grade and service branch. Results showed confidential survey administration yielded slightly higher ratings on protective factors and lower ratings on risk factors, with statistically significant but small differences. Sensitive items, such as self-reported connectedness, exhibited relatively larger effects. The methodological contribution of AIPW lies in its ability to balance compositional covariate distributions between treatment groups even

when the groups were already randomized at the unit-level registrations, thereby ensuring towards robust causal estimates in complex organizational data structures and fielding scenarios. This approach aligns with best practices in federal statistics by addressing quasi-randomized treatment assignments with potential unmeasured confounding, offering a scalable framework for causal analysis in practical applications including scenarios-based survey designs and policy evaluations under varying conditions. Future research could explore its integration into broader federal data analytics and application in contingent scenarios with relevant confounding measures, further enhancing the accuracy, reliability, relevance, and timeliness of statistical insights and decision-making analytics.

Weighting Methods for Subannual Survey Estimates

Jonathan Eggleston, *U.S. Census Bureau*

In the U.S. federal statistical system, many surveys collect data continuously throughout the year but release estimates only on a quarterly or annual basis. More frequent releases could provide timely statistics to the public without the added cost of launching a new survey. For such potential subannual estimates, one important design feature is the weighting approach for creating subannual estimates. This talk uses the American Community Survey as an example to illustrate how a weighting algorithm designed for annual or multi-year estimates can be adapted to generate monthly estimates from continuously collected data. I will describe how the weighting approach incorporates administrative data to both reduce nonresponse bias and improve the precision of monthly estimates. Additionally, I will give a high-level overview on how the weighting algorithm is able to efficiently use a large number of weighting inputs to achieve these objectives. These results will show how surveys originally designed for annual releases could be retooled in the future to produce more frequent estimates, even without making other survey design changes (such as altering sampling rates) that might compromise the quality of established annual products.

Computing Mortality-Adjusted Weights for the NLSY79

Jonathan Mendelson, *Bureau of Labor Statistics*

Keenan Dworak-Fisher, *Bureau of Labor Statistics*

Rosella Gardecki, *The Ohio State University*

The National Longitudinal Survey of Youth 1979 cohort (NLSY79) follows 12,686 American youth born from 1957–1964 who completed the base year (Round 1) survey in 1979. NLSY79 tracks the cohort's labor market experiences over several decades, starting from ages 14–22. Cross-sectional weights are computed every round that account for round-specific nonresponse via a poststratification adjustment, which adjusts the weights of that round's respondents to population benchmarks from the base year. However, as the cohort has aged, panel attrition from mortality has gradually emerged as an issue, reducing the comparability between the surveyed population and the population benchmarks.

Our paper provides a new, mortality-adjusted weighting methodology to computing cross-sectional weights in the NLSY79. The main change is in the poststratification step, wherein we adjust the population benchmarks to reflect the portion of the 1979 youth population that is still living as of the given survey round. We do so in a manner that makes use of information on mortality collected during each round's data collection, in conjunction with survey design information, to adjust the original benchmarks for mortality. The approach also accounts for reductions to the NLSY79 sample that were made in 1985 and 1991, while also adapting other nuances of the existing weighting approach (e.g., cell collapsing criteria) in a manner that aims to avoid unnecessary changes.

In addition to reviewing historical weighting procedures and providing our new approach, we will also discuss implications for data users, which is a sizable group considering that NLSY79 data have been used in at least 3,000 journal articles and thousands of other publications to date. We will also discuss the applicability of the overarching approach to subsequent NLSY cohorts and to other longitudinal surveys.

Fast Method for Dual System Estimation of Population Size under the Log-Linear Model

Zhiyuan Lu, *U.S. Census Bureau*

The dual-system estimation (DSE) is heavily used in past and present Census Bureau coverage studies. With the use of DSE, it is important to implement methods to infer the population sizes among those with missing data from one or both data sources. The use of log-linear models, with calculations performed through an EM algorithm, promises a way for providing a complete set of estimates, as displayed in Van Der Heijden et al. (2022). Unfortunately, the EM algorithm demonstrates poor scalability when multiple demographic factors are analyzed. This paper discusses an alternative method to calculating the log-linear estimates that addresses this computational bottleneck. The alternative method has been applied to multiple data-sets, yielding similar results to the EM algorithm while requiring 2000 times less computing time.

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Thursday, March 12, 2026

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Session F-8: Improving Efficiency and Tackling Rising Survey Costs

Organizers: George R. Carter III, *Dept. of Housing and Urban Development* & Jennifer M. Ortman, *U.S. Census Bureau*

Chair: Jennifer M. Ortman, *U.S. Census Bureau*

Discussant: Stephanie Coffey, *U.S. Census Bureau*

BLS Auditorium C

Analyzing the Impact of Resource Optimization Strategies on Costs and Data Quality

Graham Jones, *Bureau of Labor Statistics*

Noell Koehlinger, *Bureau of Labor Statistics*

Brian T. Nix, *Bureau of Labor Statistics*

The Consumer Expenditure (CE) Survey is a nationwide household survey program sponsored by the Bureau of Labor Statistics (BLS) and fielded by the U.S. Census Bureau. To facilitate fielding of the CE Surveys, the BLS contracts with the Census Bureau to collect the survey data and deliver them back to the BLS for processing and publication. In recent years, rising CE Survey data collection costs have necessitated the implementation of cost-reduction measures on the part of the BLS and Census in FY2022 and in FY2024. In FY2022 the BLS and Census implemented a survey cost-reduction measure that restricted interviewers from returning to households for Wave 2 - 4 interviews that were previously incomplete (Type-A) interviews in August and September of 2022, and this measure was effective for keeping the costs within the budget. The success of this measure informed the development of the FY2024 cost-reduction strategy which covers the full 2024 fiscal year and impacts both the Interview and Diary Survey. The first of the FY2024 measures restricts interviewers from returning to households for Wave 4 interviews that were previously incomplete (Type-A) interviews in all previous waves, the second puts a cap on the number of unsuccessful contact attempts that could be made by an interviewer for a single case, and the third put a cap on the average number of hours interviewers worked per case.

This work will analyze the impact of the first two measures on data collection costs, using contact attempts and mode as a proxy for cost, as well as the impact on CE Survey response rates and data quality. To accomplish this, we will be making comparisons between relevant estimates (i.e., average contact attempts per case, response rates, sample composition, total expenditures, edit rates, and use of recall aides) before and after the implementation of the cost-reduction measures.

Using Cognitive Interviews to Inform the Addition of a Web Mode to the National Health Interview Survey

Eric Stone, *U.S. Census Bureau*

Jonathan Katz, *U.S. Census Bureau*

Kristen Kohm, *U.S. Census Bureau*

Jennifer Pace, *U.S. Census Bureau*

Ian Le, *U.S. Census Bureau*

Caitlyn Keeve, *U.S. Census Bureau*

Kathleen Kephart, *U.S. Census Bureau*

Benjamin Zablotsky, *National Center for Health Statistics*

Grace Medley, *National Center for Health Statistics*

Jessica Jones, *National Center for Health Statistics*

Federal agencies continue to face concerns about declining response rates, availability of interviewers, and rising costs associated with respondent recruitment. Therefore, research is needed to understand how different modes of administration may benefit federal data collection efforts experiencing these challenges. Adding a web mode may assist federal data collection efforts to address these challenges – especially given data showing that Americans' Internet usage has risen drastically over the past two decades.

The National Health Interview Survey (NHIS), conducted by the National Center for Health Statistics (NCHS), is exploring the potential use of multi-mode data collection, including the addition of a web mode. Currently, the NHIS is an interviewer-administered survey with a national probability sample of people in the United States living in noninstitutionalized housing. However, the complexity of question topics included in the NHIS (e.g., health insurance, chronic conditions) presents concerns about respondents' ability to understand and willingness to respond to a web mode – especially without an interviewer present to help navigate the questionnaire and administer read-if-necessary help text. Research that evaluates these concerns is paramount to informing the successful addition of a web mode to an interviewer-administered survey such as the NHIS.

The Census Bureau conducted two rounds of cognitive testing of web-based adaptations of select questions from the NHIS. This project involves cognitive interviews with a total of 40 participants recruited from the general population who answered web-based adaptations of NHIS questions and provided feedback on the web instrument. In this presentation, we will present: 1.) question-specific findings on the success of adapting core NHIS content to a web mode; 2.) general findings on how testing participants perceive the web mode; and 3.) recommendations for adapting federal surveys to a web format. In doing so, this presentation will demonstrate the value of inter-agency collaborations for innovating federal surveys.

Predicting Unoccupied Addresses for the 2030 Census

Andrew Keller, *U.S. Census Bureau*

For the 2020 Census, the Census Bureau implemented models using administrative records (AR) to identify addresses with a high likelihood of an unoccupied status during the nonresponse followup (NRFU) operation. These addresses received reduced field contacts. If no response was received for these addresses, they were enumerated as vacant or non-existent (i.e., the address did not exist as a housing unit on Census Day). In total, AR data helped to enumerate 14.51 percent of addresses as vacant and 3.59 percent as non-existent during the NRFU operation.

The purpose of this analysis is twofold. First, we look at scenarios in which addresses with a high likelihood of being unoccupied were assigned an occupied status during the NRFU operation. We attempt to understand why these differences existed. Integrating that analysis, we seek to improve the modeling methodology in preparation for the 2030 Census. As part of that process, we perform model refinements by incorporating new covariates and changing the modeling framework to obtain better performance.

From Biennial to Continuous: Motivations and Implications of Methodological Changes to the American Housing Survey

George R. Carter III, *Dept. of Housing and Urban Development*

Peter Mateyka, *Dept. of Housing and Urban Development*

Jason Bladen, *Dept. of Housing and Urban Development*

Samuel Risley, *Dept. of Housing and Urban Development*

The American Housing Survey (AHS) is the most comprehensive national survey of housing in the United States. Since 1973, the AHS, sponsored by the U.S. Department of Housing and Urban Development (HUD) and collected by the U.S. Census Bureau, has provided policymakers, business leaders, researchers and the public data on housing characteristics, housing costs, housing quality, and the demographic characteristics of households. The AHS is a longitudinal survey of housing units, following a representative sample of housing units over time. The survey began as the Annual Housing Survey and shifted to biennial data collection in 1983, changing its name to the American Housing Survey. In January 2026, HUD will launch a new AHS panel and shift to continuous data collection. This shift is in response to declining response rates and increasing data collection costs over the last decade in the context of constrained survey budgets. With the continuous data collection approach HUD aims to reduce data collection costs by reducing large ramp up costs required every two years to hire a new AHS workforce. The continuous approach will establish a workforce of experienced AHS interviewers who can conduct higher quality interviews and increase respondent cooperation with less money. The shift to continuous data collection will increase the frequency of AHS estimates as HUD plans to release both 1-year and 2-year estimates allowing for more current estimates and increased comparability to data sources that collect annual data like the American Community Survey and the Current Population Survey – Housing Vacancy Survey. The presentation will review the causes of declining response rates and increasing data collection costs for household surveys and methods for addressing these challenges. The presentation will also explore the implications of and opportunities presented by methodological changes for data products, including tabular data, microdata, and regularly produced research reports.

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Thursday, March 12, 2026

10:15 AM

Session F-X: Welcome to the Federal Statistical Community: Mentorship and Networking for New Federal Staff

Organizer: Jenny Hunter Childs, *U.S. Census Bureau*
Chair: Jenny Hunter Childs, *U.S. Census Bureau*
Census Training Room 3 (T-3)

Beginning a career in the federal statistical system can be both exciting and complex. This open session is designed to welcome new federal staff to the FCSM community and provide practical guidance on building connections, finding mentors, and navigating opportunities across agencies. We will highlight the annual Interagency Council on Statistical Policy (ICSP) Mentoring Program, which pairs mentors and mentees across agencies to support career growth and strengthen interagency connections. The session will then shift to an interactive discussion focused on informal networking: how to find mentors, build cross-agency relationships, make the most of conferences like FCSM, and move from attendee to contributor. New federal staff, early-career professionals, and seasoned leaders who want to connect with the next generation are all encouraged to attend.

Join us to build relationships, gain insight, and strengthen your place in the federal statistical community.

Thursday, March 12, 2026

1:15 PM

Session G-1: Practical Approaches to Data Linkage and Disclosure Avoidance

Organizer: Brandon Kopp, *Bureau of Labor Statistics*

Chair: Elan Segarra, *Bureau of Labor Statistics*

Census Auditorium 1

A Semi-Automated PII Detection, Evaluation, and Redaction Pipeline for the NAMCS HC Component Laboratory Data

Rachel Fan, *National Center for Health Statistics*

Priyam Patel, *National Center for Health Statistics*

Zachary Peters, *National Center for Health Statistics*

Lello Guluma, *National Center for Health Statistics*

The National Center for Health Statistics conducts the National Ambulatory Medical Care Survey Health Center (NAMCS HC) Component, a nationally representative establishment survey that collects healthcare data from federally qualified health centers and look-alikes. This survey collects visit-level electronic health record (EHR) data from millions of encounters. However, some records contain personally identifiable information (PII) embedded within non-PII fields, such as in laboratory results fields, which should only include laboratory test result values and units. At the scale of this survey, manually reviewing these fields to remove PII to sufficiently reduce disclosure risk and enable their inclusion in the associated restricted-use files remains a challenge.

This analysis examines the potential of machine learning and natural language processing techniques to effectively identify and redact PII in laboratory result fields from 2023 NAMCS HC. We explored several methods as we developed and assessed a semi-automated pipeline for PII detection, evaluation, and redaction. These findings can help enable the application of these innovative methods to additional EHR survey data, significantly increasing the availability of de-identified healthcare data for research and analysis.

The People Behind the Enterprises: Linking Employer Businesses, Owners and their Characteristics Using Administrative Records

Adela Luque, *U.S. Census Bureau*

Valeska Araujo, *George Mason University*

John Earle, *U.S. Census Bureau*

Lorenz Ekerdt, *U.S. Census Bureau*

Vitaliy Novik, *U.S. Census Bureau*

Nick LaBerge, *U.S. Census Bureau*

Jared Wold, *George Mason University & Census Bureau*

Sammy Young, *Arizona State University & Census Bureau*

James Noon, *U.S. Census Bureau*

In response to declining response rates, starting in 2020 the Census Bureau began providing nonemployer demographics not through a survey, but a program that leverages existing administrative and census records to identify the business owner universe and their characteristics: the annual Nonemployer Statistics by Demographics series (NES-D). Following the successful transition from surveying nonemployers to the adrec-based NES-D, a related initiative for employers, known as the Employer Characteristics Project (EC), has been underway. While the methodology underlying EC is related to that of NES-D, employer businesses introduce novel challenges due to their more complex organizational structures. In this presentation we will cover our methodology, challenges, and comparisons of administrative record based demographics to those from surveys.

Beyond unburdening surveys and lowering costs, we expect the Employer Characteristics data to enable substantial advances in business research. The linkage between businesses and business owners will allow researchers to examine the outcomes of owners of exiting businesses, the role of owner characteristic in firm ex ante heterogeneity, and more.

Writing History with Data: Multi-Method Analysis of the Impact of Relocation on American Households

Nathaniel Reynolds, *Library of Congress*

The Federal Research Division (FRD) of the Library of Congress recently completed a national mixed-method research study for the Federal Highway Administration Office of Real Estate Services. The study focused on the impacts of federally-funded transportation projects on the quality of life of household residents who were relocated as a result. The original methodology included data collection through surveys and interviews of individual households, as well as a data analysis of relocation records and demographic datasets. These datasets were used to compare characteristics of displacement and replacement communities. The data included survey statistics from the U.S. Census Bureau and quality of life indexes compiled by government, private industry, and non-profit organizations. The focus of this presentation will be on the integrated use of these resources to compare characteristics of neighborhoods in the United States, requiring creative data cleaning and linkage techniques to match statistics aggregated at various geographic levels to administrative records of differential quality and consistency. The results of this study will inform future federal relocation projects, and also set a standard for innovative analyses of administrative data generated from government interactions with the American public.

The Experience of a Small Federal Statistical Agency with Unique Data Confidentiality Challenges

Clara Reschovsky, *Bureau of Transportation Statistics*

The Bureau of Transportation Statistics (BTS) is a small federal statistical agency with a variety of data collections that are very different from large traditional surveys, requiring different approaches to disclosure avoidance. A traditional Disclosure Review Board made up of staff is not practical in a small agency with disparate programs. Many of our current confidential data collections are part of a Precursor Safety Data Program for close-call safety events to improve safety as well as industry data collections with daily automated data collections and products. BTS also has confidential big-data sets that require disclosure avoidance analysis before products are released. With one Confidentiality Officer, it is necessary to draw on each programmatic area to work collaboratively and potentially interact with industry stakeholders to determine the best path forward to produce releasable products. Many of the challenges faced include industries with very few members which makes it difficult to maintain anonymity as well as challenges with having enough staff and time to perform sufficient data checks. An advantage of being a small agency is that the Confidentiality Officer is able to work collaboratively with program staff to figure out workable strategies since the process can be iterative, rather than prescriptive before the data is processed. Some of the challenges BTS faces are shared with all the other federal statistical agencies, and others are unique due to the types of data collections. As non-survey data collections increase, it will be incumbent on all federal statistical agencies to determine best practices to avoid disclosure.

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Session G-2: Taming the Wild Text: NLP Models for Text Classification

Organizer: Matthew Chambers, *Bureau of Economic Analysis*

Chair: Matthew Chambers, *Bureau of Economic Analysis*

Census Auditorium 2

Improving medications data quality through natural language processing

Nikki Adams, *National Center for Health Statistics*

Rihem Badwe, *National Center for Health Statistics*

Lello Guluma, *National Center for Health Statistics*

Zachary Peters, *National Center for Health Statistics*

Ashley Shi, *National Center for Health Statistics*

The National Hospital Care Survey (NHCS) and the National Ambulatory Medical Care Survey Health Center (NAMCS HC) Component are two nationally representative establishment surveys that collect health care data from hospitals and federally qualified health centers (FQHCs), respectively.

Medication data in these surveys sometimes come with no medication code and only a descriptive label. To allow more standardized medication searches, researchers normalized the data by filling in RxNorm codes based on their descriptive label. Natural language processing (NLP) techniques were used to find the best match between the survey medications labels and a standardized list of medication descriptions from the Unified Medical Language System. Two methods of finding the best match were used: the Jaro-Winkler distance and Jaccard similarity. Each method produces a score. For each method, we determined, through stratified sampling, what is the lowest score at which we can accept the proposed match. Separately, we also compared overall accuracy of the methods to each other by applying both to the same sampled dataset and having a clinician determine if the proposed matches were correct.

This presentation will highlight how NLP algorithms can be utilized to improve data quality and its potential use in other federal statistical systems. Additionally, presenters will discuss lessons learned, limitations, and areas for improvement.

Evaluating Natural Language Processing (NLP) Models for Communication related safety events.

Sreedevi (Sree) Thiagarajan, *Veteran Affairs Administration*

Effective communication is critical in hospitals. Despite stringent protocols, communication errors remain prevalent in hospitals, often leading to adverse outcomes. This study explores the development of a Natural language processing model like BERT (Bidirectional Encoder Representations from Transformers) model tailored to identify and classify communication errors in hospital settings.

Our model was trained on diverse datasets, encompassing event report summaries, and annotations curated from other articles. We employed a suite of natural language processing techniques to preprocess and annotate the data, incorporating unsupervised pretraining on a vast dataset of event summaries followed by supervised fine-tuning on manually annotated communication error instances. Evaluation metrics were rigorously defined to measure the model's precision, recall, and F1-score against established benchmarks, revealing marked improvements in identifying and classifying a wide spectrum of communication errors.

Application of Text Classification Techniques

Abigail Gaudinier, *Statistics of Income*

Jacob Dailey, *Statistics of Income*

This study explores the application of text classification techniques, including FastText and BERT models, to automate the allocation of corporate balance sheet items into standardized reporting categories. The current process relies heavily on manual review by SOI tax editors in the Ogden Service Center, thereby streamlining the existing parcel allocation workflow. Although the work is still in progress, initial findings demonstrate strong potential for operational efficiency gains and consistency improvements. Moreover, this research

establishes the methodological foundation for extending similar automated classification approaches to other domains, including categorizing sale of capital asset transactions reported on Form 1065 into appropriate asset classifications.

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Thursday, March 12, 2026

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Session G-3: Using Proprietary Data to Address Information Gaps in Food Statistics

Organizer: Abigail Okrent, *Economic Research Service*

Chair: Elina T. Page, *Economic Research Service*

Census Auditorium 3

Has consumer food demand become more price sensitive? A case study of beverages using retail- and household-based scanner data

Patrick McLaughlin, *Economic Research Service*

Abigail Okrent, *Economic Research Service*

Despite real (i.e., inflation earnings) earnings of American households increasing after economic relief programs concerning the pandemic expired, consumers reported significant dissatisfaction with the economy in part due to record high increases in food prices. To help resolve this puzzle, this paper explores the question of whether consumers have become more sensitive to price (i.e., have a price elasticity of demand) in the post-pandemic era. We use “real time” proprietary scanner data acquired from Circana on household purchases and retail sales of non-alcoholic beverages to estimate changes in the price elasticity of demand for these products, as well as the propensity to substitute to cheaper brands like private labels or store brands. We demonstrate the value of “real time” proprietary scanner data in fast-turnaround economic analyses for policy stakeholders’ timely informational needs.

State-Level Food Expenditure Estimates: Adjusting for Out-of-State Visitors and Commuters

Eliana Zeballos, *Economic Research Service*

Wilson Sinclair, *Economic Research Service*

This presentation expands the Economic Research Service’s State-level Food Expenditure Series (FES) by adding a complementary, State resident only food sales to the existing total food sales that captures both resident and nonresident expenditures. The current State-level FES captures the total value of all food sales occurring within a State, regardless of whether the purchaser is a resident, commuter, or tourist. These estimates are essential for understanding the scale and composition of a State’s food economy. However, non-resident tourists and commuters can account for a substantial proportion of some State’s food spending, masking resident food spending patterns and thus, may not be a good measure of economic welfare of a State’s residents. Using proprietary anonymized mobile location data, we estimate the share of grocery and restaurant visits made by non-residents and adjust the State-level FES per capita estimates accordingly. The result is a new set of resident-based consumption measures that, alongside the original total sales figures provide a more complete picture of State-level food demand—supporting more informed decision-making by public agencies, researchers, and industry stakeholders.

Differences in Rural and Urban Food Price and Purchasing Patterns

Scott Okrent, *Federal Reserve Board of Governors*

The Consumer Price Index provides timely information on temporal food price variation for urban areas across the United States. However, little is known about food prices in rural areas and whether rural areas face higher prices than urban areas. This research uses receipt data collected from Numerator from 2019 to 2024, and the Rural-Urban Continuum Codes produced by the Economic Research Service to compare food prices by urbanicity. Preliminary examination shows that prices are not systematically higher based on urbanicity and vary by geography and food product.

How were Extra Benefits Spent?

Sinem Hacıoğlu Hoke, *Federal Reserve Board of Governors*

This paper examines the spending effects of a pandemic-related change to the Supplemental Nutrition Assistance Program (SNAP), one of the largest U.S. aid programs. SNAP recipients received emergency allotments that significantly increased their monthly grocery funds in response to the pandemic. Using novel proprietary spending data that captures households' goods spending in full (Numerator), I show that emergency allotments increased SNAP recipients' monthly food-at-home spending by approximately \$100 compared to non-recipients. The estimated Marginal Propensity to Consume food out of extra benefits is approximately 0.5, which is larger than cash. SNAP recipients increased spending on products with longer shelf lives and higher nutrition, which might otherwise be less affordable. However they also increased spending on beverages and snacks. Various non-food goods spending measures show little to no change. I support these empirical findings with a household survey to provide a comprehensive understanding of how extra SNAP benefits influenced spending patterns during the pandemic.

U.S. Food Assistance Participation and Demand for Food

Abigail Okrent, *Economic Research Service*

Patrick McLaughlin, *Economic Research Service*

Many U.S. policymakers have implemented or introduced fiscal policies with the aim to encourage healthy eating behaviors especially among lower income households and those participating in food assistance programs. Prior research on demand for food for lower income or food assistance participants is sparse and based on older, less timely public-use surveys like the 2012-13 FoodAPS or 1996-97 National Food Stamp Program Survey. This research extends the previous literature by examining demand for foods by households on SNAP and those that are not after the COVID-19 pandemic. Because the pandemic followed by the historic food price inflation may have had long-lasting impacts on consumer preferences for food their price responsiveness, we examine demand during post pandemic, and exploit the longitudinal nature of the 2021-23 household scanner data, which previous studies had not done before. To do this, we develop longitudinal weights for the data, which allows us to examine the effects of within-household price variation on food demand, controlling for unobserved time-invariant preferences for food. We find that the longitudinal weights affect statistical significance of estimates although the effect on the magnitude of the estimates is economically small.

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Session G-4: Pilot Studies for Modernization: Methods, Findings, and Lessons Learned

Organizer: Julie Parker, *Economic Research Service*

Chair: Art Hughes, *National Center for Science and Engineering Statistics*

Census Conference Rooms 1-2

Evaluation of the Health Care Surveys (HCS) Fast Healthcare Interoperability Resource (FHIR) Pilot

Prachi Mehta, *National Center for Health Statistics*

Carol DeFrances, *National Center for Health Statistics*

Josephine M. Alford, *National Center for Health Statistics*

Danielle A. Davis, *National Center for Health Statistics*

Lello Guluma, *National Center for Health Statistics*

Joy E. Onukwufor, *National Center for Health Statistics*

Michael Seo, *Code360*

Juncong Shi, *National Center for Health Statistics*

Nicole Cummings, *National Center for Health Statistics*

The National Health Care Surveys, conducted by the National Center for Health Statistics (NCHS), are designed to answer questions of interest to healthcare policymakers, public health professionals, and researchers. The maturation of standards such as Fast Healthcare Interoperability Resource (FHIR), the U.S Core Data for Interoperability, and requirements for the certification of health information technology by the Assistant Secretary for Technology Policy have created an environment to develop scalable and reusable interoperable solutions addressing challenges related to electronic health record (EHR) data quality, timeliness, and process inefficiencies in public health reporting.

In response to these factors and building upon existing FHIR standards, CDC developed a flexible and interoperable reference architecture (RA) to enable efficient sharing of clinical data with public health agencies. The National Health Care Surveys piloted the application of this RA with a focus on two surveys that collect encounter-level EHR data in ambulatory, in-patient, and emergency department settings. The purpose of this presentation is to discuss the results of this pilot project that will inform future data modernization efforts for the National Health Care Surveys.

The technical methodology discussed will describe the relevant integrated components of the RA to facilitate real-time automated reporting. The data analytics methodology presented will focus on a mixed-method approach using both qualitative (e.g., interviews) and quantitative data (e.g., EHR data) to evaluate the difference in impact of the current reporting approach vs. using FHIR.

The lesson learned from the evaluation depict an innovative technical approach that leverages modern-day standards and infrastructure to facilitate the automation of data collection and reporting efforts from clinical care providers to public health agencies. This approach has the potential to improve current legacy standards and eliminate manual processes so that participation burden is decreased for health care providers when reporting National Health Care Surveys data.

Assessing the 2022 National Training, Education, and Workforce (NTEWS) Pilot Data to Inform Survey Modernization Efforts

Gigi Jones, *National Center for Science & Engineering Statistics*

Danielle Taylor, *National Center for Science & Engineering Statistics*

Ruiyi Li, *National Center for Science and Engineering Statistics*

Lisa Mirel, *National Center for Science and Engineering Statistics*

Vrinda Nair, *National Center for Science and Engineering Statistics*

In 2022, the National Center for Science and Engineering Statistics (NCSES) conducted a pilot study to capture critical information on the skilled technical workforce (STW)—a rare but critical subset of the U.S. Science and

Engineering (S&E) workforce. STW includes workers in S&E-related occupations who use substantial technical expertise but do not hold a bachelor's degree or higher. The pilot aimed to produce key estimates related to employment status, occupation, educational background, field of degree, work credentials, training or experience, and demographic characteristics.

As the first administration of the NTEWS, the survey was launched as a pilot, and the resulting data are considered experimental. Nonetheless, the pilot's data are valuable for informing future survey integration efforts as part of the Center's broader survey modernization work.

This assessment evaluated the pilot data by comparing key estimates in the NTEWS to other national surveys, analyzing effective sample size of key estimates based on the replicate weight construction, and measuring design effects of unequal weighting due to sample design and non-response adjustment. Results informed recommendations for sample size, stratification and revisions to data collection instruments (paper, web, and computer-assisted telephone interview). Additional analyses—covering data quality, paradata from web instrument, non-sampling bias, and write-in autocoding—guided further refinements to question wording and cognitive testing.

In sum, this presentation will highlight key findings from the assessment. It also provides a possible model for how federal statistical agencies can leverage pilot data to guide innovation and modernization in future data collections. For example, NCSES aims to use insights from the pilot's assessment to inform its workforce survey integration effort, which seeks to combine three existing national workforce survey—each targeting different workforce subpopulations—and incorporate linkages to alternative data sources.

SurveyLand: A Customizable Application for Streamlined Data Processing, Analysis, and Presentation—A Pilot with NCHS RANDS

Sarah Forrest, *National Center for Health Statistics*

Kristen Cibelli Hibben, *National Center for Health Statistics*

Paul Scanlon, *OpenAI, (former National Center for Health Statistics)*

Federal statistical agencies face growing demands to deliver timely, high- quality analyses of complex survey data to support decision-making. While advances in data science offer opportunities to enhance the efficiency, transparency, and reproducibility of survey data processes, agencies continue to face persistent challenges such as version control, non-standardized workflows, repetitive manual processing, and limited accessibility for non-technical users, all of which contribute to inefficiencies and errors. To address these challenges, we developed SurveyLand, a customizable and user-friendly application piloted with the National Center for Health Statistics' (NCHS) Research and Development Survey (RANDS). SurveyLand streamlines the end-to-end processing, analysis, and presentation of complex survey data, using open-source technologies (e.g., R, Shiny, Quarto, and git) and adhering to best practices in reproducibility and standardization. By automating frequently produced outputs and enforcing NCHS data presentation standards, the application empowers both technical and non-technical users to generate consistent, presentation-ready data tables and figures for dissemination to internal or external audiences. In this presentation we will demonstrate the key features of the tool (e.g., filtering, weighting, clustering, and customizable output options), focusing on how it enables a streamlined, reproduceable approach to dataprocessing, analysis, and presentation. The pilot implementation with NCHS RANDS demonstrates how SurveyLand can boost productivity and the timely communication of survey findings, while also making data insights more accessible. It is also adaptable to other survey products and agencies, supporting broader adoption of increased efficiency and best practices across the Federal Statistical System. As a scalable proof-of-concept, the application supports ongoing data modernization initiatives in federal agencies.

Toward a timber asset account for the United States: A discount price approach

Scott Wentland, *Bureau of Economic Analysis*

Sonia R. Bruck, *U.S. Forest Service*

Christopher Mihlar, *U.S. Forest Service*

Bin Mei, *Duke University*

Thomas Brandeis, *U.S. Forest Service*

Matthew Chambers, *Bureau of Economic Analysis*

Julie L. Hass, *Bureau of Economic Analysis*
Travis Warziniack, *U.S. Forest Service*

Timber is an important natural resource supporting the U.S. economy, yet it remains underrepresented in national economic accounts. This paper introduces a novel methodology which we call the discount timber price method to value pre-merchantable timber (commonly excluded from timber asset accounts) in a manner consistent with the System of Environmental-Economic Accounting and the System of National Accounts (SNA), updated in 2025. Unlike traditional approaches that rely on detailed age-yield curves or land valuation, our method uses observed diameter at breast height, species-specific yield data, and market stumpage prices to estimate the discounted value of immature timber. This approach enables the inclusion of pre-merchantable timber as a work-in-progress inventory asset, enhancing comparability with other asset classes on the national balance sheet and aligning with international recommendations on the treatment of biological resources in the accounts.

We demonstrate the method through a pilot timber asset account for the state of Georgia, focusing on southern yellow pine species. Using Forest Inventory and Analysis microdata, Forest Vegetation Simulator yield projections, and TimberMart South price data, we estimate the total value of Georgia's standing softwood timber at approximately \$5.9 billion in 2023. Of this, \$244 million is attributed to pre-merchantable timber. Our method is conservative, tractable, and scalable. It contributes to the growing body of U.S. natural resource pilot accounts and provides a framework for integrating timber into the national accounts, supporting the transition to the SNA 2025. Future extensions to other forest types and regions will support more comprehensive and policy-relevant assessments of U.S. natural timber resource wealth

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Thursday, March 12, 2026

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Session G-6: Indigenous Data and Tribal Sovereignty: Best Practices from Across the Federal Government

Organizer: Carla Medalia, Dee Alexander, and Patricia Goerman, *U.S. Census Bureau*

Chair: Sallie Keller, *U.S. Census Bureau*

BLS Auditorium A

U.S. Census Bureau Tribal Consultation and Coordination Best Practices

Dee Alexander, *U.S. Census Bureau*

Melissa Bruce, *U.S. Census Bureau*

The Census Bureau has a long history of working with tribal governments to collect data and produce valuable statistics to Tribal Nations. The Census Bureau also works with other federal agencies in acquiring administrative records and collaborating on projects where tribal governments have been consulted.

Executive Order No. 13175, "Consultation and Coordination with Indian Tribal Governments," (11/6/2000) requires federal agencies to have an accountable process to ensure meaningful and timely input by tribal officials in developing policies that have tribal implications. The Census Bureau recognizes the unique legal relationship as established in the Constitution, statutes, treaties and federal court decisions, between tribal governments and the federal government.

This talk will highlight the Census Bureau's tribal consultation practices—ensuring tribal sovereignty is recognized upfront in our work in improving the collection of tribal data, creating new data sources, and producing new data-driven research projects with tribes. We will discuss the Census Bureau's American and Alaska Native (AIAN) policy and the Department of Commerce's Consultation and Coordination Policy, which outlines the principles to be followed in all Census Bureau interactions with federally recognized AIAN tribal governments and reaffirms the unique government-to-government relationship that exists between AIAN tribal governments and the Census Bureau.

“Tribal Sovereignty” refers to the legal recognition in U.S. law of the inherent sovereignty of American Indian Nations. The law applied to American Indians by the U.S. government is called Federal Indian Law. Although undermined by various Acts, legal decisions, and policies, it still stands to this day. Indian reservations are recognized as “nations within.” Each has its own government and sovereign powers to make laws, tax, etc., with most having their own tribal justice system, based upon their inherent powers.

Sovereignty in Collaboration: Tribal Data-Sharing with the U.S. Census Bureau

Shadana Sultan, *U.S. Census Bureau*

Vicki McIntire, *U.S. Census Bureau*

Meagan Tydings, *U.S. Census Bureau*

Annabelle Pham, *U.S. Census Bureau*

In 2023, the U.S. Census Bureau began exploring opportunities to strengthen data quality and coverage by partnering with Tribal Nations to share administrative records. While traditional census operations relied on self-collected data, administrative records from other sources help address low-response rates, better represent historically undercounted populations, and provide additional sources for comparing and validating accuracy of data. Tribal Nations can benefit from data-sharing by working with the Census Bureau to address data processing and analytical needs that support a deeper understanding of their own communities.

Each Tribal community is unique, and no two administrative datasets are alike, and so, the Census Bureau embraces flexible and collaborative approaches that prioritize their needs. To date, two Tribal Nations have signed administrative record data-sharing agreements with the Census Bureau, laying the groundwork for protected partnerships that honor sovereignty and create shared value.

The session will walk through the steps involved in developing Tribal data-sharing agreements at the U.S. Census Bureau, demonstrate how data analysis can be conducted as a shared effort, highlight the early successes, share lessons learned thus far, and discuss future opportunities that showcase insights into Tribal populations. By working together, both the Census Bureau and Tribal Nations can pave the way for improved data accuracy and representation.

Respect for Tribal Sovereignty and Community-Engagement in Applied Research and Evaluation at the Administration for Children and Families (ACF)

Aleta Meyer, *Administration for Children and Families*

Alysia Blandon, *Administration for Children and Families*

The Administration for Children and Families (ACF) supports human services that promote health and well-being among children, families, and individuals in urban, rural, suburban, and Tribal communities across the United States. Within ACF, the Office of Planning Research and Evaluation (OPRE) partners with program offices to produce evidence and to help apply it. In our work with Native communities, OPRE strives to follow the principles of *The Roadmap for Collaborative and Effective Evaluation in Tribal Communities* (the Roadmap). The Roadmap serves as a call to action to change the narrative on federal activities related to data, research, and evaluation with Native communities. This new narrative is grounded in specific values (e.g., Community Engagement, Indigenous Ways of Knowing, Respect for Tribal Sovereignty, Strengths Focus, Cultural and Scientific Rigor, Knowledge Sharing, and Ethical Practices) and focuses on system improvement, bi-directional learning, and collaborative, culturally responsive evaluation.

To demonstrate how the Roadmap best practices can be applied by other federal agencies in their efforts to honor Tribal data sovereignty, we will provide concrete examples of how the values of Respect for Tribal Sovereignty and Community Engagement guide our approach with two on-going research projects:

- OPRE's partnership with ACF's Office of Head Start to conduct a nationally representative study describing the children, families, and programs in Region XI Head Start
- OPRE's partnership with ACF's Administration for Native Americans to design a national study of Native American language use in the United States.

We expect that by sharing the specific approaches we have enacted, such as protocols for Tribal approval of participation and data use and community work groups who provide input on each stage of the inquiry process, our colleagues across government can consider how to apply the Roadmap to their own work with Tribal communities.

Respectful Engagement with Tribal Nations on Data Privacy: Centering Sovereignty in Federal Data Practices

Crystal Moore, *Administration for Children and Families*

Aleta Meyer, *Administration for Children and Families*

This presentation introduces a practical guide developed by the Administration for Children and Families (ACF) to support respectful, lawful, and culturally appropriate data engagement with Tribal Nations. Grounded in the CARE Principles—Collective Benefit, Authority to Control, Responsibility, and Ethics—and aligned with Executive Order 14243, the guide outlines actionable strategies for federal staff to uphold Tribal data sovereignty while meeting open data and privacy mandates. The guide draws on the National Institutes of Health (NIH) definition of Indigenous Data Sovereignty (IDSov), which affirms the right of Indigenous peoples to control their data—including oral traditions, genomics, community health data, and more—across its collection, storage, and interpretation.

The session highlights key practices such as early engagement with Tribal leaders, informed consent, data minimization, and transparent communication. It emphasizes the importance of formal documentation (e.g., DSAs, DUAs, SORNs, as applicable), respectful language, and honoring Tribal governance structures. The guide also provides a framework for navigating legal requirements such as the Privacy Act and HHS Tribal Consultation Policy (TCP) – as a sample of a federal TCP.

By centering Tribal sovereignty and IDSov, this session supports a focus on a trust-based relationships that respect Indigenous Peoples and Tribal Nations rights and knowledge systems. It equips federal staff with tools to build equity and accountability in data stewardship with Native communities.

How spatial data of IHS is maintained in our medical records system

Kristina Rogers, *Indian Health Service*

This presentation will discuss how spatial data of the Indian Health Service (IHS) is maintained and utilized within our system. We will discuss the IHS Standard Code Book (SCB) that houses the standard data elements, the Reference Management Tool (RMT) that is used to update the SCB, the role of Area Statistical Officers working with Tribal nations to collect the spatial data and its use by demography staff.

Tribal Data in Practice at HUD

Jacqueline Bachand, *U.S. Department of Housing and Urban Development*

Elizabeth Rudd, *U.S. Department of Housing and Urban Development*

Blair Russell, *U.S. Department of Housing and Urban Development*

We will talk about how HUD's Office of Policy Development and Research (PD&R) uses Tribal data and engages Tribes in relation to these activities, touching on the roles of HUD's Office of Native American Programs, HUD's Tribal Intergovernmental Advisory Committee, and PD&R. We will cover collection of administrative data, uses of ACS data to support allocation of Indian Housing Block Grant funds, and data collected for HUD's most recent, major study of Indian housing published in 2017, the Assessment of American Indian, Alaska Native, and Native Hawaiian Housing Needs.

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Session G-7: From Frames to Estimates: Establishment Survey Sampling and Weighting

Organizer: Rachel Hansen, *Bureau of Justice Statistics*

Chair: Rachel Hansen, *Bureau of Justice Statistics*

BLS CTC 2A

Feasibility of Constructing a Home Care Sampling Frame for a National Federal Survey

Christine Caffrey, *National Center for Health Statistics*

Manisha Sengupta, *National Center for Health Statistics*

Jess Penn Lendon, *National Center for Health Statistics*

Priyanka Singh, *National Center for Health Statistics*

Shannon Kindilien, *National Center for Health Statistics*

This presentation will share findings from a National Center for Health Statistics (NCHS) project exploring the feasibility of constructing a home care sampling frame for the United States. Home care is a major state-regulated provider of non-medical long-term care services and supports (LTSS) for persons aging-in-place or with disabilities and chronic conditions. The support may include preparing meals, shopping, transportation, caregiver respite, socio-emotional support, and help with activities of daily living (ADLs). Since home care is not federally regulated, there is no central source for home care agencies to obtain a sampling frame for data collection or produce national-level home care data. NCHS' National Post-acute and Long-term Care (NPALS), a biennial survey of adult day services centers and residential care communities (RCCs), contacts various state agency representatives from each of the 50 states and District of Columbia to develop the RCC sampling frame. For the most recent NPALS RCC frame development, the interviews with state agency representatives also included questions asking about the availability of home care agency listings and details on the content in the listings (for example, agency name, physical address, license number). Findings suggest that building a state-based sampling frame for home care is feasible. Listings are available in a variety of ways, such as by contacting state agencies or by obtaining listings online, and include information typically available and needed in a sampling frame to conduct a national survey. NPALS will continue to examine frame development and eligibility criteria for home care agencies to potentially provide first ever source of federal home care data. This work would fill a gap in data sources covering LTSS settings to inform policy, research, and health care providers.

A Preliminary Study of Import and Export Price Indexes and Blended Variance Estimation using Census Trade Data

Daniel Yang, *Bureau of Labor Statistics*

The International Price Program (IPP) at the Bureau of Labor Statistics (BLS) produces Import and Export Price Indexes (MXPI) which consists of two components: import and export merchandise. MXPI replaced portions of the directly collected establishment survey data with Census Trade Data (CTD) starting in February 2025. CTD are administrative transaction shipment records, which includes information such as international Harmonized System (HS) product classification codes for the U.S., trade values in dollars, foreign country of import or export, shipment quantity, etc. CTD have the advantage of covering almost all U.S. trade within a month. However, the price that is captured by CTD is an average price of shipment which is different from the directly collected survey. MXPI developed a Bootstrap resampling algorithm on the directly collected survey data to evaluate the sampling error, and a Jackknife algorithm to try to address the propagation of error in CTD. In this study, we developed a Taylor Series Linearization variance approximation to reflect the variance of the blended MXPI from two data sources: an establishment survey and a census. We then explored the aggregated estimation of the blended index at the Stratum-Upper (SU) level (a higher level of aggregation of classification of products in the index calculation process) to address the covariance between Bootstrap and Jackknife replicates.

Impact of Imputation on the Variance of Estimates from the Employer Costs for Employee Compensation Program

Christopher J. Guciaro, *Bureau of Labor Statistics*

The Employer Costs for Employee Compensation (ECEC) Program estimates the mean hourly costs of wages and benefits using data from the National Compensation Survey (NCS) of the Bureau of Labor Statistics. The NCS is a sample of job quotes selected from a sample of establishments. Collected items include wages, benefit incidence, benefit costs, and hours worked. Item non-response is treated using imputation methods such as regression and random hot deck. Each imputation method relies on a donor pool of collected microdata and their sampling weights. Variance is estimated using Balanced Repeated Replication (BRR). Replicate estimates use the same formula as the full sample estimate, except the sampling weights vary by replicate. The imputed microdata, however, are left unchanged; that is, the full sample imputed values are re-used for each replicate. Yet imputation clearly contributes to the true variance because the donor pool and set of recipients both vary by sample, and for each sample, each random hot deck recipient has many candidate donors to choose from. It is important to evaluate how much the variance is underestimated by the current variance estimator. Hence, for this research, the imputation process is rerun for each replicate by letting the sampling weights in the donor pool vary by replicate. Also, for each hot deck recipient, a new, independent, donor is selected for each replicate. We compare these alternate variance estimates with the original variance estimates. As expected, a majority of the alternate variance estimates are greater than their originals.

Improving MPPS Sampling Design with PPSWOR: A New Approach for Agricultural Surveys

Yang Cheng, *National Agricultural Statistics Service*

Luca Sartore, *National Institute of Statistical Science/National Agricultural Statistics Service*

Eric Slud, *University of Maryland*

Denise A. Abreu, *National Agricultural Statistics Service*

National Agricultural Statistics Service (NASS) has employed the Multivariate Probability Proportional to Size (MPPS) sampling design since 1996. This approach enables more efficient allocation of sample units across variables, improving coverage, and cost effectiveness in multipurpose survey contexts. The traditional MPPS design consists of two steps: 1) computing the maximum Probability Proportional to Size (PPS) across all list frames, and 2) selecting a Poisson sample based on these inclusion probabilities. In this approach, the randomness of the sample size is a notable drawback that complicates survey planning and estimation. In this talk, we consider a modified design that replaces Poisson sampling by sampling Without Replacement (WOR). This alternative sampling scheme maintains the desirable properties of MPPS by ensuring a fixed sample size while improving operational control and consistency. Overall performance is assessed using data from multiple crop-specific list frames, supplemented by official statistics from the U.S. Census of Agriculture. Results show promising gains in efficiency and reliability.

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Session G-8: Optimizing contact strategies to increase response rates

Organizer: Jonathan Mendelson, *Bureau of Labor Statistics*

Chair: Jonathan Mendelson, *Bureau of Labor Statistics*

BLS Auditorium C

Automating Contact Research for Economic Survey Nonrespondents

Emily Wiley, *U.S. Census Bureau*

John Lombardi, *U.S. Census Bureau*

In the Census Bureau's economic programs, survey non-respondents are often contacted via telephone to remind them to respond. For some business establishments, contact information may be out-of-date or missing. Historically, clerks have done research to find the correct phone number for these non-respondents. Data scientists in the Economy-Wide Statistics Division developed a machine learning and web scraping pipeline to provide phone numbers via the Google API. With an establishment's name and approximate GPS coordinates, we were able to extract phone numbers, validate the phone information, and use record linkage to match results from Google to a sampling frame. The pipeline has been implemented in a number of economic programs, including the 2022 Economic Census and the 2024 Annual Business Survey. This process has saved thousands of hours of time and hundreds of thousands of dollars in cost. Our work demonstrates that opportunities exist for using data science techniques that can yield significant cost savings for field operations.

Small-Scale Testing, Big Impact: Innovative Approaches to Boosting Survey Self-Response Rates in Federal Data Collection

Anita Shagnea, *U.S. Census Bureau*

As survey response rates decline, threatening data quality and increasing costs, researchers at the U.S. Census Bureau continue to be on the forefront, testing new contact strategies such as the use of Priority Mail® envelopes, mailing dates, and Quick-Response (QR) codes in a shorter-term, cost-effective manner. The Small-Scale Response Testing (SmaRT) program uses nationally representative samples designed to pre-test new strategies and improve existing ones before including them in a major field test or the 2030 Census.

The three iterations of SmaRT this decade have leveraged the flexible SmaRT experimental design using a mixed-mode survey – which includes an internet instrument and a paper questionnaire – to gain insight into how people engage with the online survey and mail materials, as well as ways to assist responders and increase self-response. We present the experimental design, sampling methods, and results from treatments such as Priority Mail® envelopes, mailing dates on materials, and QR codes on self-response rates, as well as the specific mode of response.

A Focused Initiative to Optimize Mailing Strategies in a Survey: Sending Every Door Direct Mail to Potentially Boost Self-Response.

Ivan J Cardona-Tomassini, *U.S. Census Bureau*

Karlesha G. LeGrier, *U.S. Census Bureau*

Faraz Shaikh, *U.S. Census Bureau*

Theodore Charm, *U.S. Census Bureau*

Every Door Direct Mail, commonly referred to as EDDM, is a service offered by the United States Postal Service (USPS) that delivers non-addressed mail pieces to a specified geographic area (e.g., ZIP Code, mail route). Ideally, the EDDM reaches all addresses on a mail route, even if addresses are unknown to the Census Bureau. Current uses of EDDM were explored to determine the effect on self-response rates and estimate mailing costs, as well as expand best practices for the mail piece design and distribution. Three EDDM pieces that were sent during the 2020 Census were evaluated: (1) an EDDM to tracts with a higher likelihood to have housing units with young children, (2) an EDDM delivered on Census Day to housing units in sampled USPS mail routes, and (3) an EDDM to areas where PO Boxes were the only mailing addresses available. Additionally, two external

studies were examined for the effect of EDDM on self-response rates and costs compared to standard postcards. The results and recommendations of this research will be discussed, focusing on the impact on self-response rates and how the EDDM design influences respondents' confidence to respond.

Optimizing Mailing Cohort Assignments

Karlesha G. LeGrier, *U.S. Census Bureau*

Theodore Charm, *U.S. Census Bureau*

Ivan J. Cardona-Tomassini, *U.S. Census Bureau*

Faraz Shaikh, *U.S. Census Bureau*

During the 2020 Census, addresses that received the Internet First contact strategy were assigned to staggered mail cohorts based on operational factors and self-response rate projections. This approach was designed to manage demand on Census Questionnaire Assistance call centers and the internet instrument. Using 2020 Census self-response data, this research evaluates the effectiveness of the cohort assignments to develop recommendations for the 2030 Census. We compare actual and projected self-response rates for each mailing cohort, evaluate self-response rates and instrument login patterns, and assess the relationship between the daily self-response projections and final self-response data. We will discuss these findings, including specific recommendations for refining the allocation of addresses to cohorts and optimizing the mail delivery timeframe to improve operational efficiency.

Evaluating Contact Strategy and Bilingual Mail Material Assignments

Faraz Shaikh, *U.S. Census Bureau*

Karlesha G. LeGrier, *U.S. Census Bureau*

Ivan J. Cardona-Tomassini, *U.S. Census Bureau*

Theodore Charm, *U.S. Census Bureau*

In the 2020 Census, households assigned to Internet First areas were sent an invitation to respond online or over the phone in the first mailing, while households assigned to Internet Choice areas were sent a paper questionnaire in addition to the invitation to respond online. The bilingual mail materials were sent to areas that had relatively higher populations of Spanish speakers who did not speak English very well. Non-bilingual areas only received English language materials. This research evaluates potential refinements in how contact strategy and language of the mail materials are assigned for the 2030 Census. Actual and projected 2020 Census self-response rates were compared for each assignment. Regression models using 2020 Census data and American Community Survey (ACS) 5-year estimates were used to determine which new ACS characteristics were associated with paper self-response rates for contact strategy and Spanish self-response rates for bilingual mail materials. ACS 5-year internet access data were also used instead of Federal Communications Commission (FCC) data to retrospectively assign contact strategy and compared to the original assignments. This presentation discusses the results of the research and recommendations for the 2030 Census.

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Thursday, March 12, 2026

1:15 PM

Session G-X: Federal Committee on Statistical Methodology (FCSM) Geospatial Interest Group (GIG) Un-Session

Organizer: Ed Strocko, *Bureau of Transportation Statistics*

Chair: Ed Strocko, *Bureau of Transportation Statistics*

Census Training Room 1 (T-1)

Data is more than just numbers; it's about place. The FCSM Geospatial Interest Group (GIG) invites you to an un-session focused on the intersection of geography and federal statistics. Since 2014, we've been the collaborative hub for federal employees— experts and non-experts alike — to share ideas, tackle challenges,

and explore innovations in analyzing and visualizing geospatial data for federal statistics. No lectures here—just idea sharing, collaborative problem-solving and innovation. If you're ready to explore how geospatial insights can transform federal data, pull up a chair and let's get to work!

Thursday, March 12, 2026

3:00 PM

Session H-1: Synthetic Data: Applications Across the Federal Statistical System

Organizer: Michael Hawes, *U.S. Census Bureau*

Chair: Amy Lauger, *Bureau of Justice Statistics*

Discussant: Michael Hawes, *U.S. Census Bureau*

Census Auditorium 1

NCHS Data Linkage Program: Exploring Privacy Enhancing Technologies for Data Linkage

Cordell Golden, *National Center for Health Statistics*

The National Center for Health Statistics (NCHS) has a long-established data linkage program that links NCHS health survey data with vital and other administrative records to expand the analytic potential of both the survey and administrative data. Integrating survey and administrative data produces rich analytic resources that can be used to support public health surveillance, evidence-based policymaking, and patient-centered outcomes research. However, combining data from multiple sources can also increase disclosure risk. To mitigate this risk, disclosure avoidance strategies can be applied to the input files prior to linkage or the output files post-linkage. The NCHS Data Linkage Program has been exploring the use of privacy enhancing methodologies, such as privacy preserving record linkage (PPRL) to address input privacy and synthetic data generation to address output privacy. This talk will provide an overview of NCHS' implementation and evaluation of selected PPRL methodologies and describe a pilot project to generate statistically valid synthetic linked data files. The talk will conclude with a discussion of next steps in NCHS' exploration of privacy enhancing techniques.

Balancing Risk and Utility: The IRS's Fully Synthetic Public Use Files

Victoria Bryant, *Statistics of Income*

In collaboration with the Urban Institute and the Tax Policy Center, the IRS Statistics of Income (SOI) Division has produced fully synthetic public-use tax data files that represent the entire U.S. individual income tax population. This effort has resulted in the release of two key datasets: the Synthetic Public Use File (SynPUF), which models the tax filing population, and the Supplemental SynPUF, which includes individuals below the filing threshold. These releases mark a significant shift away from the legacy Public Use Files (PUFs) that relied on traditional statistical disclosure control (SDC) techniques. Instead, SOI now employs nonparametric machine learning methods, implemented via the "tidysynth" R package, to generate synthetic records that preserve statistical utility while protecting taxpayer confidentiality. This presentation will outline the methodology behind the synthetic file creation and the comparative advantages of this transition. A key focus will be the tradeoffs between data utility and disclosure risk, assessed using a comprehensive suite of metrics across four dimensions: synthesizer precision, copy protection, membership inference, and attribute inference. The introduction of fully synthetic data, ultimately paired with validation servers, opens the door to wider, more equitable research access to tax data. It enhances support for policy analysis, promotes transparency, and enables rigorous social science—all while safeguarding taxpayer privacy.

Fully synthetic data for the American Community Survey

Michael Freiman, *U.S. Census Bureau*

Evan Totty, *U.S. Census Bureau*

Aref Dajani, *U.S. Census Bureau*

To continue providing access to high-quality data while maintaining respondent confidentiality, the Census Bureau is researching the feasibility of a fully synthetic dataset for the American Community Survey, to be accompanied by a service allowing users to validate their results on the original internal data. Synthesis follows a sequential tree-based method, where a tree is fit on each variable based on the previously synthesized variables. A synthetic value for the variable is drawn at random from the appropriate leaf. To maintain intra-household relationships, we synthesize some “stub variables” for households and people within them together, and then the rest of each person and household record is synthesized separately. For larger households, a separate matching routine adds additional synthetic person records to the household until it reaches the appropriate size.

To assess the quality of the synthetic data, we are developing a question bank, including thousands of modeled and descriptive statistics that we have produced and compared between the original data, the current version of the synthetic data, and the existing public use microdata. We focus on whether one would draw the same statistical conclusion from the synthetic data as from the internal data. To assess the privacy of the synthetic data, we use a variation of the re-identification studies that the Census Bureau has conducted on several surveys. We simulate an attack where an external intruder has access to the unprotected internal data and matches these records to records on an external file with personally identifiable information. We assess whether the inclusion or exclusion of re-identified records in the training data for synthesis affects the frequency of records similar to the re-identified records in the synthetic data.

Protecting Output from a Synthetic Data Validation Service

Gary Benedetto, *U.S. Census Bureau*

The Census Bureau is currently researching the possibility of providing more accurate answers to custom statistical queries with more effective privacy protections and to a broader audience than traditional tiers of access have offered in the past. Pre-packaged tables can be very limiting in the types of questions they can answer. Traditional public use microdata can be very risky from a privacy perspective, and, as we discover and address more of their vulnerabilities, these public files become less accurate in ways that are not transparent to external users. Access to secure data enclaves is very resource intensive, and, therefore, very limiting in the audience it can reach.

Synthetic data coupled with a new tier of access called a validation service offers the potential to address all of these concerns. In this tier, external users can develop analyses using the synthetic data, submit their mature analyses to the validation service, and receive disclosable results from the internal data. However, a validation service also poses some significant challenges. Perhaps chief among these challenges is performing statistical disclosure limitation (SDL) on the validated output in a scalable fashion. We will discuss some of our ongoing research regarding automated SDL that would be similar in spirit to some of the protections users of Census Bureau public use microdata are accustomed, while also offering greater transparency by more properly accounting for the SDL in corresponding measures of uncertainty.

Steps to Tiered Access: Use of Synthetic Data

Lisa Mirel, *National Center for Science and Engineering Statistics*

John Finamore, *National Center for Science and Engineering Statistics*

Synthetic data generation can be used to create datasets that do not contain the exact records of the original dataset but instead retain the statistical properties of the original data. The anonymity of the original data is not compromised since synthetic data do not directly correspond with the original values. Creating synthetic data allows for a tiered access model to maximize access to data while ensuring privacy protection. However, producing synthetic data can prove challenging and resource intensive. The federal statistical community has been exploring and, at times, releasing synthetic or partially synthetic datasets for many years. With emerging needs for more access to data, particularly to help train AI/machine learning models, generating synthetic data and generating it in a way that balances privacy and utility is now at the forefront of many conversations. This

talk will focus on how the creation of synthetic data has supported the National Secure Data Service (NSDS) Demonstration Project goals to inform a governmentwide effort to strengthen data access infrastructure for data-driven decision making. The talk will provide background on the NSDS with a focus on providing solutions to data access through open source synthetic data generation. In addition the opportunities and challenges with synthesizing data and socializing their use will be highlighted. The talk will conclude with a discussion of future directions on increasing access to data while maintaining privacy protections in a shared service environment.

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Thursday, March 12, 2026

3:00 PM

Session H-2: Statisticians don't code, they prompt: Modernizing methods with AI

Organizer: Jeramiah Yeksavich, *Energy Information Administration*

Chair: Jeramiah Yeksavich, *Energy Information Administration*

Census Auditorium 2

Nowcasting Distributional National Accounts for the United States: A Machine Learning Approach

Marina Gindelsky, *Bureau of Economic Analysis*

Gary Cornwall, *Bureau of Economic Analysis*

Inequality statistics are usually calculated from high-quality, comprehensive survey or administrative microdata. Naturally, this data is typically available with a lag of at least 9 months from the reference period. In turbulent times, there is interest in knowing the distributional impacts of observable aggregate business cycle and policy changes sooner. In this paper, we use an elastic net, a generalized model that incorporates lasso and ridge regressions as special cases, to nowcast the overall Gini coefficient and quintile-level income shares. We use national accounts data starting in 2000, published by the U.S. Bureau of Economic Analysis, as features instead of the underlying microdata to produce a series of distributional nowcasts for 2020–2022. These nowcasts predict turning points with at least 85% accuracy across all metrics and minimal errors relative to naive models. We find that we can create advance inequality estimates approximately one month after the end of the calendar year, reducing the present lag by almost a year.

Generative Artificial Intelligence (Gen AI) Research: Conversion of Legacy NASS Software into Python and R

Sean Rhodes, *National Agricultural Statistics Service*

Ricardo Millan, *United States Department of Agriculture*

Noelle Dullaghan, *United States Department of Agriculture*

Darcy Miller, *National Agricultural Statistics Service*

Denise A. Abreu, *National Agricultural Statistics Service*

The United States Department of Agriculture's (USDA) National Agricultural Statistics Service (NASS) conducts hundreds of surveys each year to estimate nearly every facet of agriculture in the U.S. NASS utilizes thousands of software programs written in various programming languages to accomplish the agency's mission of providing timely and accurate statistics in use to U.S. agriculture. Programs range in complexity and scope, including data analysis, sampling, and publishing official statistics. Most programs are written in SAS and require an increasingly expensive license fee to run. As a result, NASS has set a goal to convert production SAS code to free software, such as R or Python. Due to the unique features of each programming language, this code conversion process can be complex and time-consuming. NASS is exploring the use of Generative Artificial Intelligence (Gen AI) to assist with these translations and help reduce staff burden. As part of this research effort, several production scripts have been translated along with specific metrics to quantitatively measure

translation impact. Also, a variety of testing approaches are evaluated to ensure code accuracy. This presentation discusses the utility of leveraging Gen AI for translating legacy code to test and assesses the variability in translation across different prompts and models.

Improving Translation Documentation and Expert Reviews Through AI Integration

Betsarí Otero Class, *U.S. Census Bureau*

Patricia L. Goerman, *U.S. Census Bureau*

Marcus P. Berger, *U.S. Census Bureau*

The Coverage Improvement Methods team at Census often conduct Spanish survey translation expert reviews for various surveys. The purpose of the translation expert review is to evaluate the linguistic accuracy, cultural appropriateness, meaning, and tone of the translated material matches and conveys the same meaning as the source text. Our current review processes rely on different files, varied reviewer notes, and individual institutional knowledge of past testing, making it difficult to ensure continuity and consistency in wording when team members transition or new staff participate.

This project addresses these challenges by developing a centralized, Excel-based database to document and manage Spanish survey translations and expert review recommendations, focusing on the “Documentation” component of the TRAPD method (Translation, Review, Adjudication, Pretesting, and Documentation). The database is being developed with the help of various AI tools to extract existing translations from multiple projects. AI will reduce manual workload for the team while maintaining quality through reviews throughout the compilation of these translated files. The database includes the English source text, Spanish translations for review, historical versions with revision rationale, expert review comments, and topic classifications. The database aims to improve translation consistency, streamline expert review, and preserve institutional knowledge by ensuring that all relevant materials remain accessible to current and future project teams.

Centralizing resources and integrating both human expertise and AI helps create a model for translation management that can be adapted to other multilingual survey contexts. The presentation will share the project’s methodology, early findings, and lessons learned, with a focus on practical strategies for maintaining translation documentation and continuity over time.

HRTnomaly: Historical, Relational, and Tail Anomaly-Detection Algorithms

Justin van Wart, *National Agricultural Statistics Service*

Luca Sartore, *National Institute of Statistical Sciences*

Lu Chen, *Capital One*

Andrew Dau, *National Agricultural Statistics Service*

Valbona Bejleri, *National Agricultural Statistics Service*

Data acquired through repeated surveys are characterized by a variety of complex relationships that make their review and vetting process time-consuming. Therefore, the necessity of maintaining a high standard of data quality becomes more difficult to sustain in relatively short time frames. Thus, the United States Department of Agriculture’s (USDA’s) National Agricultural Statistics Service (NASS) has implemented a semi-automated revision process to improve the accuracy of the information acquired through surveys and, consequently, the accuracy of final estimates. Although some level of human intervention will remain necessary for complex cases, an anomaly detection system based on decision rules may not detect anomalies that break linear relationships. As a result, NASS has been investigating alternative approaches to modernize its anomaly detection system with more sophisticated unsupervised learning methods. For instance, some of these methods are implemented in the R-package HRTnomaly. In this presentation, we review novel methodologies that identify four types of anomalous data entries (or cellwise outliers): format-inconsistent, historical, tail, and relational anomalies. Anomaly scores are computed using a variety of distribution-free methods for all four types of anomalies. Fuzzy logic and Bayesian approaches are successively used to combine the anomaly scores for identifying anomalous data entries. NASS survey data for livestock and corn yield are considered for comparing the proposed methods with existing anomaly detection algorithms.

Dynamic Outlier Detection using Machine Learning in the Producer Price Index

Chelsea Velic, *Bureau of Labor Statistics*

Sarah Eian, *Bureau of Labor Statistics*

Monica Moleres, *Bureau of Labor Statistics*

Brendan Hadder, *Bureau of Labor Statistics*

Arthur K Edouard, *Bureau of Labor Statistics*

When constructing the monthly Producer Price Index (PPI), economists at the Bureau of Labor Statistics (BLS) validate price changes that exceed tolerance levels set within the PPI processing system. Historically, PPI economists used prior industry knowledge to set initial industry tolerance levels when a new sample was introduced. Tolerance levels remained static across processing months unless an economist changed the values based on updated industry information. The BLS recently introduced a machine learning model to improve outlier detection in micro data for the PPI program. This model is now used to estimate and provide PPI economists with monthly-updated, data driven tolerance boundaries to validate price changes. In developing this tool, BLS compared various statistical models' estimates of industry tolerance thresholds. After assessing models based on accuracy, generalizability, computational efficiency, and interpretability, BLS selected an Auto-ARIMA model. Historical industry price data are now fed into the model, and the outputs are used to calculate monthly-updated upper and lower tolerance boundaries on an industry-by-industry basis. BLS economists review these industry tolerances through a POSIT dashboard to determine if they should be used in the upcoming month of price review. Since this additional validation process is new to the PPI processing system, BLS is working on evaluating its effect on micro data review efficiency, aggregated index data review, and resource allocation. The end goal of this project is to both improve the quality of and create operational efficiencies in outlier detection and microdata review by using a more dynamic method for setting price tolerance levels.

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Session H-4: Modernizing Survey Instruments: Evaluations of Visual Design Decisions

Organizer: Keith Finlay, *U.S. Census Bureau*

Chair: Keith Finlay, *U.S. Census Bureau*

Census Conference Rooms 1-2

Not Just Respondents: Understanding the effect of questionnaire design on interviewer behavior

Erica Yu, *Bureau of Labor Statistics*

Victoria Narine, *Bureau of Labor Statistics*

The visual display of survey questions is an ongoing area of research and innovation for self-administered surveys. Previous research has found that the visual representation of response options, including their order, layout, and visibility, influence response in self-administration (e.g., Stern, Dillman, & Smyth, 2007). The current research explores whether a similar effect is seen among interviewers in interviewer-administered surveys.

The Consumer Expenditure (CE) Interview Survey is a computer-assisted personal interviewing (CAPI) survey. Most of the questionnaire is organized into 'laundry lists,' whereby a series of questions share a question stem. Each question is designed to be asked aloud and elicit a yes/no response. The CAPI instrument displays these questions in one of two designs. In the first design, questions are grouped together on a single screen, like grids in self-administered surveys. Research has shown that grids are vulnerable to straight lining, missing data, and satisficing (e.g., Tourangeau, Couper, & Conrad, 2004), raising the possibility that laundry lists are vulnerable to interviewers skipping and combining questions to reduce the number of questions asked. In the second design, questions that share a question stem are displayed item-by-item on separate screens. These two

question types are intended to function the same way from the respondent's perspective. To an interviewer administering the questions, though, these two question designs look very different.

We used audio recordings of interview interactions to investigate how visual representation of questions affects administration. Our sample comprises over 100 interviewers, each with about 5 interviews. We listened to recordings of the two question designs within each interview and coded the question-asking style and whether each item in the series was read aloud. The findings from this research will be applied to future instrument design and can inform research on how instrument and questionnaire design affect interviewer behaviors.

Best Practices by Design: Optimizing Layouts and Response Instructions in Web Surveys

Caitlyn Keeve, *U.S. Census Bureau*

Casey Eggleston, *U.S. Census Bureau*

As more federal surveys incorporate internet self-response (ISR) as a data collection mode, it is increasingly important for survey methodologists in the federal statistical system to understand and use standardized best practices for ISR instruments. To that end, the U.S. Census Bureau has developed a set of web survey design guidelines that incorporate the available evidence about optimal design practices and govern ISR instrument development at the Census Bureau. These guidelines are updated regularly to incorporate new evidence, and research teams at the Census Bureau periodically conduct qualitative testing and survey experiments to fill in knowledge gaps related to specific practices related to web survey methodology. This presentation provides the results of two such experiments that were incorporated into the 2025 first quarter recruitment questionnaire for the Household Trends and Outlook Pulse Survey (HTOPS), a national probability panel administered by the Census Bureau. The first experiment looks at the data quality implications of including or excluding a "Select all that apply" instruction for multiple-select survey questions. The second experiment investigates the impact of presenting multiple questions with a shared set of response options in either a grid format or item-by-item for respondents completing the ISR instrument on a larger screen (like a laptop or desktop). This presentation summarizes the results of these two experiments, with the goal of providing additional evidence in support of best practices for ISR instruments.

Does Formatting Matter: Testing Visual Variations in Survey User Interface Elements

Cameron Pedretti, *U.S. Census Bureau*

Rachel Horwitz, *U.S. Census Bureau*

Leann Weyl, *U.S. Census Bureau*

Laurin Contard, *U.S. Census Bureau*

Michael Risley, *U.S. Census Bureau*

Christine Bottini, *U.S. Census Bureau*

The American Community Survey (ACS) began offering a web option in 2013. While the content of the instrument has evolved over time, its overall format and appearance have remained unchanged. As part of an effort to modernize U.S. Census Bureau surveys and ensure they align with industry best practices as well as inform web standards, the ACS conducted a fully factorial experiment. There were three treatment groups and a control group to test the impact of a standard radio button/check box format versus a response button format, where the response option and the corresponding radio button or check box is surrounded with a box that highlights when hovered over or selected. It also tested the original edit message format, which has been used since 2013 versus one designed by the U.S. Web Design System as well as the combined effect of the response and edit message format.

We hypothesize that the response buttons will be more user friendly than traditional radio buttons and check boxes because the clickable area is larger, better defined, and the highlighting makes it easier to select the intended option. We do not expect there to be an impact on data quality. Similarly, we do not expect the format of the edit messages to have an impact because the wording is not changing. To evaluate these hypotheses, we examine data quality metrics, including response distributions, edit message triggers and corresponding corrections. Also included in the analysis is item nonresponse as well as usability metrics, such as time on task and answer changes. This experiment will determine whether modern formatting will have any impacts on data quality and the user experience for the ACS and other Census Bureau surveys.

A study on grid design: Can reducing the number of screens improve respondent burden without sacrificing data quality for web-collected income data?

Elizabeth Pelletier, *U.S. Census Bureau*

Lisa Cheok, *U.S. Census Bureau*

Katherine Engel, *U.S. Census Bureau*

Melissa Kollar, *U.S. Census Bureau*

The Current Population Survey Annual Social and Economic Supplement (CPS ASEC) captures data on household income in the United States. The Census Bureau is introducing an internet self-response (ISR) data collection mode for the CPS ASEC. A concern with ISR is maintaining data quality and comparability with other modes without increasing respondent burden. Specifically, one challenge is that survey instruments may become very long. The CPS ASEC is testing a grid design to mitigate this concern. The grid design leverages the visual nature of ISR to collect multiple pieces of information on the same screen. This design reduces the number of screens a self-respondent sees compared to the item-by-item format currently in use.

In this project, we compare the two designs for income amount collection using a mixed-methods approach. First, we analyze data from a split-panel randomized design implemented on the Census Household Panel survey. Respondents were randomly assigned either the item-by-item design (the control group) or the grid design (the test group). We also report findings from cognitive interviews that gathered qualitative data on the grid design. This presentation will focus on preliminary results and next steps. The findings from these tests are being used to refine ISR questionnaire design, both for the CPS ASEC and other Census surveys.

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Thursday, March 12, 2026

3:00 PM

Session H-6: Innovation before inference – new steps along the way to estimation

Organizer: Brad Rhein, *Bureau of Labor Statistics*

Chair: Alice Yu, *Bureau of Labor Statistics*

BLS Auditorium A

Examining the Impact of New Statewide Intercensal Population Controls for Estimates of Employment and Unemployment

Andrew C. Forrester, *Bureau of Labor Statistics*

Sean B. Wilson, *Bureau of Labor Statistics*

The Local Area Unemployment Statistics (LAUS) program at the U.S. Bureau of Labor Statistics (BLS) produces monthly estimates of employment and unemployment for states and select substate areas. Each year, the LAUS program controls survey-based estimates of employment and unemployment from the Current Population Survey (CPS) to match independent population controls produced by the U.S. Census Bureau. In March 2025, LAUS introduced new intercensal population controls that reconcile decadal differences between population projections from one census count to the next. These differences, or errors of closure, presented issues when making comparisons over time, as shifts from one population vintage to the next introduced historical breaks around census years in state-level labor force indicators. We show the extent of these time series breaks historically and demonstrate how the revised population estimates improve LAUS estimates of labor force participation rates and employment-population ratios.

Missing the Lunch Rush: Downtown Workforce Dynamics and the Food-Away-From-Home Economy

Eliana Zeballos, *Economic Research Service*

The food-away-from-home (FAFH) sector—particularly in downtown areas—is closely linked to the presence of a daytime workforce. This study uses the COVID-19 pandemic as a natural experiment to examine how reductions in workplace activity reshape the FAFH economy. Drawing on anonymized, high-frequency mobility data from 2018 to 2022, we track weekly foot traffic to restaurants and other FAFH establishments across major U.S. downtowns. We define downtowns using the National Establishment Time Series, identifying census blocks with dense office agglomerations, and construct indices of workplace presence to measure daytime worker density before, during, and after pandemic disruptions. Using a difference-in-differences framework, we estimate the causal effects of reduced workplace presence on FAFH visits, comparing downtown areas with high concentrations of office employment to non-downtown areas with lower exposure to remote work.

Findings provide new evidence on how workplace dynamics drive urban food demand and how persistent shifts toward remote and hybrid work may alter the structure of the FAFH sector. This work is relevant for policymakers and industry, as FAFH accounts for a growing share of U.S. food expenditures and plays a central role in employment, food access, and agricultural demand. By linking workplace-driven changes in FAFH consumption to the agricultural supply chain, the study informs strategies for strengthening food system resilience and supporting both producers and downstream markets. Results also shed light on how evolving food purchasing behaviors influence diet quality, access to prepared foods in urban areas, and the stability of markets dependent on foodservice demand. Together, these insights advance understanding of the long-term implications of workforce shifts for the U.S. food economy.

The Seasons They Are A-Changin': A Century of Definitions and a Way Forward

Carter Bryson, *Bureau of Economic Analysis*

Gary Cornwall, *Bureau of Economic Analysis*

What is seasonality? To date, there is not yet a consensus definition, as over a dozen unique definitions of seasonality can be found in the economics and statistics literature. This lack of consistency complicates the identification of, and adjustment for, seasonal effects in economic time series. In this paper, we propose a new definition of seasonality as well as how to detect and adjust for it, ultimately aiming to build consensus around a more unified approach. We first review the literature and identify common themes across definitions with an eye towards formulating a foundational definition – expressed in both plain language and mathematical terms – that can be used by academics, data providers, policymakers, and practitioners alike. We then propose to classify a series as seasonal if, among all its possible repeating cycles, the ones associated with the fundamental periodicity are demonstrably larger than the others. In other words, we define seasonality as a measure of relative peak dominance in the spectral density function, following Granger (1978). Next, we develop a new methodology for detecting seasonality and performing seasonal adjustment that follows directly from our definition and leverages properties of the sample periodogram, the empirical counterpart to the spectral density function. We illustrate the performance of our test statistic through simulation studies and find that it is well-sized and has good power. Lastly, we introduce Stochastic Spectral Imputation (SSI), an adjustment procedure that outperforms standard methods currently used by national statistical offices when applied to prominent macroeconomic time series.

Model for Preliminary Estimation of the Chained CPI-U Using Hierarchical Echo State Networks

Kate Eckerle, *Bureau of Labor Statistics*

Erin Boon, *Bureau of Labor Statistics*

Calculation of the Chained CPI-U requires monthly item-area expenditure shares from the same time-period as item-area price index relatives. Cell-level expenditure data become available 10-12 months after the price data, however, which precludes timely estimation of the index as defined using the Tornqvist equation for aggregation. In the interim, the BLS issues a preliminary estimate of the index using (fixed) reference period expenditure share data and a Constant Elasticity of Substitution (CES) model. We propose an alternative method for preliminary estimation that retains the Tornqvist formula for aggregation and forecasts the missing

item-area expenditure data using a set of hierarchical Echo State Networks (ESNs), in an effort to improve the accuracy of early estimates. An ESN is a type of Recurrent Neural Network in which internal couplings are randomized, as well as those between neurons and inputs, and only output couplings (connecting neurons to responses) are estimated. ESNs are flexible, nonlinear, hidden variable models that can predict series with complex temporal dynamics after a relatively simple training process. We develop an iterative procedure to forecast a vector of item expenditures for a given area based on its past expenditure data (endogenous predictors) as well as past and concurrent price data (exogenous predictors), over multi-step horizons. Additionally, we include the option to supplement the ESN neuron states with discrete Fourier modes at the seasonal frequencies to improve prediction among items with strong seasonal components. We explore randomized grid search and simulated-annealing approaches to hyperparameter selection, aggregating the results from multiple validation periods to better estimate future model performance. We compare example results from our prototype models to estimates from the CES model currently used in production.

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Thursday, March 12, 2026

3:00 PM

Session H-7: Creating Accuracy: Advancing Sample Coverage, Sampling Frames, and Sample Outcomes

Organizer: Jonathan Law, *Economic Research Service*

Chair: Jonathan Law, *Economic Research Service*

BLS CTC 2A

Clopper Pearson All-or-Nothing Estimation of Population Totals

Kendal Ferguson, *Department of Veterans Affairs*

When characterizing populations, we often estimate proportions (e.g., the proportion of medical claims containing an error), related counts (e.g., the number of medical claims containing an error), and totals based on these counts and an auxiliary variable (e.g., the sum of the dollar amounts of all medical claims containing an error). We denote these totals as “all or nothing” because they include either the entire value of the auxiliary variable for a population unit (e.g., the entire value of \$100 if the claim is for \$100), or none of this value.

When constructing confidence intervals, we typically assume our sampling distribution is normal, an assumption which may be poor when sample sizes are small. Fortunately, with equal-probability sampling, we can construct conservative Clopper Pearson intervals for proportions, and related counts, based on the binomial sampling distribution. However, we are generally unable to do so when estimating totals. As such, small sample sizes may result in indefensible confidence intervals for totals.

In this paper, we explore probability-proportional-to-size (pps) sampling as a means to construct Clopper Pearson confidence intervals around totals, regardless of the sample size. This method is applicable when (1) the sampling unit can be classified binarily, (2) the total is based on the sampling unit’s measure of size, and (3) the total is based on all-or-nothing amounts. We provide a mathematical proof for our method’s validity. Although this method is rather simple, we cannot find any evidence of its presentation, application, or proof, in the literature. This method will be of interest to statisticians who support performance auditors in need of defensible estimates of all-or-nothing population totals based on small sample sizes.

Using space filling curves to design a well spread and balanced sample

John Hogland, *U.S. Forest Service*

Andrew Lister, *U.S. Forest Service*

Charles Scott, *U.S. Forest Service*

David Affleck, *University of Montana*

Hilbert space filling curves are ideally suited to partition and group m-dimensional space into locally ordered one dimensional space. Expanding on the idea of m-dimensional space filling curves, we present a new approach to creating well spread and balanced samples using Hilbert space fitting curves. Our approach

systematically samples from encoded Hilbert distance and like other spread and balance techniques such as GRTS and the local pivotal method, reduces the average nearest neighbor distance between selected locations. As a result, it significantly reduces sampling error for attributes exhibiting localized structure. Our method is easy to implement, is quick, and can be easily used to encode and decode data into Hilbert distances and back to positions in coordinate and attribute space. For a study site in Angola, Africa, we demonstrate the utility of spreading and balancing sample observations across Landsat 8 spectral band, Copernicus digital elevation model, and coordinate space values using a Hilbert space fitting curve, Python, GitHub, Jupyter notebooks, and Google's Colab.

Assessing the Relative Contribution of Demographic and Neighborhood Effects on Census Coverage

Julianne Zamora, *U.S. Census Bureau*

Each decade the U.S. Census Bureau measures the coverage of the decennial census. Since the 1980 Census, census coverage measurement has involved dual-system estimation via a post-enumeration survey. The dual-system estimates in 1980, 1990, and 2000 used post-stratification to minimize the impact of the failure of heterogeneous independence (Sekar and Deming, 1949). Post-stratification involves grouping individuals likely to have similar census inclusion probabilities. The 2010 and 2020 coverage measurement programs implemented logistic regression to replace post-stratification with the same underlying principle.

A perennial question for coverage measurement programs is which factors affect census inclusion and can be used to create post-strata. Traditionally, demographic characteristics such as race, Hispanic origin, age, sex, and housing tenure have been used to control heterogeneity. Hengartner and Speed (1993) used data from the 1990 Post-Enumeration Survey to investigate the homogeneity in demographic post-strata compared with post-strata created by the geographic block. For this paper, we revisit this line of research to develop methods using the Demographic Frame (a collection of Administrative Records) in place of a post-enumeration survey.

Evaluating the Quality of the Census Bureau's Demographic Frame

Timothy Kennel, *U.S. Census Bureau*

The U.S. Census Bureau's Demographic Frame, a composite frame derived from administrative records, is a critical input for many Census Bureau activities, including the 2026 Census Test. Ensuring its quality is paramount for numerous enhancements and efficiency gains to under consideration for the Census Bureau. This presentation introduces a comprehensive set of summary indicators designed to evaluate the overall quality of this frame.

Our methodology focuses on estimating content and coverage errors. This presentation will describe the methods and metrics we will use to estimate Demographic Frame coverage at national, state, county, and tract levels. We will discuss our framework for estimating and analyzing content error on the Demographic Frame. These coverage and content indicators will not only highlight the strengths and limitations of the Demographic Frame, but also guide efforts to improve the Demographic Frame.

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Session H-8: Engaging Respondents: Tools to Improve Participation and Data Quality

Organizer: Natalie Young, *National Center for Health Statistics*

Chair: Natalie Young, *National Center for Health Statistics*

BLS Auditorium C

Comparing English and Spanish Language Respondents on the Rapid Surveys System

Valerie Ryan, *National Center for Health Statistics*

Katherine Irimata, *National Center for Health Statistics*

Jim Dahlhamer, *National Center for Health Statistics*

The National Center for Health Statistics (NCHS) Rapid Surveys System (RSS) is a series of web-based surveys conducted using commercial survey panels. Questionnaires are offered to panelists in English or Spanish based on the language preference of the respondent. Each round of RSS typically has between 3-4% of completed questionnaires conducted in Spanish. We investigated differences between English and Spanish language respondents on RSS rounds 1-4 to understand how Spanish language respondents differ from English language respondents and how those differences may impact estimates. We also compared RSS estimates to those of English and Spanish language respondents on the National Health Interview Survey (NHIS). Descriptive analyses of RSS data showed that Spanish language respondents had lower education levels, higher unemployment, higher poverty rates, and higher food insecurity, compared to English language respondents. Modeling was performed to further evaluate differences between RSS English and Spanish language respondents on selected health outcomes while adjusting for sociodemographic features. Our presentation summarizes results of these models as well as the comparisons to health estimates on the NHIS.

Using a Time Capsule Question to Increase Survey Engagement

Bonnie Moore, *U.S. Census Bureau*

Could the inclusion in the 2030 Census of an optional time-capsule question for respondents to write an open-ended message increase self-response and engagement, as well as provide valuable and interesting information to the public upon release 72 years later? A time-capsule question can provide a space for the public to share stories and make the Decennial Census have a more personalized feel. Multiple data collections provided insights into the feasibility and efficacy of this idea for the 2030 Census. We present findings on the effects of this new question on key metrics such as self-response rate, completion time, item nonresponse, and duplicate submissions. The results demonstrate how an element like the time capsule can enhance self-response and public interest with minimal disruption to data collection.

Using Images, Motivational Messages, and Additional Information to Probe Don't know or Skipped Answers on the Web

Maxwell Hope, *U.S. Census Bureau*

Renee Stepler, *U.S. Census Bureau*

While interviewer-administered surveys commonly allow "Don't know" or "Refused" answers, many web surveys use these types of responses sparingly instead allowing respondent to skip the question without providing an answer. In an interviewer-administered environment, the interviewer can use probes to motivate the respondent to report, encourage them to estimate if they are unsure, or can provide additional help information to enable the respondent to provide an answer. However, this interaction can be challenging to replicate on the web. Prior work has shown motivational follow-ups are useful in encouraging valid answers to replicate this interviewer-respondent interaction on the web.

As major household Census Bureau surveys begin to integrate a web response mode, whether to add a "Don't Know" option and how to encourage respondents to provide information when they are less certain are vital. Many of these surveys contain technical information and detailed questions and allow proxy reports which may

require providing more information to respondents. Using the Census Household Panel, this study tested using an explicit “Don’t know” option against a control group without a “Don’t know” option. We test follow-up messages to encourage respondents who skipped or selected “Don’t know” to select an option. The follow-up probes include messages to remind respondents that the question is important for the survey, additional information to help the respondent select an answer, or images to assist the respondent in selecting the most optimal option. A “Don’t know” option is an available option on the follow-up screen for both the experimental and control group.

Our results show that providing images and additional information reduces item nonresponse for both those who received a “Don’t Know” in the initial questions and those who did not. Therefore, these findings contribute a concrete method of how to enhance data quality in federal statistics.

Do I Live Here or Not Live Here? That is the Question: A Split Panel Experiment to Examine the Use of Behavioral Language to Define Household Membership

Kathleen Kephart, *U.S. Census Bureau*

Eric Stone, *U.S. Census Bureau*

Maxwell Hope, *U.S. Census Bureau*

Rostering is critical for surveys that need to collect data on all people who live at the sampled address. For many households, rostering is a simple exercise, but for those who live with unrelated families, people staying for undefined periods of time, and other complex living situations, rostering procedures can be difficult. Past research has shown inconsistent interpretations of words common to rostering procedures, such as “family”, “household”, “live,” and “stay.” A previous qualitative study (Piccirillo, Kephart, Stone, and Kohm 2024) tested the use of objective behavioral measures, such as sleeping and receiving mail to help respondents decide who to include in a household roster.

For this research, we conducted a split-panel experiment fielded via the Census Bureau’s Household Panel. This is a probability-based nationwide panel that periodically fields surveys (or “topicals”) to panel members. The main objective of the experiment was to compare a rostering procedure that asks about a defined behavior (i.e. sleeping) to a standard rostering procedure commonly used by the U.S. Census Bureau. Additionally, we compared asking about “living” to asking about “living or staying” in various rostering question stems to assess how the wording differences impact response distributions. A “measure of truth” was also obtained by a series of follow-up questions about additional people with any connection to the address such as having a government ID listing the address, receiving mail at the address, having a key to the address, etc. Follow-up questions were also asked about how often these people stayed at the address and their general living situations.

Findings from this research comparing the different rostering procedures will be presented and recommendations will be made that may be beneficial to federal surveys that rely on household rostering.

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